

ACOPOSmicro

User's Manual

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Model number: **MAACPMICRO1-ENG**

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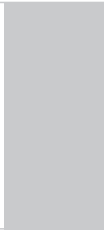
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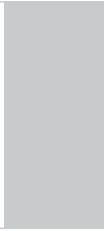
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Chapter 1 • General information

1. Manual history

Information:

B&R does its best to keep the printed versions of its user's manuals as current as possible. However, newer versions of the User's Manual are always available first for download in electronic form (PDF) from the B&R homepage www.br-automation.com.

Version	Date	Comment
0.10	November 2009	First edition

Table 1: Manual history

2. Safety guidelines

Information:

These safety guidelines, the connection specifications (type plate and documentation) and limit values listed in the technical data are to be read carefully before installation and commissioning and must be observed.

These safety notices must be kept and included with the device if it's passed on to anyone else (e.g. sold, rented, etc.).

The user is responsible for the observance of all relevant international and national electrical standards.

2.1 Organization of safety notices

Safety notice	Description
Danger!	Disregarding the safety regulations and guidelines can be life-threatening.
Warning!	Disregarding the safety regulations and guidelines can result in severe injury or major damage to material.
Caution!	Disregarding the safety regulations and guidelines can result in injury or damage to material.
Information:	Important information for preventing errors.

Table 2: Description of the safety notices used

2.2 General information

B&R drive systems and motors have been designed, developed and manufactured for conventional use in industry. They were not designed, developed, and manufactured for any use involving serious risks or hazards that could lead to death, injury, serious physical damage, or loss of any kind without the implementation of exceptionally stringent safety precautions. Such risks include in particular the use of these devices to monitor nuclear reactions in nuclear power plants, flight control systems, flight safety, the control of mass transportation systems, medical life support systems and the control of weapons systems.

Danger!

Drive systems and motors can have bare parts with voltages applied (e. g. terminals) or hot surfaces. Additional sources of danger result from moving machine parts. Improperly removing the required covers, inappropriate use, incorrect installation or incorrect operation can result in severe personal injury or damage to property.

All tasks, such as transport, installation, commissioning and service, are only permitted to be carried out by qualified personnel. Qualified personnel are persons familiar with transport, mounting, installation, commissioning and operation of the product and have the respective qualifications (e. g. IEC 60364). National accident prevention guidelines must be followed.

The safety guidelines, connection descriptions (type plate and documentation), and limit values listed in the technical data are to be read carefully before installation and commissioning and must be observed.

Danger!

Handling drive systems and motors incorrectly can cause severe personal injury or damage to property!

2.3 Intended use

Motor drives and motors are components designed to be installed in electrical systems or machines. They are only permitted to be operated if the machine meets regulation 2006/42/EG (machine directive) as well as regulation 89/336/EWG (EMC regulation).

Danger!

Before connecting the power lines, it's necessary to check if the B&R drive system being used is suitable for the respective mains. It's absolutely necessary to comply with the specifications and restrictions in the documentation for the respective device series!

When used in living areas, shops and small businesses, additional measures must be implemented by the user.

Technical data as well as connection and environmental specifications can be found on the type plate and in the user's manual. The connection and environmental specifications must be met!

Danger!

**Electronic devices are generally not failsafe.
If the drive systems fails, the user is responsible for making sure that the connected motors are placed in a secure state.**

2.4 Protection against electrostatic discharges

Electrical components that are vulnerable to electrostatic discharge (ESD) must be handled accordingly.

2.4.1 Packaging

Electrical components with housing do not require special ESD packaging, but must be handled properly.

2.4.2 Guidelines for proper ESD handling

- Do not touch the connector contacts on connected cables.
- Do not touch the contact tips on the circuit boards.

2.5 Transport and storage

During transport and storage, devices must be protected from excessive stress (mechanical load, temperature, humidity, aggressive atmospheres, etc.).

Drive systems contain components sensitive to electrostatic charges which can be damaged by inappropriate handling. It is therefore necessary to provide the required safety precautions against electrostatic discharges during installation or removal of drive systems.

2.6 Handling and installation

Warning!

Therefore, during handling and installation of B&R drive systems or motors, there's danger of personal injury or damage to property (shearing, impact, cutting or crushing). Suitable protective equipment (e.g. safety glasses, protective gloves, safety shoes, etc.) should be used when necessary!

Installation must take place according to the user's manual using suitable equipment and tools.

Devices must be installed without voltage applied and by qualified personnel. Before installation, voltage to the switching cabinet should be switched off and prevented from being switched on again.

The general safety regulations and national accident prevention guidelines (e. g. VBG 4) must be observed when working with high voltage systems.

The electrical installation must be carried out according to the relevant guidelines (e. g. line cross section, fuse, protective ground connection, also see chapter 5 "Dimensioning").

2.7 Operation

2.7.1 Protection against touching electrical parts

Danger!

To operate drive systems, it is necessary for certain parts to carry dangerous voltages over 42 VDC. A life-threatening electrical shock could occur if you come into contact with these parts. This could result in death, severe injury or material damage.

Before turning on a drive system, make sure that the housing is properly connected to ground (PE rail). The ground connection must be made, even when testing the drive system or when operating it for a short time!

Before turning the device on, make sure that all parts with voltage applied are securely covered. During operation, all covers and switching cabinet doors must remain closed.

Control and high power contacts can have voltage applied, even when the motor is not turning. Touching the contacts when the device is switched on is not permitted.

Before working on drive systems, they must be disconnected from the power mains and prevented from being switched on again.

Danger!

Risk of electric shock!

Remove the supply voltage, wait 5 minutes and make sure that the voltage on terminal X1 has dropped to below 42 VDC before beginning maintenance work.

The drive systems are labeled with the following warning:

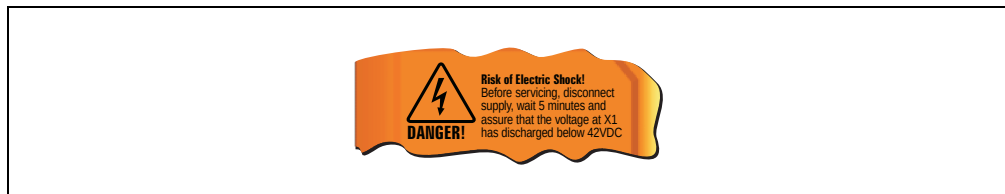


Figure 1: Warning on the drive system

ACOPOSmicro units are provided with labels in different languages for multi-language identification (see page 60).

The connections for the signal voltages (5 to 30 V) found on the drive system are isolated circuits. Therefore, the signal voltage connections and interfaces are only permitted to be connected to devices or electrical components that have sufficient isolation according to IEC 60364-4-41 or EN 50178 and that correspond to SELV / PELV.

Never remove the electrical connections from the drive system with voltage applied. In unfavorable conditions, arcs can occur causing personal injury and damage to contacts.

2.7.2 Protection from dangerous movements

Danger!

Incorrect control of motors can cause unwanted and dangerous movements! Such incorrect behavior can have various causes:

- **Incorrect installation or an error when handling the components**
- **Incorrect or incomplete wiring**
- **Defective devices (drive system, motor, position encoder, cable, brake)**
- **Incorrect control (e. g. caused by software error)**

Some of these causes can be recognized and prevented by the drive system using internal monitoring. However, it is generally possible for the motor shaft to move every time the device is switched on! Therefore protection of personnel and the machine can only be guaranteed using higher level safety precautions.

The movement area of machines must be protected to prevent accidental access. This type of protection can be obtained by using stabile mechanical protection such as protective covers, protective fences, protective gates or photocells.

Removing, bridging or bypassing these safety features and entering the movement area is prohibited.

A sufficient number of emergency stop switches are to be installed directly next to the machine. The emergency stop equipment must be checked before commissioning the machine.

On free running motors, remove shaft keys or prevent them from being catapulted.

The holding brake built into the motors cannot prevent hoists from allowing the load to sink.

2.7.3 Protection against burns

The surfaces of drive systems and motors can become very hot during operation.

Therefore, drive systems and motors must be indicated using the included labels (see page 60):



Figure 2: Label for "Hot surface" (3 pcs. are included in delivery of the ACOPOSmicro)

3. Terminology

Term	Explanation
SG3	System Generation 3 (SG3) - CPUs with Motorola processors. The following CPUs belong to this series: CP260, IF161, IP161, XP152, CP100, CP104, CP152, CP153, CP200, CP210, CP430, CP470, CP474, CP476, CP770, CP774, PP15, PP21, PP35, PP41
SG4	System Generation 4 (SG4) - CPUs with Intel processors. The following CPUs belong to this series: CP1483, CP1483-1, CP1484, CP1484-1, CP1485, CP1485-1, CP1486, CP3484, CP3484-1, CP3485, CP3485-1, CP3486, PP45, PP100/200, PP300/400, MP100/200, CP340, CP360, CP380, CP382, CP570, EC20, EC21, AC140, AC141, AR000, AR010, AR102, AR105, APC620, APC700, APC810
SGC	System Generation Compact CPUs (SGC) - CPUs with Motorola processors (Embedded μP). The following CPUs belong to this series: CP0201, CP0291, CP0292, XC0201, XC0202, XC0292

Table 3: Definition of terms

Chapter 2 • System characteristics

1. The ACOPOSmicro drive system

1.1 Flexibility

Ever-changing fields of application and the necessity to integrate different types of drives in one machine often creates accordingly large challenges for the developers of the application. Creating projects with B&R Automation Studio makes it possible to operate a wide range of drive designs using the same tools. That means that flexible drive architectures can be implemented linked together or even independent of one another.

It is also possible to include various machine versions in the configuration phase and to account for different models with alternating drive types. A combination of stepper motors and servos is easy to implement. This gives the machine manufacturer new levels of freedom in regard to flexibility.

1.2 Smallest dimensions

The compactness of the module is clearly evident in the two-channel version. The ACOPOSmicro requires a base surface of less than 50 cm² per axis.

This not only saves space in the switching cabinet for applications with multiple axes, but also provides advantages in wiring because the bus and supply voltage connection is only needed for every second motor.

A continuous current of 10 A and a 15 A peak current are possible for each stepper motor. This power rating is sufficient to run even the largest stepper motors. The module is protected against over-current, over-temperature and other prohibited operating conditions. The respective warnings and error messages can be output via the fieldbus.

1.3 Variable rated voltage ranges

To achieve higher torque values at high speeds, the first voltage variation was designed for a rated voltage of 80 VDC. A withstand capability is provided up to 100 VDC. However, the drive can also be utilized in the low voltage range with nearly no limitations. This means that the ACOPOSmicro product line can be operated down to a voltage of 18 VDC.

1.4 Integrated I/O

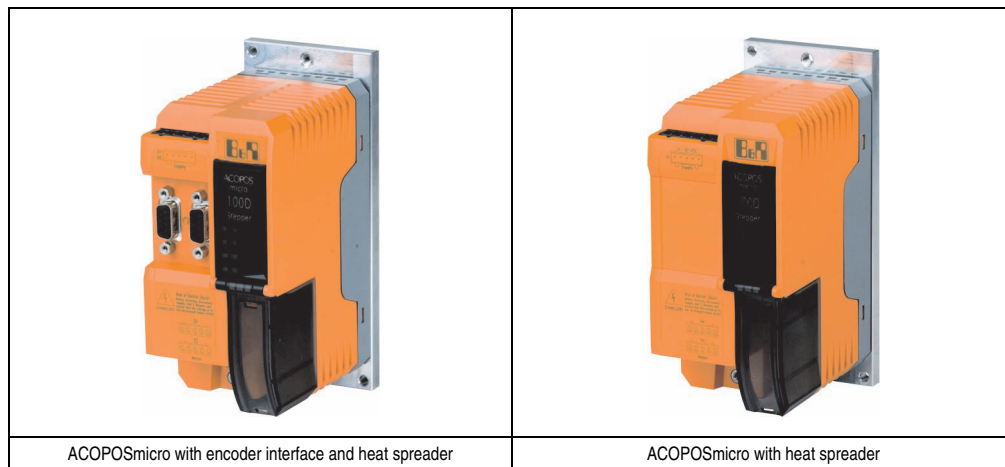
In addition to trigger inputs, a 24 VDC output is also provided. This can be used, among other things, to control external brakes and is available for other tasks.

If required, additional options can be added via an optional module at a minimum cost and without taking up extra space.

This makes it possible to implement even extraordinary customer-specific requirements. The optional support of many different encoder systems, even on the stepper motor variation, illustrates this product line's high degree of flexibility.

1.5 Highlights of the ACOPOSmicro drive system

- Innovative drive concept
- Wide range of use
- Compact construction
- Flexible functionality
- Optional expansions



1.6 Areas of application for stepper motor control

Whereas earlier stepper motors were mainly reserved for simpler tasks, today they are used to meet even complex CNC demands. The particular advantage offered by these motors is their high level of torque in the low to middle speed range that can be implemented starting with very small motor designs. The use of this technology is an economical solution within the torque limits.

1.7 Highest resolution

Depending on the stepper motor being used, the ACOPOSmicro can handle a resolution in up to 102,400 partial steps. This is possible due to the 256 micro-steps supported by the system. The basis stepping angle is automatically divided into the maximum possible micro-steps according to the speed. In addition to the possibility for increased positioning accuracy, this also enables much smoother operation. The common problem of accumulating resonance frequencies is significantly minimized by the fine grading of current changes. The current controller's high frequency also plays a role in this. Optional encoder feedback can help stepper motors achieve highly accurate positioning under a wide range of load torques.

1.8 Ease of handling

Of course no switches are needed for selecting the motor current on the device. All module settings are software-based. The default values of all existing parameters are quickly and easily adapted to the project and overwritten at runtime. Resolution of the current settings to approximately one percent the rated current meets all needs in regard to selecting a current value. The module's main feature is that holding current, rated current and maximum current (boost current) can each be defined separately. This keeps thermal loss in the stepper motor to a minimum and the maximum torque is available exactly when it is needed. All of this leads to noticeable savings in energy and reduces power loss in the motor, which further results in a longer life for all components.

1.9 Coding and identification

The possibility of using coded connection terminals is useful, especially on the two-channel version. In particular, this eliminates fatal errors that can occur by connecting the wrong drive axis during commissioning. The possibility to easily label the ACOPOSmicro also helps to avoid errors.

1.10 Modular cooling design

The excessive heat generated in the device is mostly emitted to the heat spreader integrated in the ACOPOSmicro. This excessive heat can then be dissipated in various way such as through a conventional heat sink on the cooling medium of an external (water or oil-cooled) cold-plate or passed on to machine parts.

1.11 Operational and connection elements

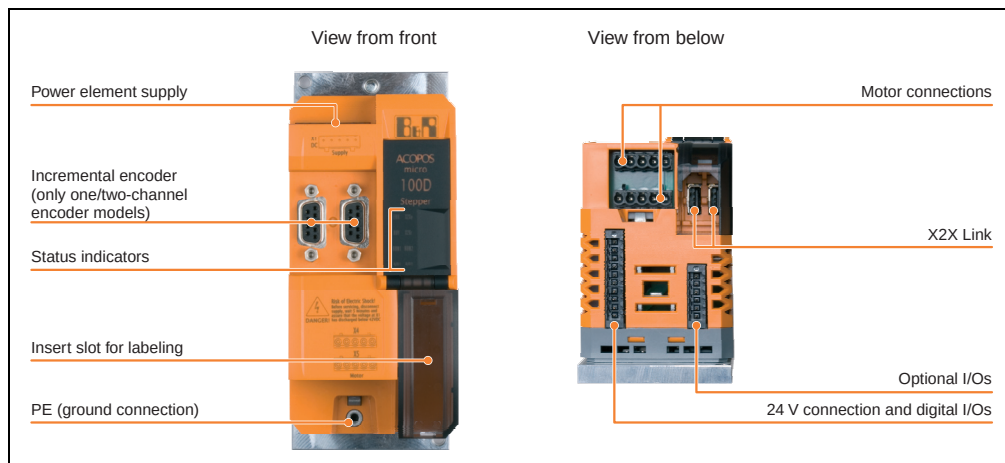


Figure 3: Operating and connection elements

1.12 Dimensions

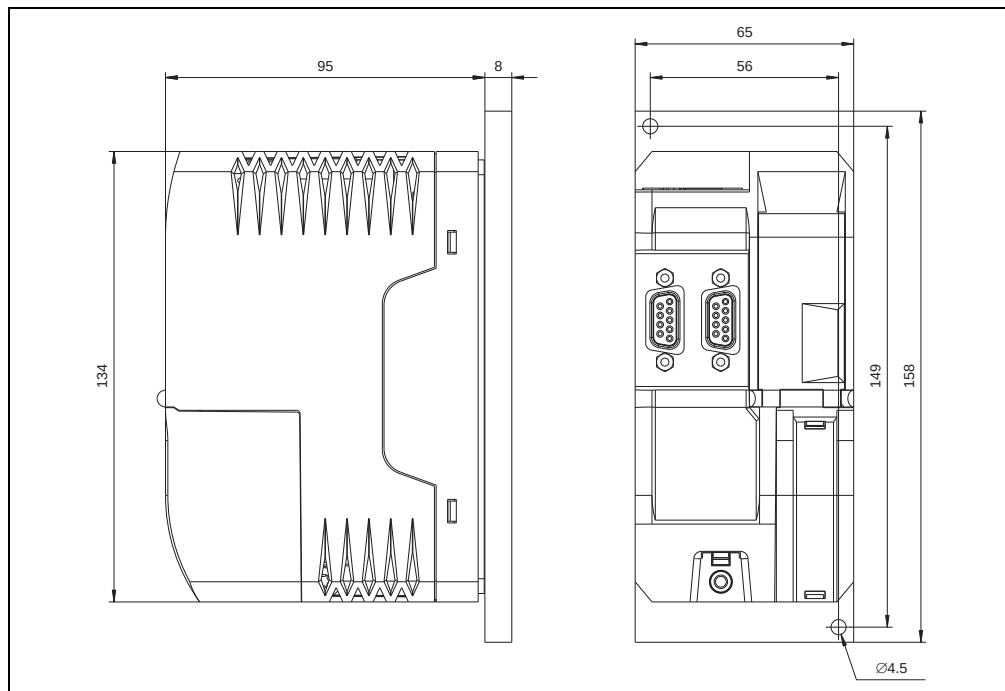


Figure 4: Dimensions

1.13 Software

1.13.1 Standardized programming

The creation of PLCopen motion control function blocks ensures the support of a standard that can handle positioning tasks quickly, easily, and efficiently. They can be programmed in the proven IEC 61131 standard programming languages Structured Text, Instruction List, Ladder Diagram, or Sequential Function Chart. In addition to these languages, B&R also supports programming in B&R Automation Basic and C. All motor types supported by the ACOPOSmicro drive system can be controlled with these PLCopen function blocks.

PLCopen function blocks	
	The universal availability of PLCopen function blocks for all B&R products makes it possible to optimize the component selection to match the performance demands of every application. The PLCopen library is included in the Automation Studio package. Selecting this library in the project automatically imports it and makes the function blocks available for programming.

1.13.2 Full integration in the CNC system

With B&R Automation Studio, commissioning is easy as pie. Stepper motors can be operated in single-axis mode, with an axis group, or connected to other axes. The B&R Soft CNC is able to handle up to 100 axes per interface. The interfaces within a CNC channel can be mixed as needed.

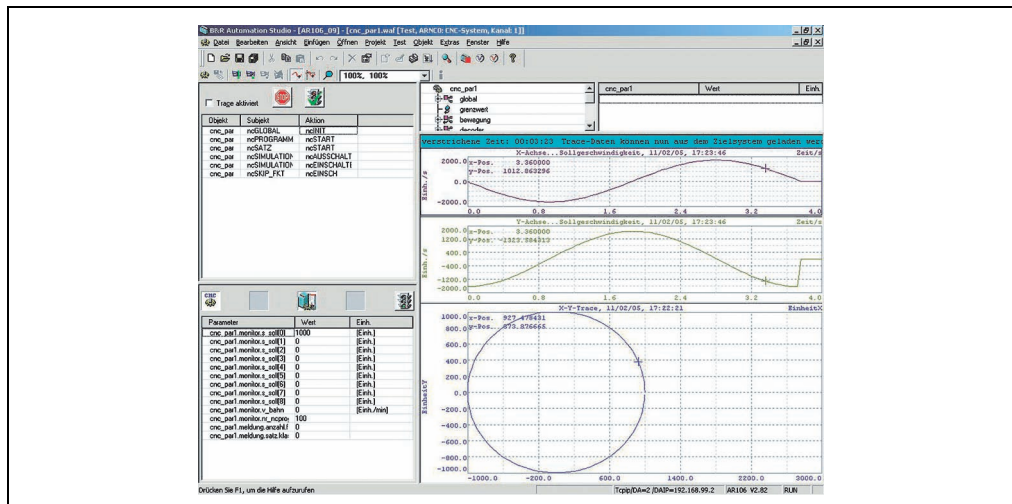


Figure 5: Complete integration in Automation Studio

2. Topologies

2.1 Compact solution for small and mid-sized machines

The ACOPOSmicro is connected directly to a B&R controller system. This is the most compact solution for remote distributed I/O systems. The ACOPOSmicro, X67 system, Compact I/O system, X20 system and simple operating panels can be operated on the same line.

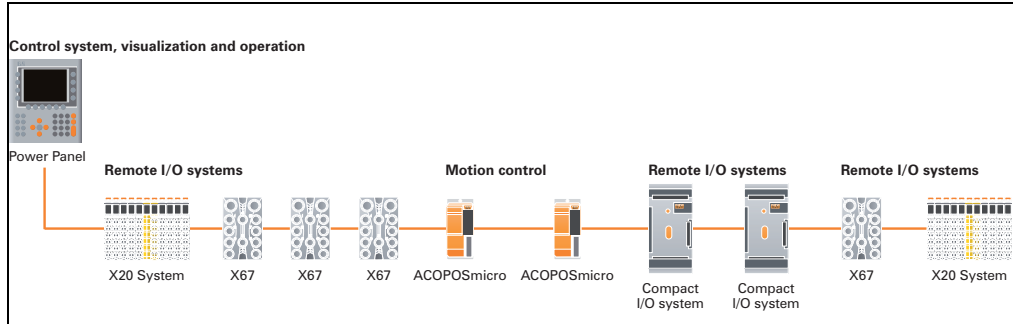


Figure 6: Topology - Compact solution for small and mid-sized machines

Components and technologies	
Control system	Power Panel: Integrated control, operation, and visualization
Visualization and operation	Power Panel: Integrated control, operation, and visualization
Motion control	ACOPOSmicro: Stepper motor drive system Stepper motors
Remote I/O systems	X20 System: Slice-based I/O and control system X67 System: Remote I/O with IP67 protection Compact I/O system and valve connections: Economical usage of peripheral space

Table 4: Topology - Compact solution for small and mid-sized machines

2.2 Custom-fit system combinations

Optimized machine designs require customized implementation of fieldbus systems. With the flexibility of the X67 system and the openness of B&R's system components, the automation can adapt ideally to the cost and performance demands of the application.

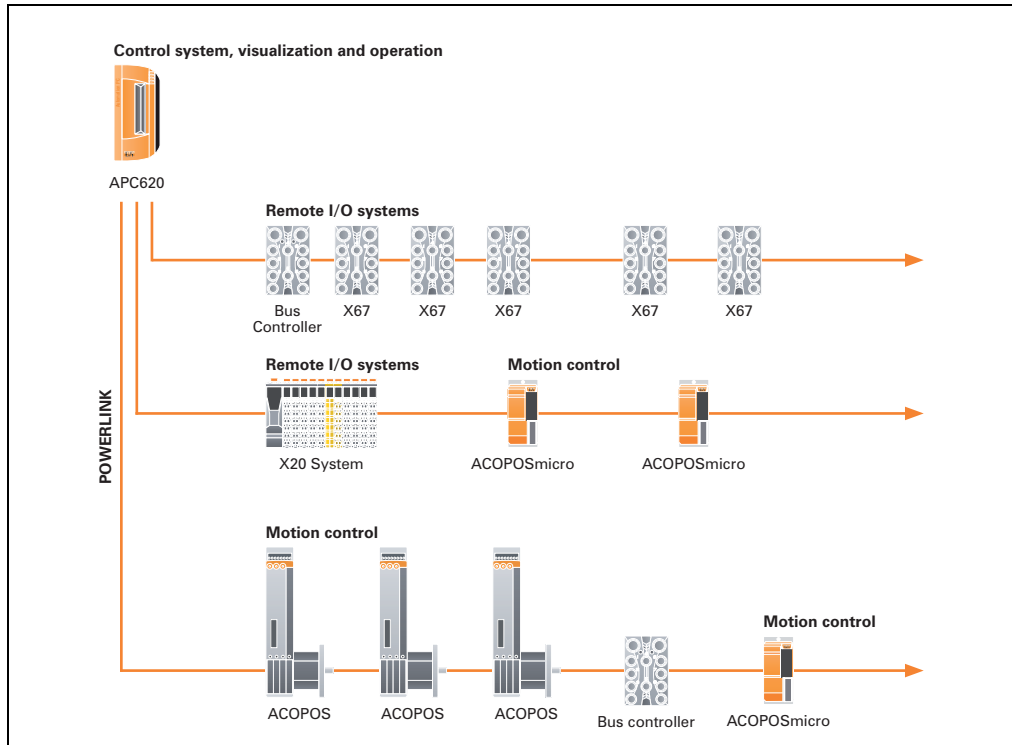


Figure 7: Topology - Tailored system combinations

Components and technologies	
Control system	Automation PC APC620: The new industrial PC generation Panel PC: Operation and PC integrated
Visualization and operation	Panel PC: Operation and PC integrated Automation Panel: A new dimension in machine visualization
Motion control	ACOPOSmicro: Stepper motor drive system Stepper motors ACOPOS: Intelligent servo drives Synchronous Motors: Dynamic precision drives
Remote I/O systems	X20 System: Slice-based I/O and control system X67 System: Remote I/O with IP67 protection Compact I/O system and valve connections: Economical usage of peripheral space
Network and fieldbus modules	Various fieldbus modules

Table 5: Topology - Tailored system combinations

2.3 APC620 embedded for central control, visualization and operation

The control program runs on the APC620 embedded. The visualization project is integrated with Visual Components. A display unit is connected to the PC. The PC is networked via Ethernet TCP/IP; additional Power Panel-based operator terminals can also be connected via Ethernet. Fieldbus systems (CAN bus, X2X Link) are used to handle communication to I/O systems with axis control.

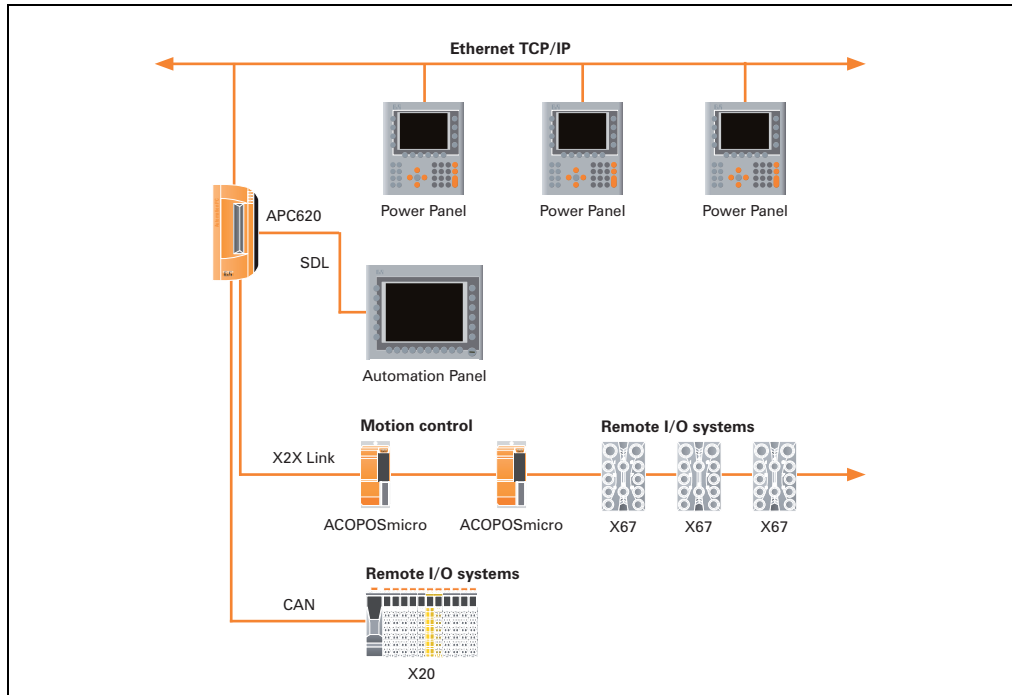


Figure 8: Topology - APC620 embedded for central control, visualization and operation

Components and technologies	
Control system	APC620: Automation PC
Visualization and operation	Power Panel: Integrated control, operation, and visualization Automation Panel 800: Modular operation and visualization Automation Panel 900: Compact operation and visualization
Motion control	ACOPOSmicro: Stepper motor drive system Stepper motors
Remote I/O systems	X20 System: Slice-based I/O and control system X67 System: Remote I/O with IP67 protection

Table 6: Topology - APC620 embedded for central control, visualization and operation of the drive technology

Chapter 3 • Technical data

1. General information

ACOPOSmicro stepper motor modules are used to control up to two stepper motors with a rated voltage of 24 - 64 VDC $\pm 25\%$ at a motor current up to 10 A (15 A peak).

By individually adjusting the coil currents, the motor is only operated with the current it actually needs. This simplifies the selection of the available motors and prevents unnecessary heating. Because the latter reduces the energy consumption and thermal load, the effects are positive on the lifespan of the entire system. Complete flexibility is achieved by using the values for holding, rated and maximum current (boost current), which are completely independent of each other. The current for the microsteps is automatically adjusted to the configured current values.

The automatic motor detection system is an enormous help when standstills occur. The stepper motor modules can identify the connected motors using their coil characteristics and generate feedback in the form of an analog value. This makes it possible to detect not only wiring errors, but incorrect motor types being used mistakenly as well.

- 1 or 2 stepper motors, 24 - 64 VDC $\pm 25\%$, 10 A (15 A peak)
- Holding, rated and maximum current (boost current) configured independent of each other
- 38.5 kHz PWM frequency
- Integrated motor detection
- 256 micro-steps
- Complete integration in Automation Studio and CNC
- Ramp function model is based on the CANopen communication profile DS402.

2. ACOPOSmicro models

In addition to the basic components, an ACOPOSmicro module also consists of various other components such as those for a stepper motor and encoder. As a result, different models are available for users:

Model number	Stepper motor components	Encoder components	Additional inputs/outputs
80SD100XD.C044-01	2	2 x incremental encoder	-
80SD100XD.C04X-13	2	1 x incremental encoder	4 digital inputs, 24 V 2 digital outputs, 24 V 0.5 A
80SD100XD.C0XX-01	2	-	-
80SD100XD.C0XX-21	2	-	2 analog inputs, ± 10 V, 13 bit
80SD100XS.C04X-01	1	1 x incremental encoder	-
80SD100XS.C0XX-01	1	-	-

Table 7: Overview of available ACOPOSmicro models

The following section provides technical data for the individual components.

2.1 Order data

Model number	Short description	Figure
80SD100XD.C044-01	ACOPOSmicro stepper motor module, X2X Link connection, 24-64 VDC $\pm 25\%$ supply, 2 motor connections, 10 A, 2 incremental encoders 24 V, 2 digital inputs 24 VDC sink (can be used as trigger inputs), 1 digital output 24 VDC, 1.0 A, LED status indicators	
80SD100XD.C04X-13	ACOPOSmicro stepper motor module, X2X Link connection, 24-64 VDC $\pm 25\%$ supply, 2 motor connections, 10 A, 1 incremental encoder 24 V, 2 digital inputs 24 VDC sink (can be used as trigger inputs), 4 digital inputs 24 VDC sink (can be used as incremental encoders), 1 digital output 24 VDC, 1.0 A, 2 digital outputs 24 VDC, 0.5 A, LED status indicators	
80SD100XD.C0XX-01	ACOPOSmicro stepper motor module, X2X Link connection, 24-64 VDC $\pm 25\%$ supply, 2 motor connections, 10 A, 2 digital inputs 24 VDC sink (can be used as trigger inputs), 1 digital output 24 VDC, 1.0 A, LED status indicators	
80SD100XD.C0XX-21	ACOPOSmicro stepper motor module, X2X Link connection, 24-64 VDC $\pm 25\%$ supply, 2 motor connections, 10 A, 2 digital inputs 24 VDC sink (can be used as trigger inputs), 1 digital output 24 VDC, 1.0 A, 2 analog inputs ± 10 V, 13 bit, LED status indicators	
80SD100XS.C04X-01	ACOPOSmicro stepper motor module, X2X Link connection, 24-64 VDC $\pm 25\%$ supply, 1 motor connections, 10 A, 1 x 24 V incremental encoder, 2 digital inputs 24 VDC sink (input 1 can be used as trigger input), 1 digital output 24 VDC, 1.0 A, LED status indicators	
80SD100XS.C0XX-01	ACOPOSmicro stepper motor module, X2X Link connection, 24-64 VDC $\pm 25\%$ supply, 1 motor connection, 10 A, 2 digital inputs 24 VDC sink (input 1 can be used as trigger input), 1 digital output 24 VDC, 1.0 A, LED status indicators	
Accessories included in delivery		
- Accessory set for ACOPOSmicro X2X - ACOPOSmicro multi-language warning sticker		
Required accessories		
Terminal block and X2X Link cable must be ordered separately. See Section 2.2 "Required accessories" on page 38.		

Table 8: ACOPOSmicro stepper motor modules - Order data

2.2 Required accessories

The following accessories are required depending on the ACOPOSmicro model and how it will be used:

Model number	Short description	For terminal X	80SD100XD.C044-01	80SD100XD.C04X-13	80SD100XD.C0XX-01	80SD100XD.C0XX-21	80SD100XS.C04X-01	80SD100XS.C0XX-01	Page
0TB1106.8010	Accessory terminal block (3.5), 6-pin, screw clamp, 1.5 mm ² , protection against vibration with the screw flange	X7		●		●			52
0TB1106.8110	Accessory terminal block (3.5), 6-pin, cage clamp, 1.5 mm ² , protection against vibration with the screw flange	X7		●		●			52
0TB1110.8010	Accessory terminal block (3.5), 10-pin, screw clamp, 1.5 mm ² , protection against vibration with the screw flange	X2	●	●	●	●	●	●	53
0TB1110.8110	Accessory terminal block (3.5), 10-pin, cage clamp, 1.5 mm ² , protection against vibration with the screw flange	X2	●	●	●	●	●	●	53
0TB1310.3100	Accessory terminal block (3.5), 3 x 10-pin cage clamp, 1.5 mm ² , protected against vibration by the screw flange	X2	●	●	●	●	●	●	53
0TB2105.4021	Accessory terminal block (5.08), 5-pin, AX1-coded, M.P. screw clamp 2.5 mm ²	X4	●	●	●	●	●	●	54
0TB2105.4022	Accessory terminal block (5.08), 5-pin, AX2-coded, M.P. screw clamp 2.5 mm ²	X5	●	●	●	●			54
0TB2105.4121	Accessory terminal block (5.08), 5-pin, AX1-coded, M.P. cage clamp 2.5 mm ²	X4	●	●	●	●	●	●	54
0TB2105.4122	Accessory terminal block (5.08), 5-pin, AX2-coded, M.P. cage clamp 2.5 mm ²	X5	●	●	●	●			54
0TB2105.9021	Accessory terminal block (5.08), 5-pin, DC-coded, M.S. screw clamp 2.5 mm ²	X1	●	●	●	●	●	●	55
0TB2105.9121	Accessory terminal block (5.08), 5-pin, DC-coded, M.S. cage clamp 2.5 mm ²	X1	●	●	●	●	●	●	55
X20CA0X48.xxxx	X2X Link device attachment cable	X3	●	●	●	●	●	●	56
X20CA0X68.xxxx	X2X Link device connection cable	X3	●	●	●	●	●	●	56

Table 9: Required accessories - Order data

2.3 Basic components

2.3.1 Technical data

Product ID	Basic components
Short description	
Stepper motor components	Connection for one or two stepper motors (each 2-phase bipolar, full bridge)
Digital inputs	
Amount	2
Can be used as trigger input 2-channel model 1-channel model	2 1
Rated voltage	24 VDC
Input filter Hardware	<10 µs
Input circuit	Sink
Enable input	
Amount	1
Rated voltage	24 VDC
Input circuit	Sink
X2X Link interface	
Design	4-pin plug
Minimum cycle time on the X2X Link	250 µs
24 VDC supply	
Input voltage	24 VDC ±25%
Power consumption CPU X2X Link supply	6.0 W + power consumption of other components 0.0 W (generated internally from the CPU supply)
Voltage monitoring	Yes, <18 VDC

Table 10: Basic components - Technical data

Product ID	Basic components
General information	
Status indicators	Stepper motor function for each channel, supply voltage, bus function
Diagnostics Module run/error X2X Link Motor status Module / cooling unit temperature	Yes, with status LED and software status Yes, with status LED Yes, with status LED and software status Yes, with software status
Electrical isolation from the CPU X2X Link Digital inputs Enable input Motor holding brake 24 VDC supply Power supply Incremental Encoders ¹⁾ Additional ¹⁾ inputs/outputs	Yes Yes Yes No Yes Yes No No
Certification	CE, C-UL-US (in development), GOST-R
Operating conditions	
Operating temperature	0 to +45°C
Relative humidity	5 to 85%, non-condensing
Mounting orientation	Vertical
Degree of pollution according to EN 60664-1	2 (non-conductive material)
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of ambient temperature by 0.5°C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 to +55°C
Relative humidity	5 to 95%, non-condensing
Mechanical characteristics	
Dimensions (W x H x D)	65 x 134 x 95 mm (without mounting plate or cold-plate)
Weight	650 - 700 g ²⁾
Mounting / cooling	Screw mounting with heat spreader on mounting plate or cold-plate
Backplane	ACOPOSmicro must always be mounted to a conductive (and grounded) backplane
Comment	Please order terminal block and X2X Link cable separately. See Section 2.2 "Required accessories" on page 38.

Table 10: Basic components - Technical data (cont.)

1) If present in the ACOPOSmicro model

2) Depends on the ACOPOSmicro model

2.3.2 Additional technical data

Product ID	Basic components
24 VDC supply	
Integrated protection Reverse polarity protection	Yes
Digital inputs	
Input voltage	24 VDC $\pm$ 25%
Input current at rated voltage	Typ. 10 mA
Input resistance	Typ. 2.2 k Ω
Switching threshold Low High	<5 V >15 V
Isolation voltage between input and bus	500 V _{eff}
Enable input	
Input voltage	24 VDC $\pm$ 25%
Input current at rated voltage	<100 mA
Switching threshold Low High	<5 V >15 V
Isolation voltage between input and bus	500 V _{eff}
General information	
B&R ID codes	
80SD100XD.C044-01	\$A781
80SD100XD.C04X-13	\$A78D
80SD100XD.C0XX-01	\$A4A4
80SD100XD.C0XX-21	\$A78E
80SD100XS.C04X-01	\$A780
80SD100XS.C0XX-01	\$A5AE

Table 11: Basic components - Additional technical data

2.3.3 Status LEDs

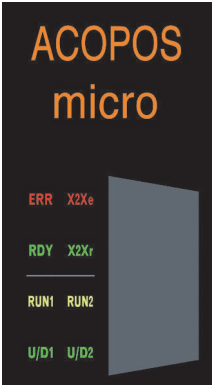
Figure	Description			
	LED	Color	Status indicators for module functionality	
	ERR	Red	Off	Module not supplied or ready for operation
			On	Error or reset state module
	RDY	Green	Off	Module supply not connected
			Single flash	Reset mode
			Double flash	Update mode (firmware)
			Blinking	Preoperational mode
			On	Run mode
	X2Xr Green	X2Xe Red	Status indicators for X2X connection	
	Off	Off	X2X not supplied	
	On	Off	Communication OK	
	Off	On	Communication not OK	
	On	On	Preoperational: Module not initialized	
	LED	Color	Status indicators for motors	
	RUN	Yellow	Motor supplied with power	
	U/D	Yellow / green	Motor's specified direction of rotation	
			Yellow	Direction of rotation ("UP")
			Green	Opposite direction of rotation ("DOWN")
1-channel model: RUN and U/D 2-channel model: RUN1, RUN2, U/D1 and U/D2				

Table 12: Status LEDs

2.3.4 Monitor for 24 VDC module supply

The module supply voltage is continually monitored. The status can be read. The error "Module Power Supply Error" occurs when the voltage falls below 18 V.

2.4 Stepper motor components

2.4.1 Technical data

Product ID	Stepper motor components
Motor connection	
Rated voltage	24 - 64 VDC $\pm 25\%$
Rated current / motor	10 A
Maximum current / motor	15 A (2 s)
Controller frequency	38.5 kHz
DC bus capacitance	940 μF
Step resolution	256 micro-steps per full-step
Motor cable length	Max. 25 m
Motor holding brake connection	
Continuous current	1.0 A
Maximum internal resistance	0.32 Ω
Rated voltage	24 VDC
Reference potential	GND for module supply (terminal X2, pin 10)
Protective measures / safeguards	Overload / short-circuit protection
Maximum switching frequency	100 Hz
Protective measures / safeguards	
Overload and short circuit protection	Yes
Cable breakage monitoring	No
Undervoltage monitoring	Yes
Maximum over-current limitation	8 A
Response threshold for undervoltage monitoring	18 VDC $+0\%$ / -10%
Power supply	
Input voltage	24 - 64 VDC $\pm 25\%$
Undervoltage cut-off	<18 VDC
Overvoltage cut-off	>95 VDC
Power consumption of power element	Max. 0.5 W
Idle (no motor supplied with current)	
Full load	See diagram in chapter 5 "Dimensioning" section 1 "Power consumption" on page 65
Voltage measurement	Yes
Line protection	Must be implemented externally See chapter 5 "Dimensioning" section 2 "Fuse protection" on page 67
Reverse polarity protection	Yes
Starting current limitation	Yes
Design	PTC (min. 9 Ω), bridged during operation using relay contact
Starting current	Max. 10 A
General information	
Electrical isolation from the CPU	Yes

Table 13: Stepper motor components - Technical data

2.4.2 Additional technical data

Product ID	Stepper motor components
Motor brake connection	
Input voltage for motor holding brake	
Rated	24 VDC
Maximum	30 VDC

Table 14: Stepper motor components - Additional technical data

2.4.3 Monitoring the power supply

The power element's supply module is monitored. The status can be read. The error "Module Power Supply Error" occurs when the voltage (U_{DC}) falls below 18 V or rises above 95 V.

Over-voltage shut-off

If the supply voltage on the module exceeds 95 V (e.g. through feedback during generator operation), then the motor output is switched off.

Also true in regard to the standard function model:

The motor output is reactivated as soon as the supply voltage is back in the valid range.

2.4.4 Over-temperature shut-off

Module temperature:

If the module temperature reaches or exceeds ¹⁾ 85°C, ...

- ... the application is notified via the "Overtemperature" error bit
- ... the motor outputs are disabled

Also true in regard to the standard function model:

The error-bits are automatically cleared by the module and the outputs become operational again after the module temperature sinks to 83°C.

Heat sink temperature (temperature on the power element):

If the module temperature on the power element reaches or exceeds ²⁾ 100°C, ...

- ... the application is notified via the "Overtemperature" error bit
- ... the motor outputs are disabled

The error-bits are automatically cleared by the module and the outputs become operational again after the module temperature on the power element sinks to 98°C.

1) See Register description 2.2.14 "Temperature" on page 96 (for the standard function model) and 3.2.11 "Temperature" on page 115 (for the ramp function model).

2) See Register description 2.2.17 "Heat sink temperature" on page 97 (for the standard function model) and 3.2.10 "Temperature of output stage" on page 115 (for the ramp function model).

2.5 Encoder components

Information:

Some ACOPOSmicro models have one and some have two encoder inputs. The technical data is based on models with one encoder channel.

2.5.1 Technical data

Product ID	Encoder components (per channel)
Incremental encoder	
Encoder inputs	24 V, asymmetrical
Counter size	16-bit
Input frequency	Max. 50 kHz
Counter frequency	200 kHz
Signal form	Square wave pulse
Evaluation	4x
Encoder supply	Module-internal, max. 40 mA per channel
General information	
Electrical isolation from the CPU	No
Additional power consumption CPU	1.0 W per channel

Table 15: Encoder components - Technical data

2.5.2 Additional technical data

Product ID	Encoder components
Incremental encoder	
Input current at rated voltage	Approx. 4 mA
Switching threshold Low High	<5 V >15 V
Output for encoder supply	24 VDC (from the 24 V CPU supply)
Protection of the encoder supply output	Short-circuit and overload protection via PTC Important: 1 PLC for both encoder channels (see 2.6.1 "X6 - Input circuit diagram for incremental encoder" on page 80)

Table 16: Encoder components - Additional technical data

2.6 Additional digital inputs/outputs

2.6.1 Technical data

Product ID	Additional digital inputs/outputs
Additional digital inputs	
Amount	4
Rated voltage	24 VDC
Input filter Hardware	$\leq 10 \mu\text{s}$
Input circuit	Sink
Additional functions for inputs	Configurable as ABR incremental encoder, event counter, period or gate measurement
Additional digital outputs	
Amount	2
Rated voltage	24 VDC
Rated output current	0.5 A
Total current	1.0 A
Output circuit	Source
Output protection	Overload and short circuit protection

Table 17: Additional digital inputs/outputs - Technical data

2.6.2 Additional technical data

Product ID	Additional digital inputs/outputs
Additional digital inputs	
Input voltage	24 VDC $\pm 25\%$
Input current at 24 VDC	Typ. 3.1 mA
Input resistance	Typ. 7.8 k Ω
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage between. input and bus	500 V _{eff}
ABR incremental encoder	
Counter size	16-bit
Input frequency	Max. 25 kHz
Evaluation	4x

Table 18: Additional digital inputs/outputs - Advanced technical data

Product ID	Additional digital inputs/outputs
Event counter	
Number of counters	1
Signal form	Square wave pulse
Input frequency	Max. 50 kHz
Counter size	16-bit
Evaluation	configurable
Gate measurement	
Number of gate measurements	1
Signal form	Square wave pulse
Evaluation	configurable
Pulse length	$\geq 10 \mu\text{s}$
Length of pauses between pulses	$\geq 10 \mu\text{s}$
Internal counter frequency	configurable
Counter size	16-bit
Additional digital outputs	
Design	HighSide FET positive switching
Switching voltage (min. / nom. / max.)	18 / 24 / 30 VDC
Supply voltage	From the 24 VDC module supply
Continuous current per output	0.5 A
Continuous current of all additional outputs	1.0 A
Switching on after overload cutoff	e.g.: ca. 10 ms (depends on the module temperature)
Switching delay 0 → 1 1 → 0	Max. 250 μs Max. 270 μs
Switching frequency Resistive load Inductive load	100 Hz See the section 2.6.3 "Switching inductive loads" on page 48 (at 90% duty cycle)
Braking voltage when switching off inductive loads	47 V

Table 18: Additional digital inputs/outputs - Advanced technical data (cont.)

2.6.3 - Switching inductive loads

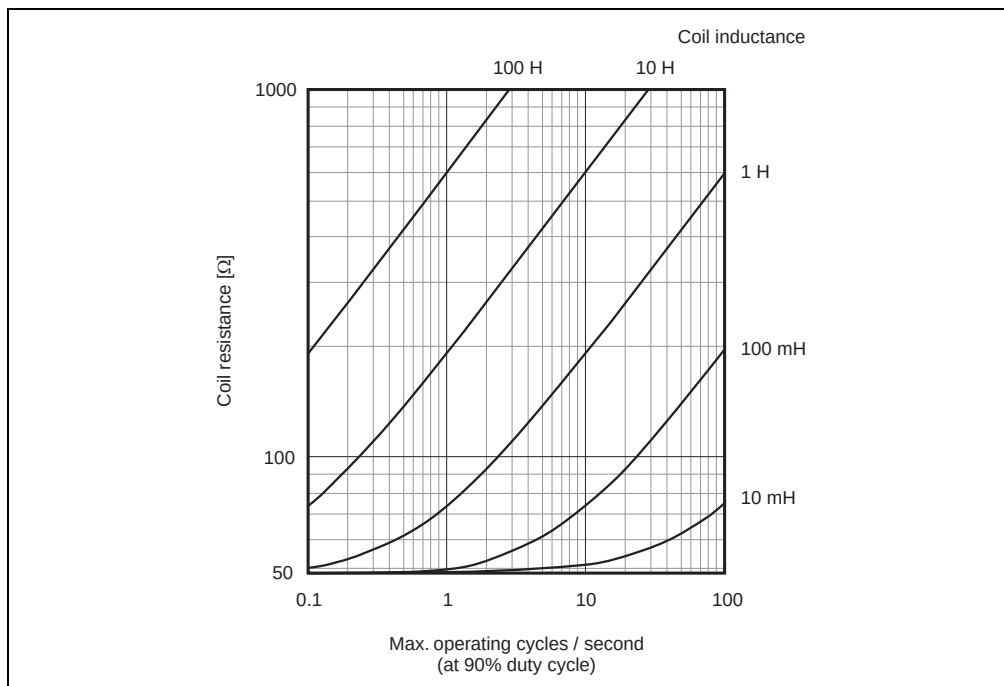


Figure 9: Additional digital inputs/outputs - Switching inductive loads

2.7 Additional analog inputs

2.7.1 Technical data

Product ID	Additional analog inputs
Additional analog inputs	
Amount	2
Input	$\pm 10\text{ V}$
Input type	Differential input
Digital converter resolution	13-bit
Conversion time	50 μs for both channels
Output format	INT
Input impedance in signal range	20 M Ω
Input protection	Protection against wiring with supply voltage (24 VDC supply)

Table 19: Additional analog inputs - Technical data

2.7.2 Additional technical data

Product ID	Additional analog inputs
Additional analog inputs	
Permitted input signal	Max. $\pm 30\text{ V}$
Output format	INT \$8001 - \$7FFF 1 LSB = \$0001 = 305.176 μV
Output of digital value under... Overload conditions Above upper limit Below lower limit	\$7FFF \$8001
Conversion method	SAR
Input filter	Hardware - Low pass 3rd order, cut-off frequency 10 kHz"
Common-mode rejection DC 50 Hz	70 dB 70 dB
Synchronized zone	$\pm 12\text{ V}$

Table 20: Additional analog inputs - Advanced technical data

2.8 Power consumption

2.8.1 Module supply

The module's total power consumption (CPU) results from the following components:

Model number	Basic components	Encoder components	Additional inputs/outputs	Total power consumption
80SD100XD.C044-01	6.0 W	2 x 1.0 W	-	8.0 W
80SD100XD.C04X-13	6.0 W	1 x 1.0 W	-	7.0 W
80SD100XD.C0XX-01	6.0 W	-	-	6.0 W
80SD100XD.C0XX-21	6.0 W	-	-	6.0 W
80SD100XS.C04X-01	6.0 W	1 x 1.0 W	-	7.0 W
80SD100XS.C0XX-01	6.0 W	-	-	6.0 W

Table 21: Power consumption of module supply

2.8.2 Power supply

The power required for the stepper motor controller is generated externally and fed-in via the power supply connection.

Further information can be found in the chapter 5 "Dimensioning" in the section 1 "Power consumption" on page 65.

When using a low-cost transformer with downstream, unregulated rectification and smoothing via electrolytic capacitor, make sure that the maximum output voltage is greater than the AC output voltage by a factor of $\sqrt{2}$ (square root of 2). Furthermore, voltage fluctuations from the mains also directly affect the output voltage.

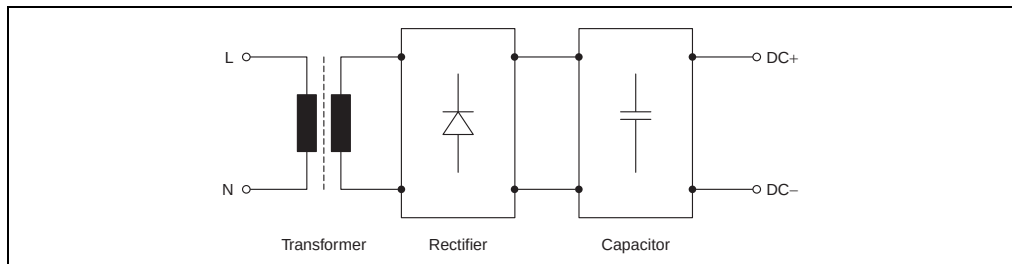


Figure 10: Power supply - Transformer

Example: 48 VAC output voltage from the transformer is rectified via a bridge connection. The output voltage increases under the conditions of ideal diodes, lossless lines and no load current:

$$48 \text{ V} \times \sqrt{2} = 67.9 \text{ VDC}$$

3. Accessories

3.1 Terminal blocks

Model number	Short description	Number of poles	Coding	Usage	Page
0TB1106.8010	Accessory terminal block (3.5), 6-pin, screw clamp, 1.5 mm ² , protection against vibration with the screw flange	6	-	Connection of additional I/O	52
0TB1106.8110	Accessory terminal block (3.5), 6-pin, cage clamp, 1.5 mm ² , protection against vibration with the screw flange	6	-		
0TB1110.8010	Accessory terminal block (3.5), 10-pin, screw clamp, 1.5 mm ² , protection against vibration with the screw flange	10	-	Connection of I/O and CPU supply	53
0TB1110.8110	Accessory terminal block (3.5), 10-pin, cage clamp, 1.5 mm ² , protection against vibration with the screw flange	10	-		
0TB1310.3100	Accessory terminal block (3.5), 3 x 10-pin cage clamp, 1.5 mm ² , protected against vibration by the screw flange	30	-		
0TB2105.4021	Accessory terminal block (5.08), 5-pin, AX1-coded, M.P. screw clamp 2.5 mm ²	5	AX1	Motorconnection	54
0TB2105.4022	Accessory terminal block (5.08), 5-pin, AX2-coded, M.P. screw clamp 2.5 mm ²	5	AX2		
0TB2105.4121	Accessory terminal block (5.08), 5-pin, AX1-coded, M.P. cage clamp 2.5 mm ²	5	AX1		
0TB2105.4122	Accessory terminal block (5.08), 5-pin, AX2-coded, M.P. cage clamp 2.5 mm ²	5	AX2		
0TB2105.9021	Accessory terminal block (5.08), 5-pin, DC-coded, M.S. screw clamp 2.5 mm ²	5	DC	Connection of powersupply	55
0TB2105.9121	Accessory terminal block (5.08), 5-pin, DC-coded, M.S. cage clamp 2.5 mm ²	5	DC		

Table 22: Terminal blocks - Overview

3.1.1 Terminal block for connecting additional inputs/outputs

Order data

Model number	Short description	Figure
OTB1106.8010	Accessory terminal block (3.5), 6-pin, screw clamp, 1.5 mm ² , protection against vibration with the screw flange	
OTB1106.8110	Accessory terminal block (3.5), 6-pin, cage clamp, 1.5 mm ² , protection against vibration with the screw flange	

Table 23: Terminal block for connecting additional inputs/outputs - Order data

Technical data

Product ID	OTB1106.8010	OTB1106.8110
Number of pins	6	6
Type of terminal	Screw clamps	Cage clamps
Distance between contacts	3.5 mm	3.5 mm
Contact resistance	≤4.2 mΩ	≤4.2 mΩ
Rated voltage	300 V	300 V
Rated current ¹⁾	10 A / contact	10 A / contact
Connection cross section		
AWG wire	28 - 14 AWG	26 - 14 AWG
Solid wire line	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Fine wire line without wire tip sleeves	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Fine wire line with wire tip sleeves	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Wire tip sleeves with plastic covering	0.20 - 1.50 mm ²	0.20 - 1.00 mm ²
Holding torque (min./max)	0.2 / 0.25 Nm	-
Cable type	Copper wires only (no aluminum wires!)	
Comment	Rated values according to UL, Protected against vibration by the screw flange	

Table 24: Terminal block for connecting additional inputs/outputs - Technical data

1) Important - keep in mind the data limits for the I/O modules.

3.1.2 Terminal blocks for connecting I/O and CPU supply

Order data

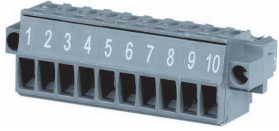
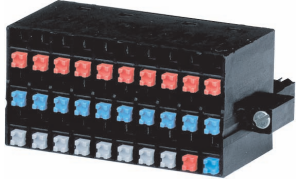
Model number	Short description	Figure
0TB1110.8010	Accessory terminal block (3.5), 10-pin, screw clamp, 1.5 mm ² , protection against vibration with the screw flange	
0TB1110.8110	Accessory terminal block (3.5), 10-pin, cage clamp, 1.5 mm ² , protection against vibration with the screw flange	
0TB1310.3100	Accessory terminal block (3.5), 3 x 10-pin cage clamp, 1.5 mm ² , protected against vibration by the screw flange	

Table 25: Terminal blocks for connecting I/O and CPU supply - Order data

Technical data

Product ID	0TB1110.8010	0TB1110.8110	0TB1310.3100
Number of pins	10	10	30
Type of terminal	Screw clamps	Cage clamps	Cage clamps
Distance between contacts	3.5 mm	3.5 mm	3.5 mm
Contact resistance	≤4.2 mΩ	≤4.2 mΩ	≤4.2 mΩ
Rated voltage	300 V	300 V	300 V
Rated current ¹⁾	10 A / contact	10 A / contact	2 A / contact
Connection cross section			
AWG wire	28 - 14 AWG	26 - 14 AWG	26 - 14 AWG
Solid wire line	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Fine wire line without wire tip sleeves	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Fine wire line with wire tip sleeves	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Wire tip sleeves with plastic covering	0.20 - 1.50 mm ²	0.20 - 1.00 mm ²	0.20 - 1.50 mm ²
Holding torque (min./max)	0.2 / 0.25 Nm	-	-
Cable type	Copper wires only (no aluminum wires!)		
Comment	Rated values according to UL, Protected against vibration by the screw flange		Rated values according to UL

Table 26: Terminal blocks for connecting I/O and CPU supply - Technical data

1) Important - keep in mind the data limits for the I/O modules.

3.1.3 Terminal blocks for the connection of motors

Model number	Short description	Figure
0TB2105.4021	Accessory terminal block (5.08), 5-pin, AX1-coded, M.P. screw clamp 2.5 mm ²	
0TB2105.4022	Accessory terminal block (5.08), 5-pin, AX2-coded, M.P. screw clamp 2.5 mm ²	
0TB2105.4121	Accessory terminal block (5.08), 5-pin, AX1-coded, M.P. cage clamp 2.5 mm ²	
0TB2105.4122	Accessory terminal block (5.08), 5-pin, AX2-coded, M.P. cage clamp 2.5 mm ²	

Table 27: Terminal blocks for the connection of motors - Order data

Technical data

Product ID	0TB2105.4021	0TB2105.4022	0TB2105.4121	0TB2105.4122
Number of pins	5	5	5	5
Coding	AX1	AX2	AX1	AX2
Type of terminal	Screw clamps	Screw clamps	Cage clamps	Cage clamps
Distance between contacts	5.08 mm	5.08 mm	5.08 mm	5.08 mm
Contact resistance	≤2 mΩ	≤2 mΩ	≤5 mΩ	≤5 mΩ
Rated voltage	300 V	300 V	300 V	300 V
Rated current ¹⁾	15 A / contact	15 A / contact	15 A / contact	15 A / contact
Connection cross section				
AWG wire	24 - 12 AWG	24 - 12 AWG	26 - 12 AWG	26 - 12 AWG
Solid wire line	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²
Fine wire line without wire tip sleeves	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²
Fine wire line with wire tip sleeves	0.25 - 2.50 mm ²	0.25 - 2.50 mm ²	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Wire tip sleeves with plastic covering	0.25 - 2.50 mm ²	0.25 - 2.50 mm ²	0.20 - 1.50 mm ²	0.20 - 1.50 mm ²
Holding torque (min./max)	0.4 / 0.5 Nm	0.4 / 0.5 Nm	-	-
Cable type	Copper wires only (no aluminum wires!)			
Comment	Rated values according to UL			

Table 28: Terminal blocks for the connection of motors - Technical data

1) Important - keep in mind the data limits for the I/O modules.

3.1.4 Terminal blocks for the connection of the power supply

Model number	Short description	Figure
OTB2105.9021	Accessory terminal block (5.08), 5-pin, DC-coded, M.S. screw clamp 2.5 mm ²	
OTB2105.9121	Accessory terminal block (5.08), 5-pin, DC-coded, M.S. cage clamp 2.5 mm ²	

Table 29: Terminal blocks for the connection of the power supply - Order data

Technical data

Product ID	OTB2105.9021	OTB2105.9121
Number of pins	5	5
Coding	DC	DC
Type of terminal	Screw clamps	Cage clamps
Distance between contacts	5.08 mm	5.08 mm
Contact resistance	≤2 mΩ	≤5 mΩ
Rated voltage	300 V	300 V
Rated current ¹⁾	15 A / contact	15 A / contact
Connection cross section		
AWG wire	24 - 12 AWG	26 - 12 AWG
Solid wire line	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²
Fine wire line without wire tip sleeves	0.20 - 2.50 mm ²	0.20 - 2.50 mm ²
Fine wire line with wire tip sleeves	0.25 - 2.50 mm ²	0.20 - 1.50 mm ²
Wire tip sleeves with plastic covering	0.25 - 2.50 mm ²	0.20 - 1.50 mm ²
Holding torque (min./max)	0.4 / 0.5 Nm	-
Cable type	Copper wires only (no aluminum wires!)	
Comment	Rated values according to UL	

Table 30: Terminal blocks for the connection of the power supply - Technical data

1) Important - keep in mind the data limits for the I/O modules.

3.2 X2X Link cable

3.2.1 Overview

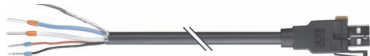
Model number	Short description	Page	Figure
	ACOPOSmicro connection cable X2X to X2X		
X20CA0X68.xxxx	X2X Link device connection cable	58	
	ACOPOSmicro attachment cable X2X - open		
X20CA0X48.xxxx	X2X Link device attachment cable	59	

Table 31: X2X Link connection and attachment cable - Overview

Information:

The digits "xxxx" in the model numbers X20CA0X68.xxxx and X20CA0X48.xxxx correspond to the length of the cable in decimeters [dm].

The following sections list the cable lengths that are available for delivery:

- 3.2.3 "X20CA0X68.xxxx" on page 58
- 3.2.4 "X20CA0X48.xxxx" on page 59

3.2.2 Technical data

Product ID	X2X Link cable	
General information	Supply lines	Data cables
Cable cross section mm ² AWG	2 x 0.34 mm ² 2 x 22 AWG	2 x 0.24 mm ² 2 x 24 AWG
Durability	Flame resistant according to VW-1 and FT1	
RoHS compliant ¹⁾	Yes	
Lines (wires)	Supply lines	Data cables
Signal lines	0.34 mm ² (22 AWG), tinned Cu wire	0.24 mm ² (24 AWG), tinned Cu wire
Wire insulation	SR-PVC	FOAM-PE
Wire colors	Red, black	Blue, white
Cable structure	Supply lines	Data cables
Signal lines Stranding Shield	- -	Twisted pair wires Paired shielding / aluminum foil / MYLAR 25% surface coverage
Entire shielding Interior Exterior	With additional wire 28 AWG (7 x 36 AWG) between interior/exterior shielding Aluminum foil / MYLAR, 25% surface coverage Tinned, braided, Cu shield, ≥85% surface coverage	
Outer sheathing Material Color Labeling	PVC Black (P001) B&R X20CA0Xxx.xxxx REV.xx	
Electrical characteristics	Supply lines	Data cables
Conductor resistance (loop)	≤55 Ω/km at 20°C	≤93.2 Ω/km at 20°C
Insulation resistance	≥100 MΩ/km at 20°C and 500 VDC	≥10 MΩ/km at 20°C and 500 VDC
Sweeping voltage test	≤500 VAC / 0.15 s	
Dielectric Strength	≤750 VAC / 1 s	
Installed voltage	Max. 30 V	
Plugs		
Type	4-pin shielded blade connector	
Contact resistance	≤3 Ω	
Mechanical characteristics		
Protection type	IP20	
Temperature range	80°C	
Outer diameter	7.0 mm ± 0.19 mm	
Flex radius	≥56 mm	

Table 32: X20CA0Xxx X2X Link cable - Technical data

- 1) RoHS (Restriction of the use of certain Hazardous Substances) limits the use of the following substances in electrical and electronic devices: Lead, mercury, cadmium, chrome VI as well as the flame-retardant polybrominated biphenyl (PBB) and polybrominated diphenyl ether (PBDE).

3.2.3 X20CA0X68.xxxx

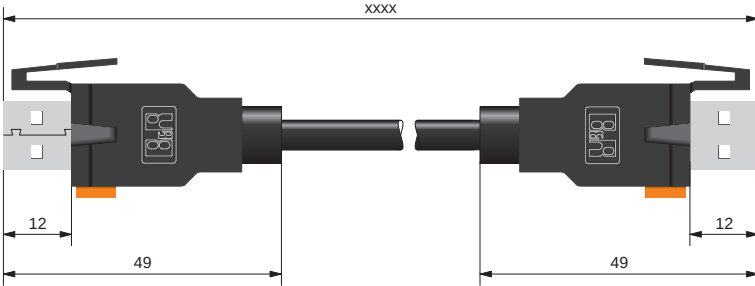
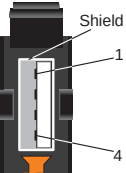
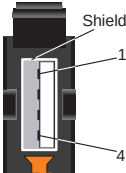
Dimensions				
				
Pin assignments				
Plugs	Pin	Name	Wire colors	Plugs
 Coding for X3B	1	X2X+	Red	 Coding for X3A
	2	X2X	White	
	3	X2X\	Blue	
	4	X2X⊥	Black	
	Plugs	SHLD	Shield	

Table 33: X20CA0X68 X2X Link device connection cable - Measurements and pin assignments

Cable lengths			
Model number	Standard length [dm]	Actual length [cm]	Tolerance [cm]
X20CA0X68.0003	3	30	±2.0
X20CA0X68.0010	10	100	±3.0
X20CA0X68.0020	20	200	±5.0
X20CA0X68.0050	50	500	±10.0
X20CA0X68.0100	100	1000	±10.0

Table 34: X20CA0X01 X2X Link connection cable - Cable lengths

3.2.4 X20CA0X48.xxxx

Dimensions				
Pin assignments				
Open	Pin	Name	Wire colors	Plugs
For custom wiring	1	X2X+	Red	
	2	X2X	White	
	3	X2X\	Blue	
	4	X2X⊥	Black	
	Plugs	SHLD	Shield	

Table 35: X20CA0X48 X2X Link attachment cable - Measurements and pin assignments

Cable lengths			
Model number	Standard length [dm]	Actual length [cm]	Tolerance [cm]
X20CA0X48.0010	10	100	±2.5
X20CA0X48.0020	20	200	±3.0
X20CA0X48.0050	50	500	±5.0
X20CA0X48.0100	100	1000	±10.0
X20CA0X48.0200	200	2000	±10.0

Table 36: X20CA0X48 X2X Link attachment cable - Cable lengths

Information:

This cable is coded as attachment cable for the ACOPOSmicro (connection to X2X Link input X3A). In order to use this cable on the X2X Link output X3B of the ACOPOSmicro, the coding must be changed accordingly (see chapter 6 "Wiring" in the section 2.4 "X3A/X3B - X2X Link connections" on page 75).

3.3 Accessories included in delivery

Brief description of the accessories included in delivery.
Accessory set for ACOPOSmicro X2X: for electromagnetic compatibility of the installation of the X2X Link cable and the PE connection. Other information in this manual:
- Usage of the accessory set in chapter 6 "Wiring" section 2.7 "Protective ground connection PE" on page 81 - Electromagnetic Compatibility of the Installation See chapter 6 "Wiring" on page 69
ACOPOSmicro multi-language warning sticker

Table 37: Accessories included in delivery

3.3.1 ACOPOSmicro multi-language warning sticker



Figure 11: Multi-language warning sticker

For warning sticker usage, please refer to chapter 1 "General information" in the sections 2.7.1 "Protection against touching electrical parts" on page 21 and 2.7.3 "Protection against burns" on page 24.

Chapter 4 • Installation

1. General information

Make sure that the ACOPOSmicro modules are installed on a flat surface which is correctly dimensioned. The dimensional diagram lists the number and type of mounting screws to be used.

ACOPOSmicro modules can only be installed in an environment which corresponds to a pollution degree of 2 (non-conductive material). When installing the device, make sure that the specifications for maximum operating temperature and protection level listed in the technical data are met (see Chapter3 "Technical data" on page 35).

2. Dimension diagrams and installation dimensions

Installation dimensions are the same for all ACOPOSmicro modules.

2.1 Dimension diagram for ACOPOSmicro

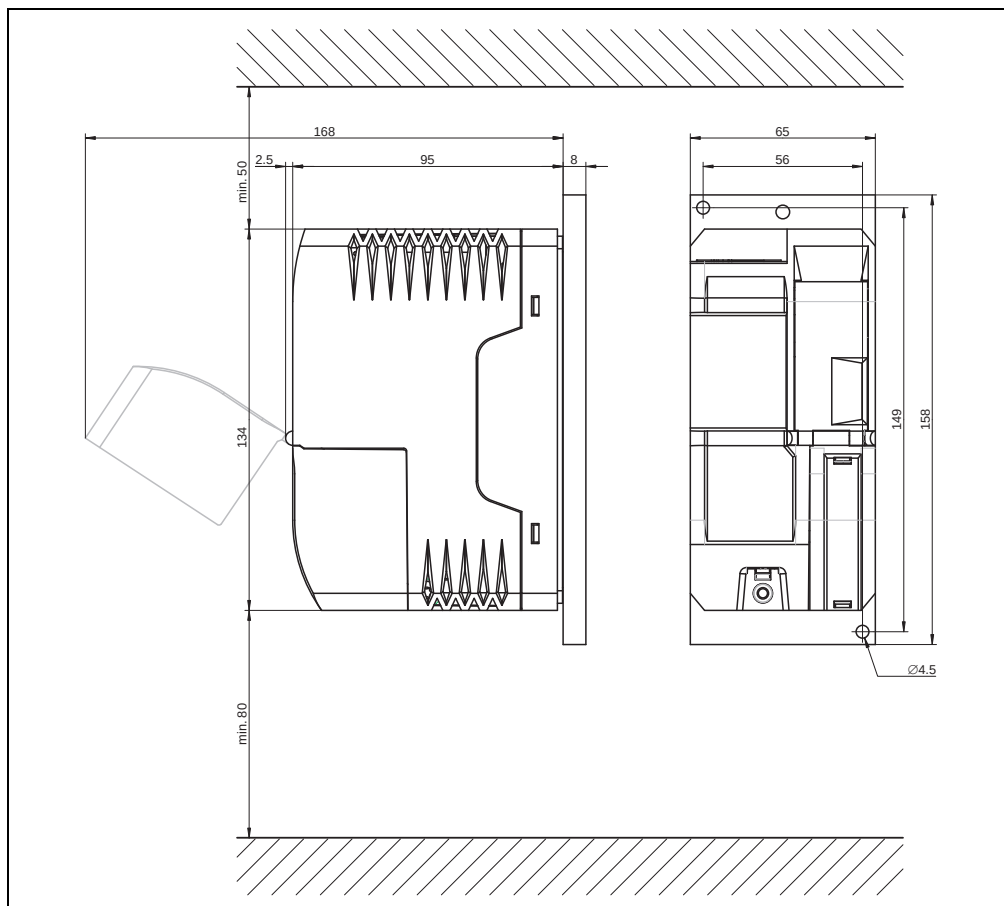


Figure 12: Dimension diagram and installation dimensions for ACOPOSmicro modules

Keep the swivel range of the connector covers on the front side of the ACOPOSmicro modules free when installed to prevent cabling problems with ACOPOSmicro modules.

For proper air circulation, free space must be left above and below the ACOPOSmicro modules (at least 50 mm above, and at least 80 mm below). The ACOPOSmicro modules can be arranged directly beside one another.

3. Cold plate mounting

The mounting surface for the mounting plate or for the cold-plate must provide sufficient stability and also be non-flammable, level and free of contaminants.

The distances that must be used for mounting and ventilation of ACOPOSmicro modules can be found in the sections 1 "General information" on page 61 and 2 "Dimension diagrams and installation dimensions" on page 62.

Caution!

It is especially important that the mounting surface is level so that the entire surface of the heatspreader sits flat against it.

Mounting on uneven surfaces can reduce heat dissipation from the ACOPOSmicro modules!

Information:

When mounting ACOPOSmicro modules, be sure not to scratch the heatspreader. This would impair thermal dissipation to the mounting plate.

3.1 Disassembly

If heat conductive paste was used when mounting the ACOPOSmicro to a cold-plate, then the drive module can be separated from the cold-plate using an M6 screw.

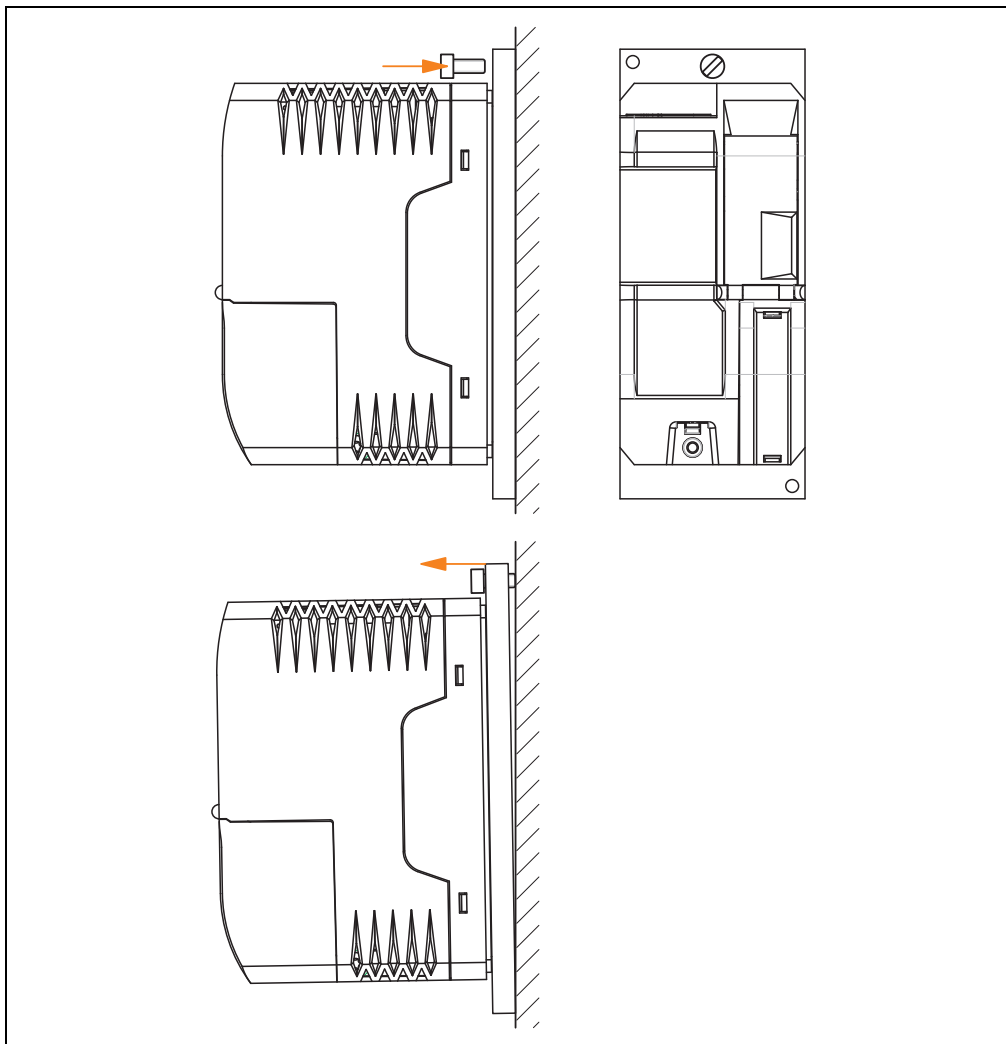


Figure 13: Removing an ACOPOSmicro from the cold-plate

Chapter 5 • Dimensioning

1. Power consumption

The maximum total dissipated power for an ACOPOSmicro module results from the power consumed by the CPU (see in Chapter 3 "Technical data" the Section 2.8 "Power consumption" on page 50) and the power element.

The following diagram displays the dissipated power of the power element for each axis provided with current, depending on the power supply voltage and motor current:

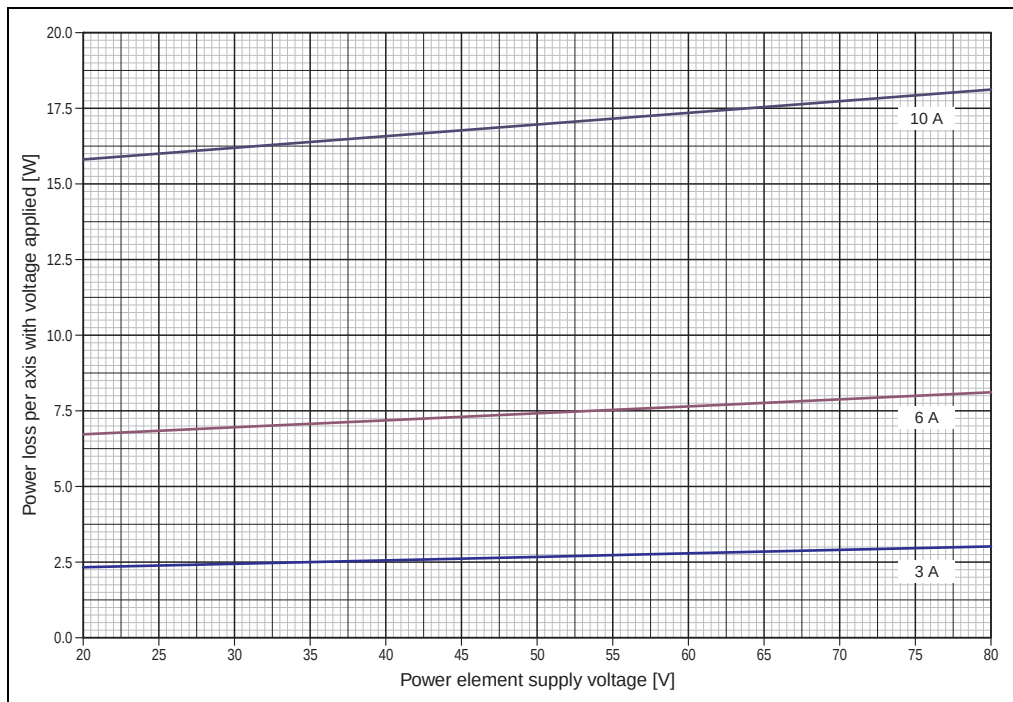


Figure 14: Dimensioning - Dissipated power from the power element for each axis provided with current

This diagram can be used to determine the expected maximum power dissipation from the ACOPOSmicro for a wide range of applications.

A majority of the dissipated heat from the power element is emitted to the ACOPOSmicro's heat spreader.

Example 1:

- ACOPOSmicro 2-channel model without encoder (80SD100XD.C0XX-01)
- 60 VDC power supply
- 10 A applied to motor 1
- 3 A applied to motor 2

Calculation:

Power consumption for ...	Power consumption
CPU (including Enable input)	6.0 watts
Power element when idle	0.5 watts
Motor 1 power element	17.3 watts
Motor 2 power element	2.8 watts
Sum	26.6 watts

Example 2:

- ACOPOSmicro 1-channel model with encoder (80SD100XS.C04X-01)
- 60 VDC power supply
- 3 A applied to motor

Calculation:

Power consumption for ...	Power consumption
CPU (including Enable input)	7.0 watts
Power element when idle	0.5 watts
Motor power element	2.8 watts
Sum	10.3 watts

Example 3:

- ACOPOSmicro 2-channel model with encoder (80SD100XD.C044-01)
- 60 VDC power supply
- 10 A applied to motor 1
- 3 A applied to motor 2

Calculation:

Power consumption for ...	Power consumption
CPU (including Enable input)	8.0 watts
Power element when idle	0.5 watts
Motor 1 power element	17.3 watts
Motor 2 power element	2.8 watts
Sum	28.6 watts

2. Fuse protection

The power supply line should be protected by a circuit breaker or a fuse. In general, dimensioning of the supply line and the over-current protection depends on the structure of the power supply (ACOPOSmicro devices can be connected individually or in groups).

Information:

The effective current for the power supply depends on the load, but is always less than the motor current. Make sure the maximum rated current of 15 A per pin is not exceeded on the power supply terminals of the power element.

When choosing a suitable fuse, the user must also account for properties such as aging effects, temperature derating, overcurrent capacity and the definition of the rated current, which can vary by manufacturer and type. Furthermore, the fuse that is selected must also be able to handle application-specific aspects (e.g. overcurrents that occur in acceleration cycles).

The cross section of the power mains and the rated current of the over-current protection used are chosen according to current load so that the maximum current load for the cable cross section selected (based on type of layout, see table) is greater than or equal to the current load in the power mains. The rated current of the over-current protection must be less than or equal to the maximum current load for the cable cross section selected (based on type of layout, see table):

$$\begin{array}{ccccc} I_{\text{mains}} & \leq & I_b & \leq & I_Z \\ \text{Network} & \leq & \text{Fuse} & \leq & \text{Line/cable} \end{array}$$

Line cross section [mm ²]	I_Z / I_b according to type of installation in an ambient air temperature of + 40°C in accordance to IEC 60204-1 ¹⁾			
	B1	B2	C	E
1.5	13.5 / 13.0	13.1 / 10.0	15.2 / 13.0	16.1 / 16.0
2.5	18.3 / 16.0	16.5 / 16.0	21.0 / 20.0	22.0 / 20.0
4.0	24.0 / 24.0	23.0 / 20.0	28.0 / 25.0	30.0 / 25.0
6.0	32.0 / 32.0	29.0 / 25.0	36.0 / 32.0	37.0 / 32.0

Table 38: Cable cross-section of the mains supply line depending on the type of layout

1) I_Z [A] ... Current carrying capacity of the cable cross-section

I_b [A] ... Rated current for over-current protection

Type of layout according to IEC60204-1	Description
B1	Wire in conduit or cable duct
B2	Cable in conduit or cable duct
C	Cables or wires on walls
E	Cable or wires on open cable tray

Table 39: Type of layout for the mains supply line

Chapter 6 • Wiring

1. Electromagnetic Compatibility of the Installation

1.1 General information

If the guidelines for electromagnetic compatibility of the installation are followed, the ACOPOSmicro drive system meets the following guidelines:

- EMC guidelines 89/336/EWG
- Low-voltage guidelines 73/23/EWG.

ACOPOSmicro drive systems also meet the requirements for harmonized EMC product standard IEC61800-3:1996 + A11:2000 for industry (second environment).

Additional EMC measures must be implemented by the manufacturer of machines or systems if the product standards for the machine has lower limits or if the machine should conform to generic standard IEC 61000-6-4. Proof of conformity to the necessary limits must be provided according to the documentation for use of the EMC guidelines from the manufacturer or distributor of the machine or system.

Additional EMC measures are needed when operating ACOPOSmicro drive systems in a living area or when connecting ACOPOSmicro drive systems to a low voltage system which supplies buildings in living areas without an intermediate transformer (first environment).

1.2 Installation notes

- 1) The switching cabinet or the system must be constructed appropriately.
- 2) To prevent the effects of disturbances, the following lines must be properly shielded:
 - motor lines
 - encoder cables
 - control lines
 - data cables
- 3) Inductive switching elements such as contactors or relays are to be equipped with corresponding suppressor elements such as varistors, RC elements or damping diodes.
- 4) All electrical connections are to be kept as short as possible.
- 5) Cable shields are to be attached to the designated shield terminals and the plug housing.
- 6) Shielded cables with copper mesh or tinned copper mesh are to be used. Twisting or extending the protective mesh using single conductors is not allowed.
- 7) Unused cable conductors are to be grounded on both sides if possible.

2. Description of connections

2.1 Overview

The following connections are provided according to the ACOPOSmicro model being used:

Connection		Short description	80SD100XD.C044-01	80SD100XD.C04X-13	80SD100XD.C0XX-01	80SD100XD.C0XX-21	80SD100XS.C04X-01	80SD100XS.C0XX-01	Page
1-channel model	2-channel model								
X1		Power element supply	●	●	●	●	●	●	72
X2		Supply for CPU and digital inputs/outputs	●	●	●	●	●	●	73
X3A		X2X Link input	●	●	●	●	●	●	75
X3B		X2X Link output	●	●	●	●	●	●	75
X4		Motor 1	●	●	●	●	●	●	77
-	X5	Motor 2	●	●	●	●			77
X6	X6A	Incremental encoder 1	●	●			●		79
-	X6B	Incremental encoder 2	●						79
X7		Additional digital inputs-/outputs		●					82
X7		Additional analog inputs				●			82

Table 40: ACOPOSmicro connections - Overview

2.2 X1 - Power element supply

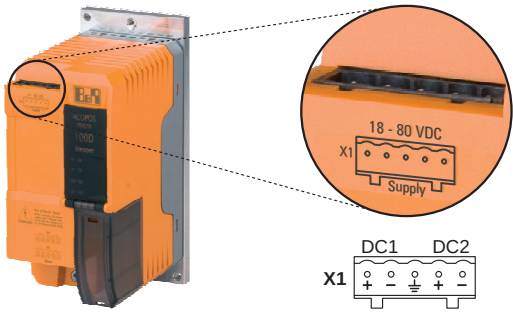
Figure	Pin assignments	
	Pin	Name
	+	DC1+
	-	DC1-
	⏏	PE (ground)
	+	DC2+
	-	DC2-
	The supply connection for the power element is made using the terminal block 0TB2105.9021 or 0TB2105.9121. Information about this terminal block can be found in the section 3.1.4 "Terminal blocks for the connection of the power supply" on page 55 in the chapter 3 "Technical data".	

Table 41: X1 connection - Power element supply

The connections DC1- and DC2- as well as DC1+ and DC2+ are connected together inside the ACOPOSmicro. Therefore, the supply connections can be used as follows:

- DC1 as feed and DC2 to route the DC voltages.
- DC1 and DC2 parallel to the feed at very high loads.

Information about safeguarding the lines can be found in the section 2 "Fuse protection" on page 67 in the chapter 5 "Dimensioning".

2.2.1 X1 diagram - Power supply

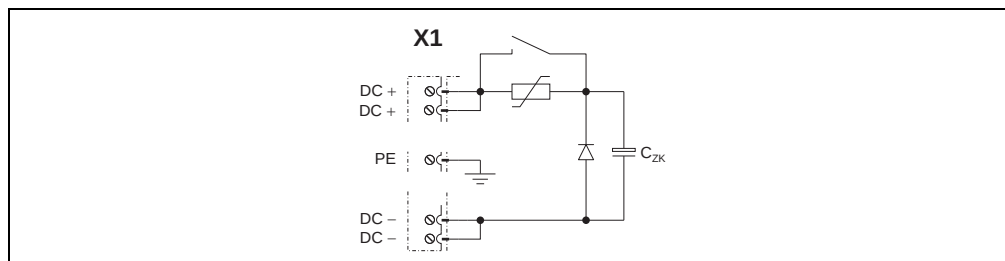


Figure 15: X1 diagram - Power supply

2.3 X2 - Supply for CPU and digital inputs/outputs

Figure	Pin assignments	
	Pin	Name
	1	Digital input 1
	2	Digital input 1 GND
	3	Digital input 2
	4	Digital input 2 GND
	5	24 VDC brake supply voltage ¹⁾
	6	24 VDC brake output
	7	Enable
	8	Enable GND
	9	CPU supply +24 VDC
	10	CPU supply GND
The supply connection for the CPU and the digital inputs/outputs is made using a terminal block (OTB1110.8010, OTB1110.8110 or OTB1310.3100). Information about these terminal blocks can be found in the section 3.1.2 "Terminal blocks for connecting I/O and CPU supply" on page 53 in the chapter 3 "Technical data".		

Table 42: X2 connection - Supply for CPU and digital inputs/outputs

1) Reference potential: CPU supply GND (pin 10)

The digital inputs 1+2 can be used as trigger inputs (see section 2.3.1 "Technical data" on page 39).

OTB1110 terminal blocks are single-row terminal blocks. In contrast, the OTB1310.3100 terminal has three rows (3 x 10 pins), which makes it particularly well-suited for connecting the inputs/outputs and acting as jumper terminal.

Information:

The red and blue rows of pins on the OTB1310.3100 are connected internally to simplify wiring. The user only has to create two connections so that the GND potential (blue) and the +24 VDC potential (red) can also supply the ACOPOSmicro CPU component on both rows of pins (see following figure).

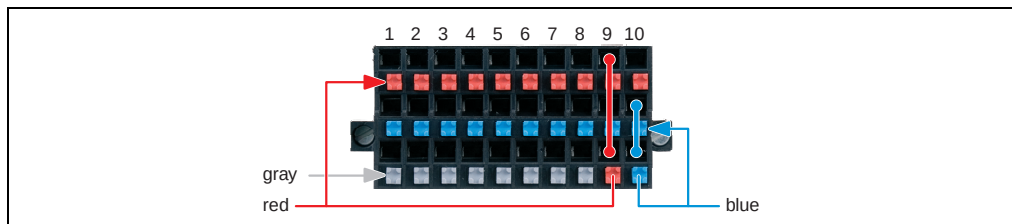


Figure 16: X2 - CPU supply via the terminal OTB1310.3100

2.3.1 X2 input/output circuit diagram

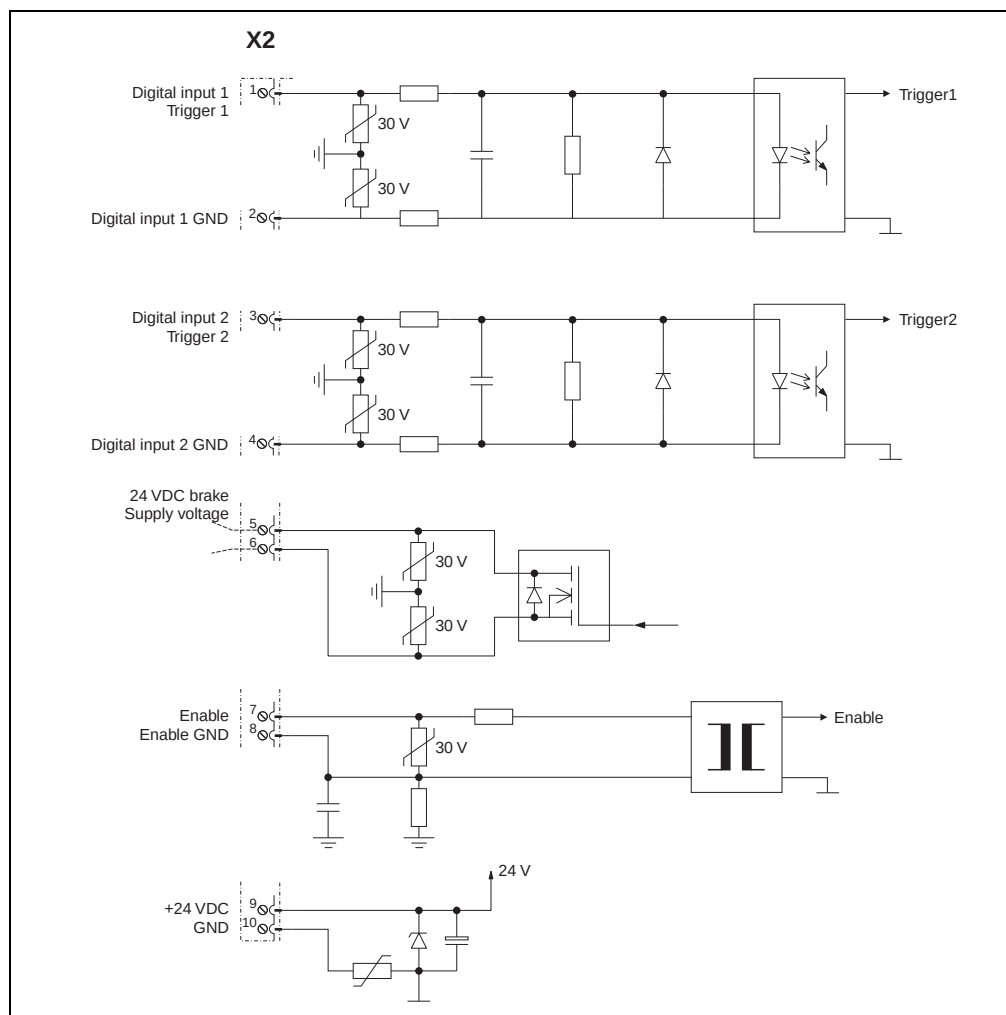


Figure 17: X2 - Input/output circuit diagram

Danger!

The brake controller integrated in the ACOPOSmicro modules and the holding brake integrated in the B&R standard motors are sufficient for the Safety Category B in accordance to EN 954-1.

Additional measures must be taken to achieve higher safety categories.

2.4 X3A/X3B - X2X Link connections

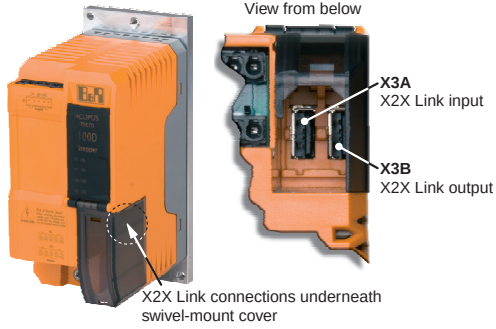
Figure	X2X Link connections
 View from below X3A X2X Link input X3B X2X Link output X2X Link connections underneath swivel-mount cover	The X2X Link connection is made using pre-assembled cables. Information about these cables can be found in the section 3.2 "X2X Link cable" on page 56 in the chapter 3 "Technical data". Information: Make sure that the X2X Link cable is properly connected and that input and output are not mixed up mistakenly. Sockets and plugs are accordingly coded for this purpose.

Table 43: X3A/X3B connections - X2X Link

When using the pre-assembled X2X Link device attachment cable (X20CA0X48), the following assignments for the individual wires must be taken into consideration:

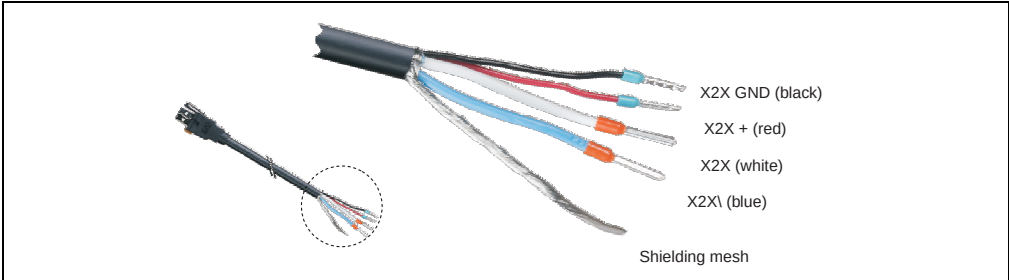


Figure 18: X2X Link device attachment cable wire colors

The X2X Link supply for the ACOPOSmicro is created **internally** from the 24 VDC supply, which means that it does not have to be supplied via an upstream power supply (e. g. X67PS1300 or X20BT9400). For subsequent modules, the X2X Link supply (X2X+, red wire on the X2X cable) is fed through internally from X3A to X3B (bridged).

Information:

By default, the X2X Link device attachment cable (see chapter 3 "Technical data" in the section 3.2.4 "X20CA0X48.xxxx" on page 59) is coded for the connection to the X2X Link input X3A. In order to use this cable on the X2X Link output X3B of the ACOPOSmicro, the coding must be changed accordingly.

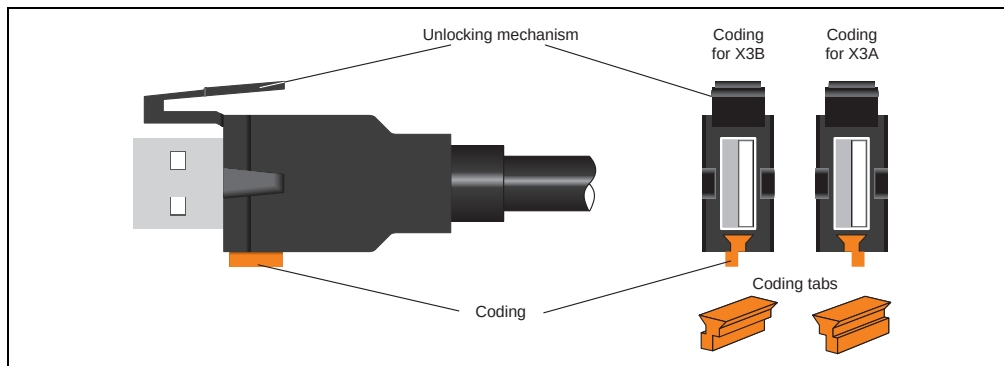


Figure 19: X2X Link plug with coding tabs

2.5 X4/X5 - Connection for motor 1/2

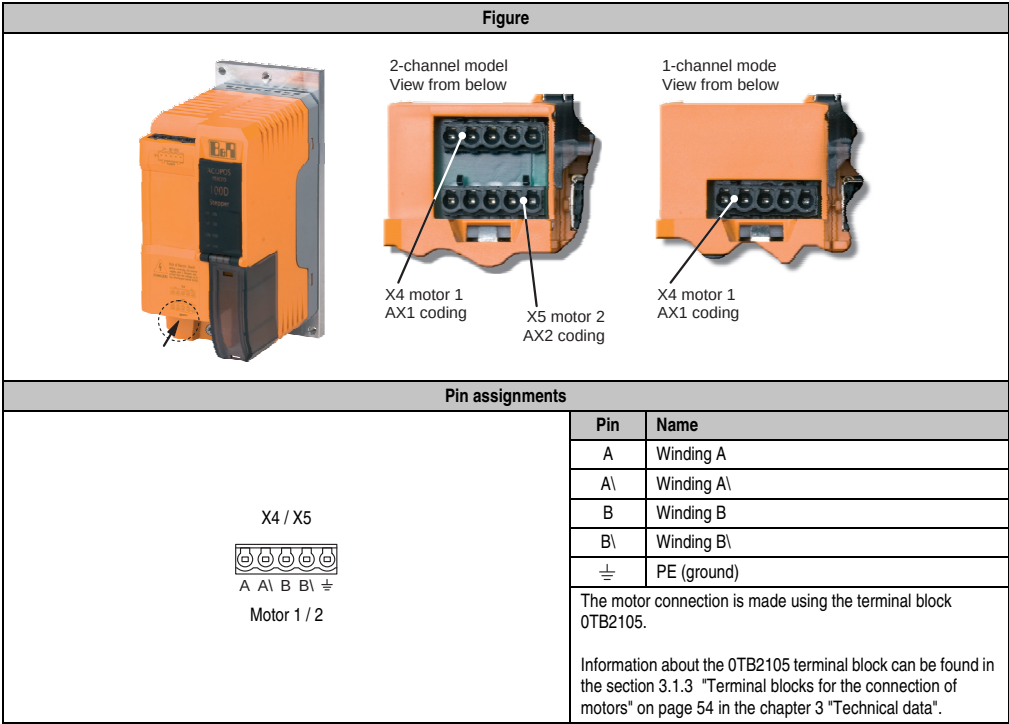


Table 44: X4/X5 connections - Motor 1/2

Information:

Since the sockets for the motor connections are coded, take care that the corresponding terminal blocks with the right coding are used. For more information about this, please refer to section 3.1.3 "Terminal blocks for the connection of motors" on page 54 in chapter 3 "Technical data".

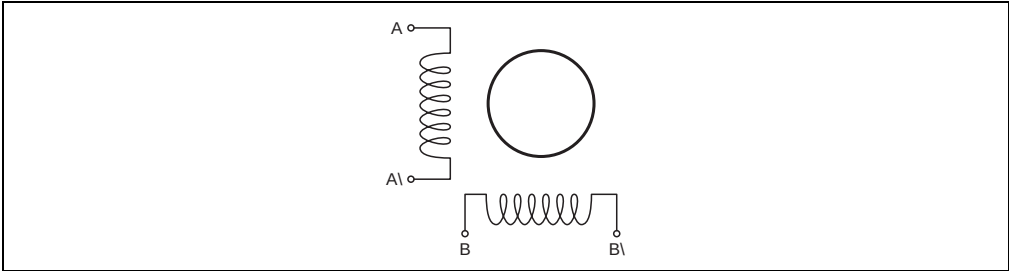


Figure 20: Stepper motor - Diagram

2.5.1 X4/X5 output circuit diagram for motor connection

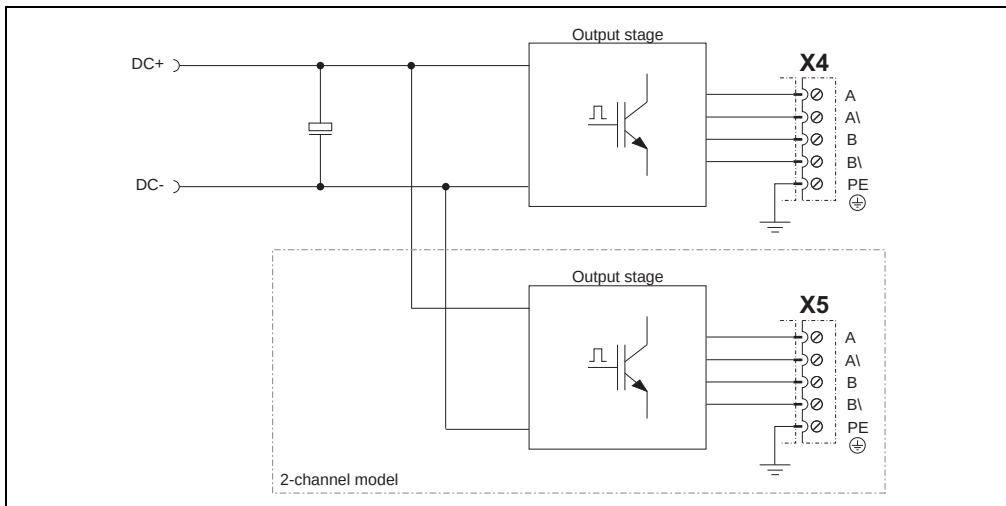


Figure 21: X4/X5 - Output circuit diagram for motor connection

2.6 X6 - Incremental encoder inputs 1/2

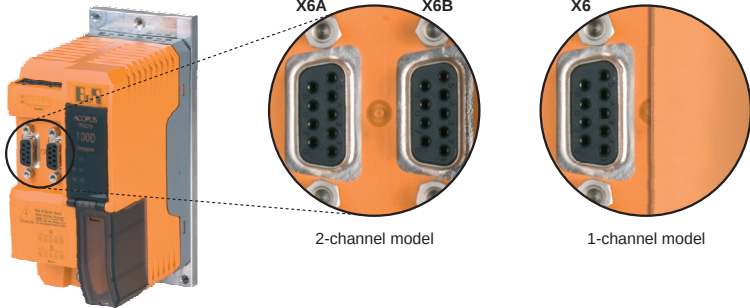
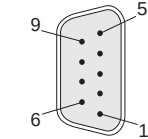
Figure		
		
Pin assignments		
 Incremental encoder 1 / 2	Pin	Name
	1	A
	2	NC
	3	B
	4	NC
	5	R
	6	NC
	7	NC
	8	+24 VDC encoder supply output
	9	GND = GND CPU supply

Table 45: X6A/X6B connections - Incremental encoder 1/2

2.6.1 X6 - Input circuit diagram for incremental encoder

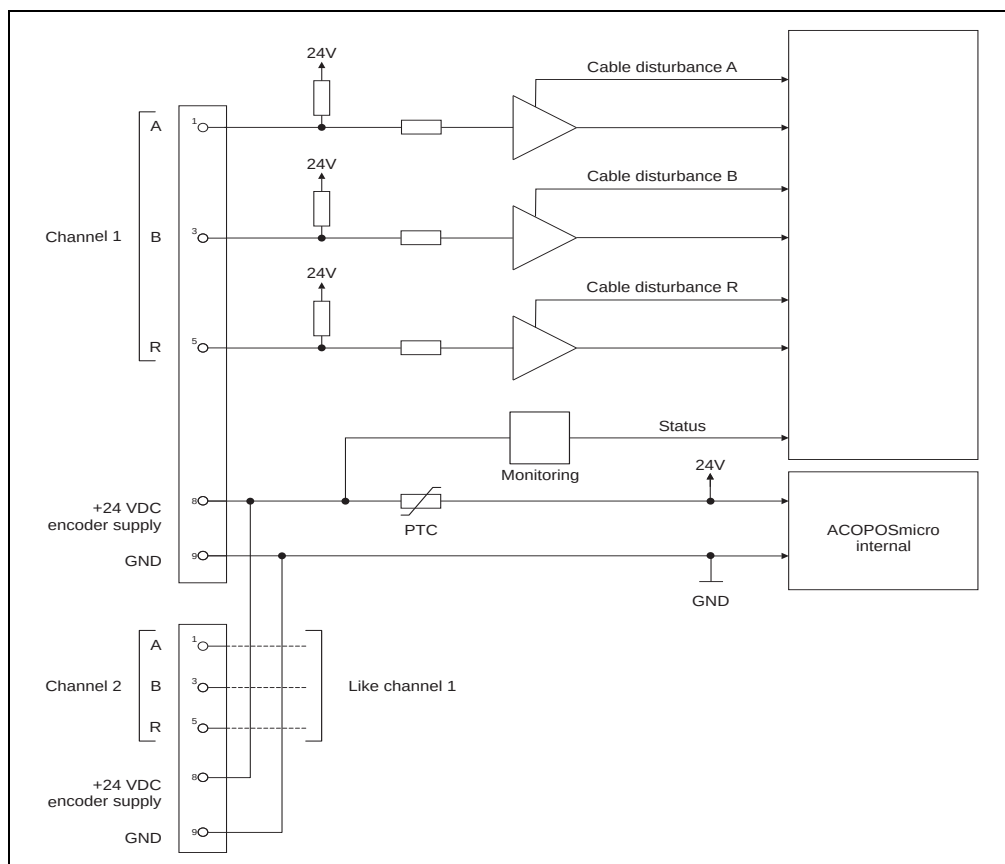


Figure 22: X6 - Input circuit diagram for incremental encoder

Information:

The +24 VDC encoder supply is safeguarded using a PTC for both encoder supply lines. Therefore, the load must be completely separated after the PTC has been tripped (e. g. due to a short-circuit or overload) to allow the PTC to cool down.

2.7 Protective ground connection PE

The wire cross section for the protective ground conductor is based on the selected cross-section of the "external line" (DC+ and DC-). In addition to the connection of the first protective ground conductor (terminal X1/PE, see section 2.2 "X1 - Power element supply" on page 72) a second, fixed (immobile) protective ground conductor might have to be connected to the respective protective ground connection on the ACOPOSmicro (in the event of a >42 VDC power supply). This must be done using a suitable M4 cable lug on the respective PE-connection.

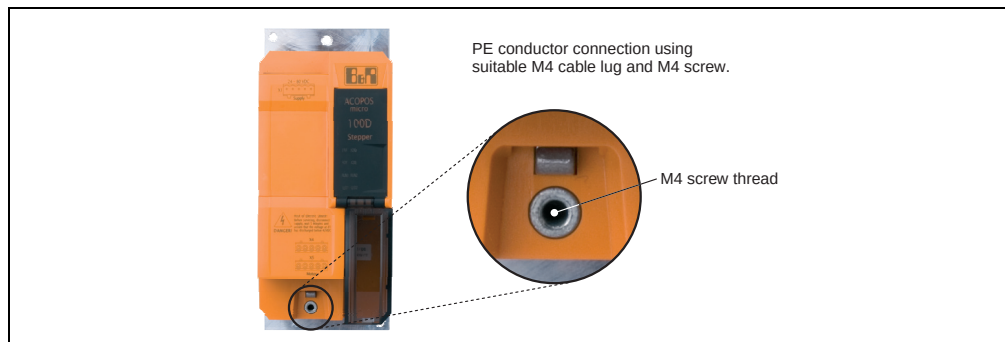


Figure 23: Connection of the PE conductor

Information:

To ensure electromagnetic compatibility of the installation, the X2X Link cable and the PE connection must be fastened using the mounting set contained in the accessory set that was included in delivery.

If the included shield plate is used, then the cable lug must be installed over it!

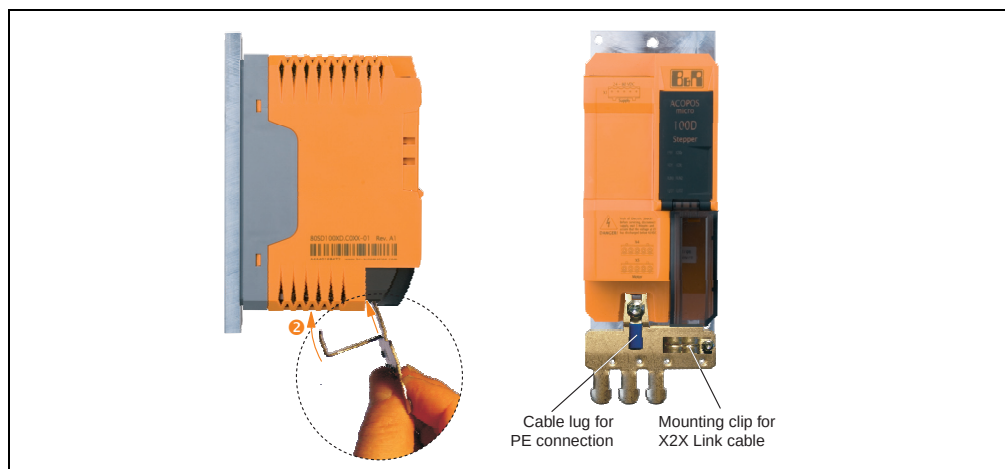


Figure 24: Electromagnetically compatible X2X Link connection

2.8 X7 - Additional inputs/outputs

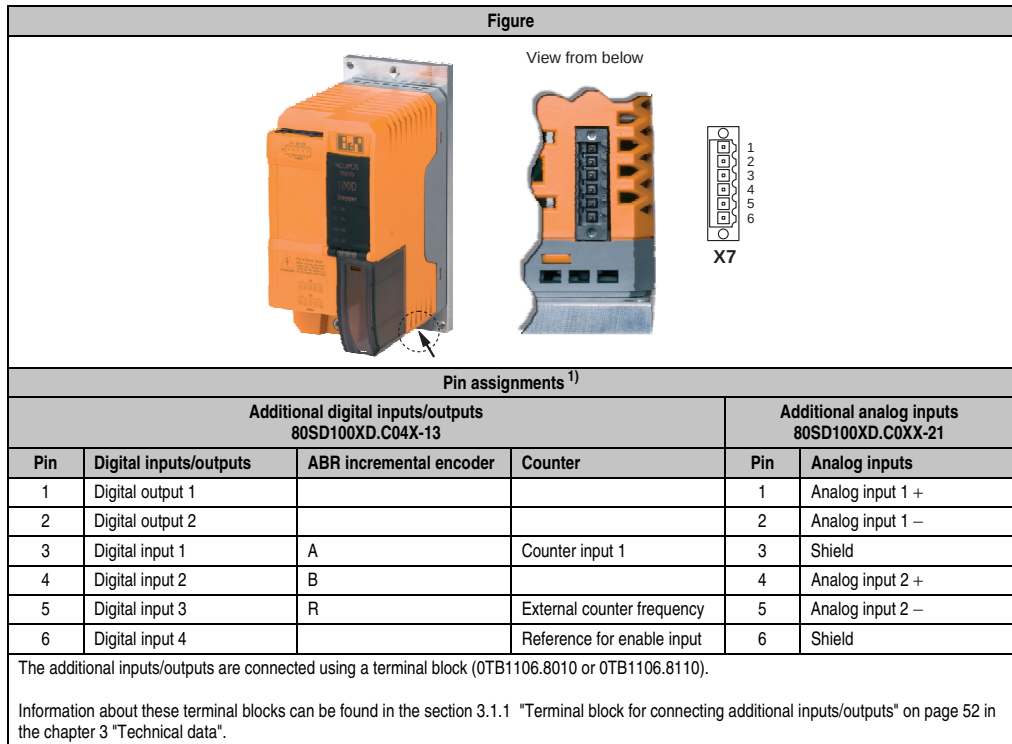


Table 46: X7 connection - Additional inputs/outputs

1) Pin assignments according to ACOPOSmicro model

2.8.1 Input circuit diagram for additional digital inputs

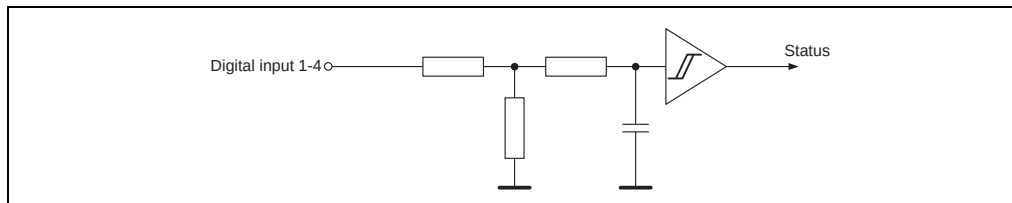


Figure 25: X7 - Input circuit diagram for additional digital inputs

2.8.2 Output circuit diagram for additional digital outputs

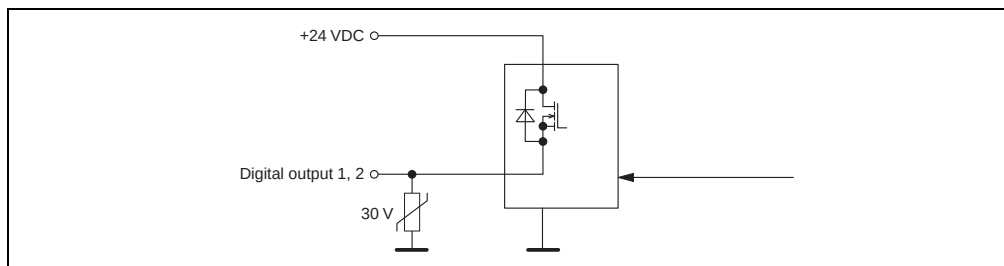


Figure 26: X7 - Output circuit diagram for additional digital outputs

2.8.3 Input circuit diagram for additional analog inputs

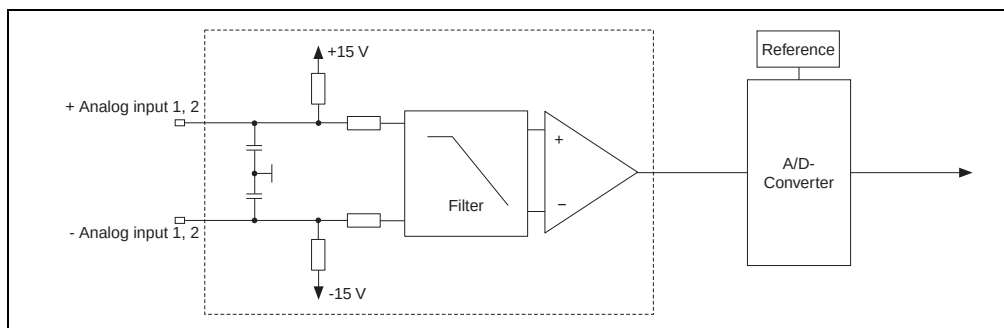


Figure 27: X7 - Input circuit diagram for additional analog inputs

Chapter 7 • Register description

1. Function models

A function model describes the registers for the module (memory model) which are available for the application. Only these registers are processed on the module during each cycle and transferred cyclically/acyclically using the bus. The function model is used to determine the module's functionality (standard, ramp).

1.1 - Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error or malfunction occurring. It should be noted that very fast cycles reduce the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time	
Standard function model	250 μ s
Ramp function model	250 μ s

Table 47: Minimum cycle time

1.2 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time	
Standard function model	250 μ s
Ramp function model	250 μ s

Table 48: Minimum I/O update time

1.3 Operation with bus controller

Information:

The ramp function model is used if the module ACOPOSmicro is operated on a bus controller.

2. Standard function model

2.1 Register overview ¹⁾

Register (dec.)		Description	Page	Configuration					
Channel 1	Channel 2			DT	L	R	W	C	A
0	2	Position	88	INT	1	●		●	●
10		Error/Status	89	UINT	1	●		●	●
12	14	Motor ID	91	UINT	1	●			●
16	24	Motor configuration MotorStep1	92	UINT	1		●	●	
18	26	Motor configuration MotorStep2	92	UINT	1		●	●	
20	28	Motor configuration MotorStep3	92	UINT	1		●	●	
22	30	Motor configuration MotorStep4	92	UINT	1		●	●	
33	36	Holding current	93	USINT	1		●		●
34	37	Rated current	93	USINT	1		●		●
35	38	Maximum current	93	USINT	1		●		●
46		Module configuration	93	UINT	1		●		●
54		Error acknowledgement / Digital output motor brake	94	USINT	1		●	●	
60	62	Position latch	94	INT	1	●		●	●
68	70	usSinceTrigger	94	UINT	1	●		●	●
72		Latch trigger status	94	USINT	1	●		●	●
73		Life cycle counter	95	SINT	1	●		●	
76		Latch configuration	95	USINT	1		●	●	
78		Trigger configuration	96	USINT	1		●	●	
80		Temperature	96	USINT	1	●			●
81		Motor ID trigger	97	USINT	1		●		●
84	86	Full step threshold	97	UINT	1		●		●
88		Heat sink temperature	97	SINT	1	●			●
90		DC bus voltage	97	UINT	1	●			●

Table 49: Standard function model

1) Key: DT ... Data type, L ... Length, R ... Read, W ... Write, C ... Cyclic, A ... Acyclic

Register description • Standard function model

Register (dec.)		Description	Page	Configuration						
Channel 1	Channel 2			DT	L	R	W	C	A	
Register for ACOPOSmicro with encoder										
130	144	ABR counter	98	INT	1	●		●		
134	148	ABR counter latch	98	INT	1	●		●		
154		ABR control register	98	USINT	1		●	●		
155		ABR configuration	99	USINT	1		●		●	
158		ABR advanced configuration	100	USINT	1		●		●	
Register for ACOPOSmicro models with additional digital inputs/outputs										
192		Status of digital inputs and outputs	100	USINT	1	●		●	●	
193		Set digital outputs / control register counter	101	USINT	1		●	●		
194		Counter	101	INT	1	●		●		
196		Counter latch	101	INT	1	●		●		
198		Counter configuration	102	UINT	1		●		●	
Register for ACOPOSmicro models with additional analog inputs										
1409		Status of the analog inputs	103	USINT	1	●		●		
1414		Analog input 1	103	INT	1	●		●		
1418		Analog input 2	103	INT	1	●		●		
Register for ACOPOSmicro models with additional analog inputs										
1038	1166	Trace control word	104	UINT	1		●	●		
1090	1218	Trace status	104	UINT	1	●		●		
1094	1222	Trace buffer status	104	UINT	1	●		●	●	
1098	1226	Number of triggers executed	105	IN	1	●		●	●	
1102	1230	Number of failed triggers	105	IN	1	●		●	●	
1345		Trace trigger	105	USINT	1		●	●		

Table 49: Standard function model (cont.)

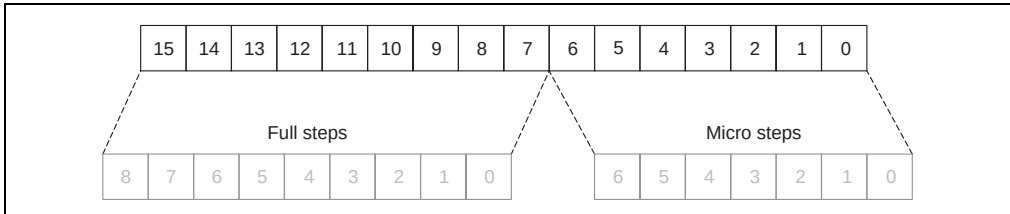
2.2 Register description

2.2.1 Position

This register is used to read the internal position counter of a motor axis. This is the equivalent to the position calculated by the ACOPOSmicro (set position). Each of these is a cyclic 16 bit counter for each channel.

The lowest 5 - 8 bits represent micro steps and the highest 8 - 11 bits represent full steps (depending on bit 5 + 6 of the "Module configuration" on page 93 register).

Example of the internal position counter format (7 bit microsteps, i.e. set bit 5 + 6 of the module configuration to binary 10):



The contents of the ABR counter and of the internal position counter can be replaced on ACOPOSmicro models with incremental encoders by setting the bits 3 and/or 7 in the ABR configuration register (see section "ABR configuration" on page 99).

Register	ABR configuration	
	Bit 3 (channel 1) / Bit 7 (channel 2) = 0	Bit 3 (channel 1) / Bit 7 (channel 2) = 1
Position	Internal position counter	ABR counter
ABR counter	ABR counter	Internal position counter

2.2.2 Error/Status

Each bit indicates an error (E) or status information (S):

Bit	Description ¹⁾	
0	0 ... Reserved	
1	E 0 ... No error 1 ... Error: Overtemperature	Motor 1
2	E 0 ... No error 1 ... Error: Set current not reached	
3	E 0 ... No error 1 ... Error: Overcurrent	
4	0 ... Reserved	
5	E 0 ... No error 1 ... Error: Overtemperature	Motor 2
6	E 0 ... No error 1 ... Error: Set current not reached	
7	E 0 ... No error 1 ... Error: Overcurrent	
8	S Status of drive 1: 0 ... Not ready 1 ... Ready	Motor 1
9	S Status of drive 2: 0 ... Not ready 1 ... Ready	Motor 2
10	S Status of ABR encoder 1: 0 ... Not ready 1 ... Ready	Encoder
11	E Encoder supply error 0 ... 24 VDC OK 1 ... Error: Encoder supply not in the required range	
12	S Status of ABR encoder 2: 0 ... Not ready 1 ... Ready	
13	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage	
14	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK	
15	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)	

1) On ACOPOSmicro 1-channel models, none of the bits that affect motor 2 have any function (0 ... reserved).
Depending on the ACOPOSmicro model, the bits that affect the encoder have no function (0 ... reserved).

If an error (indicated by "E" in the table) is detected, the corresponding error bit remains set until the error is acknowledged (see 2.2.7 "Error acknowledgement / Digital output motor brake" on page 94). Status bits (S) are not affected by the error acknowledgement.

Error "Set current not reached" (bit 2 + 6)

This error bit always occurs if the required current cannot be applied to the motor windings. This can (but doesn't have to be) caused by a broken connection. At higher speeds (depending on the motor), this error message can also appear even when a broken connection has not occurred if the desired current can no longer be provided in the motor windings. Because of the Back-EMF on the motor this bit is set at low speeds if the motor is operated with no load compared with full or part loads.

Status of "Drive X" (bit 8 + 9)

These bits have the value 0 if at least one of the following conditions is present:

- the supply voltage on the power element is not in the valid range
- the module is not enabled (Enable input)
- over-temperature error is present
- the corresponding motor is not being supplied with current

Status of "Status ABR encoder X" (bit 10 + 12)

These bits have the value 0 if at least one of the following conditions is present:

- the encoder supply (bit 11) is not OK
- broken connection on the ABR encoder (if encoder monitoring ¹⁾ is enabled)

1) Important: Encoder monitoring is only available starting with a specific hardware revision and firmware version:

80SD100XD.C044-01: Hardware revision B0 and firmware version 6

80SD100XS.C04X-01: Hardware revision B0 and firmware version 6

80SD100XD.C04X-13: Hardware revision B0 and firmware version 4

Register description • Standard function model

2.2.3 Motor ID

This register is used for practical differentiation and identification of the connected motor types (for service purposes).

The value of this register equals the amount of time (in μs) needed to trigger a current jump of $\Delta I = 1 \text{ A}$ in a motor winding. This time depends on several factors:

- Amplitude of operating voltage (DC bus voltage)
- Inductance and resistance of the motor winding

Notes																			
1.	To achieve reproducible results, the measurement must be made under the following defined conditions: a) At a standstill b) Motors must be in a half-step position (phase A with full current, phase B with no current). I.e. the internal step counter of the ACOPOSmicro (see the section 2.2.1 "Position" on page 88) must appear as follows: + Full steps are divisible by 4 + Microsteps = 0																		
2.	The condition 1.b is present after a ACOPOSmicro reset/power-up. Immediately afterwards, when the holding current is applied to the motor for the first time (at a standstill), the duration for applying the current is measured. This is also a suitable time to read the motor identification register in the application.																		
3.	As operating range for determining the motor ID, a current setting of at least 15% of the module's rated current (10 A) is recommended for the rated current register (see section 2.2.5 "Holding current, rated current, maximum current" on page 93) due to the lower relative inaccuracy at higher currents.																		
4.	Value range of the motor ID register																		
	<table> <tr> <th>Motor ID (Register 12, 14)</th><th>Meaning</th></tr> <tr> <td>0</td><td>No motor identifier available (after PowerUp as long as the measurement conditions are not met)</td></tr> <tr> <td>1 ... 32767</td><td>Valid value range of the <i>motor ID</i> register</td></tr> <tr> <td>65504 ... 65519</td><td>Ground fault: Measurement not possible</td></tr> <tr> <td>65531</td><td>Over-current: Measurement not possible</td></tr> <tr> <td>65532</td><td>Wire break: Measurement not possible</td></tr> <tr> <td>65533</td><td>Motor position incorrect: Measurement not possible</td></tr> <tr> <td>65534</td><td>Invalid value: Overflow</td></tr> <tr> <td>65535</td><td>Busy: Measurement in progress</td></tr> </table>	Motor ID (Register 12, 14)	Meaning	0	No motor identifier available (after PowerUp as long as the measurement conditions are not met)	1 ... 32767	Valid value range of the <i>motor ID</i> register	65504 ... 65519	Ground fault: Measurement not possible	65531	Over-current: Measurement not possible	65532	Wire break: Measurement not possible	65533	Motor position incorrect: Measurement not possible	65534	Invalid value: Overflow	65535	Busy: Measurement in progress
Motor ID (Register 12, 14)	Meaning																		
0	No motor identifier available (after PowerUp as long as the measurement conditions are not met)																		
1 ... 32767	Valid value range of the <i>motor ID</i> register																		
65504 ... 65519	Ground fault: Measurement not possible																		
65531	Over-current: Measurement not possible																		
65532	Wire break: Measurement not possible																		
65533	Motor position incorrect: Measurement not possible																		
65534	Invalid value: Overflow																		
65535	Busy: Measurement in progress																		

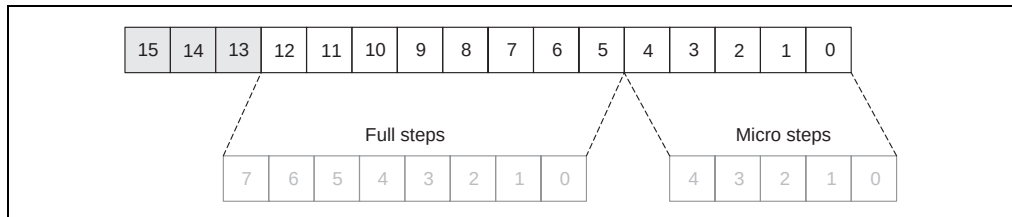
Table 50: Notes about motor identification

2.2.4 Motor configuration

These four registers (MotorStep1 to MotorStep4) are used to specify the amount and direction of steps that must be carried out by the module during the next X2X cycle, and to select the motor current (also see the section 2.2.5 "Holding current, rated current, maximum current" on page 93). Depending on the required resolution and maximum speed setting, the Module configuration can be used to specify which bit position is used as the 1s position for full steps (see bit 5+6 from the "Module configuration" on page 93 register).

Bit	Description
0 - 12	Number of steps for the module to move during the next X2X cycle.
13	Direction of movement: 0 ... Positive 1 ... Negative
14 - 15	Motor current selection: 00 ... Motor without current 01 ... Holding current 10 ... Rated current 11 ... Maximum current

Example for 5 bit microsteps (set bit 5+6 of the module configuration to binary 00):



The number of transfer values per X2X cycle is specified in the Module configuration by bit 3+4 (see the section 2.2.6 "Module configuration" on page 93). If only one transfer value (bit 3+4 = 00) is specified, then the motor is advanced by MotorStep1 until the next X2X cycle. If 2 or 4 transfer values are specified, then the X2X cycle is divided respectively.

Example: X2X cycle = 1 ms (1000 μ s)

Time	Number of transfer values (see 2.2.6 "Module configuration" on page 93)		
	1 (bit 3 - 4 = 00)	2 (bit 3 - 4 = 01)	4 (bit 3 - 4 = 10)
0 - 250 μ s	Motor step 1	Motor step 1	Motor step 1
250 - 500 μ s			Motor step 2
500 - 750 μ s		Motor step 2	Motor step 3
750 - 1000 μ s			Motor step 4

Register description • Standard function model

2.2.5 Holding current, rated current, maximum current

The holding current, rated current and maximum current registers are used for configuring the desired motor current.

- Units: Percent of the module rated current (100% is equal to 10 A)

During runtime, its possible to switch between the preconfigured current values using bits 14+15 in the "MotorStepX" registers (see section 2.2.4 "Motor configuration" on page 92).

It would make sense if:

- Holding current < rated current < maximum current

The motor rated current is entered in the rated current register according to the motor data sheet.

Register	Description
Holding current	In situations requiring less torque (e.g.: at a standstill), the device should be switched to holding current. This reduces heating of the motor.
Rated current	Current during "normal operation". Due to the lower relative inaccuracy at higher currents, a current setting of at least 15% of the module's rated current (10 A) is recommended for the rated current register.
Maximum current	Maximum current should be selected if a higher motor torque is required during acceleration phases (short-term).

2.2.6 Module configuration

Bit	Description
0 - 2	0 ... Reserved
3 - 4	Number of transfer values per X2X cycle (see description of the registers MotorStep1 to MotorStep4 in section 2.2.4 "Motor configuration" on page 92): 00 ... 1 x $\Delta s / \Delta t$ (Transfer values: MotorStep1) 01 ... 2 x $\Delta s / \Delta t$ (Transfer values: MotorStep1 - MotorStep2) 10 ... 4 x $\Delta s / \Delta t$ (Transfer values: MotorStep1 - MotorStep4) 11 ... Reserved Information: This setting applies to both channels in 2-channel models.
5 - 6	Resolution of microsteps for "MotorStepX" and "Position" 00 ... 5 bits (Bit 0 - 4) 01 ... 6 bits (Bit 0 - 5) 10 ... 7 bits (Bit 0 - 6) 11 ... 8 bits (Bit 0 - 7) See the sections 2.2.4 "Motor configuration" on page 92 and 2.2.1 "Position" on page 88
7 - 15	0 ... Reserved

2.2.7 Error acknowledgement / Digital output motor brake

Bit	Description
0	Error acknowledgement - Motor 1
1	Error acknowledgement - Motor 2
2 - 6	0 ... Reserved
7	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

2.2.8 Position latch

The position counter (internal position counter) will be applied to this register at the next latch event (see 2.2.10 "Latch trigger status" on page 94).

On ACOPOSmicro models with incremental encoders, the contents of this register can be replaced with the register ABR counter latch (see page 98) by setting the bits 3 and/or 7 in the ABR configuration register (see page 99).

Register	ABR configuration	
	Bit 3 (channel 1) / Bit 7 (channel 2) = 0	Bit 3 (channel 1) / Bit 7 (channel 2) = 1
Position latch	Internal position counter latch	ABR counter latch
ABR counter latch	ABR counter latch	Internal position counter latch

2.2.9 usSinceTrigger

This register indicates the time (in μ s) that has passed since the trigger event occurred (see 2.2.10 "Latch trigger status" on page 94).

2.2.10 Latch trigger status

Bit	Description ¹⁾	
0	Latch input: Digital input for the latch event (level)	Stepper motor 1
1	LatchDone: State changes each time the counter state is successfully latched (reset value = 0)	
2	Latch input: Digital input for the latch event (level)	Stepper motor 2
3	LatchDone: State changes each time the counter state is successfully latched (reset value = 0)	
4	Trigger 1: Trigger input (level)	
5	Trigger 2: Trigger input (level)	
6	LatchDone: State changes each time ABR counter 1 is successfully latched (reset value = 0) See ABR control register on page 98 and the ABR configuration register on page 99	ABR counter 1
7	LatchDone: State changes each time ABR counter 2 is successfully latched (reset value = 0) See ABR control register on page 98 and the ABR configuration register on page 99	ABR counter 2

1) On ACOPOSmicro 1-channel models, none of the bits that affect stepper motor 2 and trigger 2 have any function (0 ... reserved).
Depending on the ACOPOSmicro model, the bits that affect the encoder have no function (0 ... reserved).

Register description • Standard function model

2.2.11 Life cycle counter

This value is incremented with each X2X cycle (the counter starts back at zero if an overflow occurs) and is used to detect a module error, module failure or network failure.

2.2.12 Latch configuration

Bit	Description ¹⁾	
0	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 1
1 - 2	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 1 10 ... Latch position at negative edge on trigger input 1 11 ... Reserved	
3	0 ... Reserved	
4	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 2
5 - 6	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 2 10 ... Latch position at negative edge on trigger input 2 11 ... Reserved	
7	0 ... Reserved	

1) On ACOPOSmicro 1-channel models, none of the bits that affect channel 2 have any function (0 ... reserved).

2.2.13 Trigger configuration

Bit	Description ¹⁾
0	0 ... Trigger 1: Trigger edge = positive 1 ... Trigger 1: Trigger edge = negative
1	Activate trigger 1 (when changes occur)
2	0 ... Trigger 2: Trigger edge = positive 1 ... Trigger 2: Trigger edge = negative
3	Activate trigger 2 (when changes occur)
4 - 7	0 ... Reserved

1) On ACOPOSmicro 1-channel models, none of the bits that affect trigger 2 have any function (0 ... reserved).

Trigger function procedure:

- Select the desired trigger edge using bit 0 (Trigger 1) or bit 2 (Trigger 1)
- Activate the trigger function by changing the state of bit 1 (Trigger 1) or bit 3 (Trigger 1). When this bit changes, usSinceTrigger (µs counter) will be cleared.
- When the trigger event occurs, the µs counter usSinceTrigger is started.
- The counter usSinceTrigger cannot overrun, i.e. the counter is stopped at $2^{16} - 1$ and retains this value until the next time the trigger function is activated.

The trigger function can be re-activated at any time by changing the state of bit 1 or bit 3, regardless of if a trigger event has occurred or if usSinceTrigger has reached the maximum value.

2.2.14 Temperature

The internal module temperature (on the controller board) is output in °C.

Measurement range: -40 to +125°C

Overtemperature shut-off (at +85°C)

If the module temperature reaches or exceeds +85°C, ...

- ... the application is notified via the "Overtemperature" error bit.
- ... all of the module's outputs are disabled (turned off).

The error-bits are automatically cleared by the module and the outputs are re-enabled (turned on) after the module temperature sinks to +83°C.

The module temperature will not reach the cutoff limit when operating the module within the specified environmental conditions and the permissible output-current range. The temperature cutoff serves only to protect the module hardware and must not be misused for operating the module beyond the specified range as this can of course lower the life-span.

Register description • Standard function model**2.2.15 Motor ID trigger**

An asynchronous measurement of the motor ID can be made by this register. The application must ensure that the conditions for a measurement are fulfilled (see table 50 "Notes about motor identification").

Bit	Description ¹⁾
0	0 ... No effect 1 ... Positive edge triggers motor ID measurement for motor 1
1	0 ... No effect 1 ... Positive edge triggers motor ID measurement for motor 2
2 - 7	0 ... Reserved

1) On ACOPOSmicro 1-channel models, none of the bits that affect motor 2 have any function (0 ... reserved).

2.2.16 Full step threshold

This register configures the speed (unit: steps/second). When this defined speed has been reached, the drive will automatically change from micro-steps to full-step mode. This makes it possible to optimize the torque at higher speeds, while micro-step mode ensures optimum concentricity at lower speeds.

It does not make sense to change to full-step mode when at a standstill because fine positioning would then no longer be possible. This is why the value "0" does not make sense in the Full Step Threshold register and is interpreted as deactivation of full-step mode (i.e. the motor will always be operated in micro-step mode).

Example

Micro-step mode should change to full-step at 500 steps/seconds. On a motor with 200 steps per revolution, this would be equal to a speed of:

$$T^{-1} = \frac{500 \text{ Steps/Second}}{200 \text{ Steps/Revolution}} = 2.5 \frac{\text{Revolutions}}{\text{Second}} = 150 \text{ 1/min}$$

2.2.17 Heat sink temperature

This register indicates the temperature of the power board.

- Units: °C
- Measurement range: +10 to +125°C

2.2.18 DC bus voltage

Contains the supply voltage in the power element (in volts).

2.2.19 Register for ACOPOSmicro with encoder

ABR counter

This counter is a cyclic 16 bit counter for each encoder input. The relationship between this counter and the internal position counter depends on the resolution of the ABR encoder and the defined microsteps of the internal position counter.

The contents of the ABR counter and of the internal position counter can be replaced on ACOPOSmicro models with incremental encoders by setting the bits 3 and/or 7 in the ABR configuration register (see page 99).

Register	ABR configuration	
	Bit 3 (channel 1) / Bit 7 (channel 2) = 0	Bit 3 (channel 1) / Bit 7 (channel 2) = 1
Position	Internal position counter	ABR encoder counter state
ABR counter	ABR encoder counter state	Internal position counter

ABR counter latch

The ABR counter will be applied to this register at the next latch event (see 2.2.10 "Latch trigger status" on page 94).

On ACOPOSmicro models with incremental encoders, the contents of this register can be replaced with the ABR counter register by setting the bits 3 and/or 7 in the ABR configuration register (see page99).

Register	ABR configuration	
	Bit 3 (channel 1) / Bit 7 (channel 2) = 0	Bit 3 (channel 1) / Bit 7 (channel 2) = 1
Position latch	Internal position counter latch	ABR counter latch
ABR counter latch	ABR counter latch	Internal position counter latch

ABR control register

Bit	Description ¹⁾
0	0 ... Deactivate referencing of ABR counter 1 1 ... Activate referencing of ABR counter 1 (positive edge activates referencing)
1 - 3	0 ... Reserved
4	0 ... Deactivate referencing of ABR counter 2 1 ... Activate referencing of ABR counter 2 (positive edge activates referencing)
5 - 7	0 ... Reserved

1) On ACOPOSmicro models with an encoder, the bit 4 for ABR counter 2 does not have any function (0 ... reserved)

ABR configuration

Bit	Description ¹⁾	
0 - 1	Configure homing of ABR counter 1 00 ... No homing 01 ... Homing at positive edge on the R signal 10 ... Homing at negative edge on the R signal 11 ... Homing immediately at rising edge of bit 0 on the ABR control register (see page 98)	Encoder 1
2	Homing mode: 0 ... Copy ABR counter to ABR counter latch 1 ... Set ABR counter to zero	
3	0 ... No effect 1 ... Replace contents of the following register from channel 1: Position ↔ ABR counter Position latch ↔ ABR counter latch	
4 - 5	Configure homing of ABR counter 1 00 ... No homing 01 ... Homing at positive edge on the R signal 10 ... Homing at negative edge on the R signal 11 ... Homing immediately at rising edge of bit 0 on the ABR control register	Encoder 2
6	Homing mode: 0 ... Copy ABR counter to ABR counter latch 1 ... Set ABR counter to zero	
7	0 ... No effect 1 ... Replace contents of the following register from channel 2: Position ↔ ABR counter Position latch ↔ ABR counter latch	

1) On ACOPOSmicro models with an encoder, the bits 4 - 7 for ABR counter 2 does not have any function (0 ... reserved)

Register	ABR configuration	
	Bit 3 (channel 1) / Bit 7 (channel 2) = 0	Bit 3 (channel 1) / Bit 7 (channel 2) = 1
Position	Internal position counter	ABR counter
Position latch	Internal position counter latch	ABR counter latch
ABR counter	ABR counter	Internal position counter
ABR counter latch	ABR counter latch	Internal position counter latch

The following is true, regardless of this configuration (bit 3 or bit 7):

Homing of the ABR counter will always be configured with the bits 0 - 2 and 4 - 6 of the ABR configuration register and enabled at the positive edge of bit 0 and bit 4 in the ABR control register.

The latch function for the internal position counter (for the drive) will always be checked and enabled using the bits 0 - 2 and 4 - 6 of the Latch configuration register.

ABR advanced configuration

Bit	Description ¹⁾	
0 - 1	Encoder monitoring ²⁾ for ABR encoder 1: 00 ... No encoder monitoring 01 ... Encoder monitoring of signals A and B 10 ... Encoder monitoring of signals A, B and R 11 ... Reserved	Encoder 1
2 - 3	0 ... Reserved	
4 - 5	Encoder monitoring ²⁾ for ABR encoder 2: 00 ... No encoder monitoring 01 ... Encoder monitoring of signals A and B 10 ... Encoder monitoring of signals A, B and R 11 ... Reserved	Encoder 2
6 - 7	0 ... Reserved	

1) On ACOPOSmicro models with an encoder, the bits 4 - 5 for ABR encoder 2 does not have any function (0 ... reserved)

2) Important: Encoder monitoring is only available starting with a specific hardware revision and firmware version:

80SD100XD.C044-01: Hardware revision B0 and firmware version 6

80SD100XS.C04X-01: Hardware revision B0 and firmware version 6

80SD100XD.C04X-13: Hardware revision B0 and firmware version 4

2.2.20 Register for ACOPOSmicro models with additional digital inputs/outputs

Status of digital inputs and outputs

Bit	Description		
0	Status of digital input 1	A	Counter input 1
1	Status of digital input 2	B	
2	Status of digital input 3	R	External counter frequency for counter input 1
3	Status of digital input 4		Reference for enable input
4	Status of digital output 1		
5	Status of digital output 2		
6	Overflow of counter 1 (only valid if gate or period measurement is set in the counter configuration with bit 3 - 6): 0 ... Counter value is within the counting range (0 to \$7FFF). The value is only valid if bit 0 of the control register counter is set. 1 ... Counter overflow This bit is reset with Bit 0 of the control register counter.		
7	LatchDone: State changes each time the counter state is successfully latched (reset value = 0)		

Set digital outputs / control register counter

Bit	Description
0	0 ... Disable detection of counter overflow and reset the overflow bit on counter 1 1 ... Enable detection of counter overflow
1	0 ... Reserved
2	Referencing (only valid in AB(R) mode, see Bit 8+9 in counter configuration) 0 ... Disable referencing 1 ... Enable referencing
3	0 ... Reset counter (to zero) 1 ... Start counter Note: Only set after counter configuration is complete!
4 - 5	0 ... Reserved
6	Set digital output 1
7	Set digital output 2

Counter

This register contains the present counter state. Different counters (event counter, ABR counter, etc) are available depending on the configuration

Counter latch

The counter will be applied to this register at the next latch event (see ABR mode in "Counter configuration" on page 102).

Counter configuration

Bit	Description
0 - 1	Counter frequency (only valid if set in gate or period measurement): 00 ... Internal counter frequency 4 MHz 01 ... Internal counter frequency 31.25 kHz 10 ... External counter frequency (digital input 3) 11 ... Reserved
2	0 ... Reserved
3 - 6	Counter mode: 0000 ...No counter 0001 ...Event counter, positive edge 0010 ...Event counter, negative edge 0011 ...Event counter, both edges 0100 ...Period measurement between two consecutive negative edges 0101 ...Period measurement between two consecutive positive edges 0110 ...Gate measurement of the LOW level 0111 ...Gate measurement of the HIGH level 1000 ...AB counter without homing 1001 ...ABR counter with homing Counter set to zero: at positive edge of R 1010 ...ABR counter with homing Counter set to zero: at positive edge of R 1011 ...ABR counter with immediate homing Counter set to zero: at positive edge of bit 2 of the counter control register 1100 ...AB counter without homing 1101 ...ABR counter with homing Counter state is applied to the counter latch register: at positive edge of R 1110 ...ABR counter with homing Counter state is applied to the counter latch register: at positive edge of R 1111 ...ABR counter with immediate homing Counter state is applied to the counter latch register: at positive edge of bit 2 of the counter control register
7	0 ... Reserved
8 - 9	Configuration of reference enable input (digital input 4) 00 ... The digital input 4 (reference enable input) does not affect the R input. The R input is always enabled. 01 ... Reserved 10 ... R input is enabled if reference enable input is LOW. 11 ... R input is enabled if reference enable input is HIGH.
10 - 15	0 ... Reserved

Register description • Standard function model

2.2.21 Register for ACOPOSmicro models with additional analog inputs

Status of the analog inputs

Bit	Description	
0	Analog input 1	0 ... OK 1 ... Outside the valid range or open circuit
1	Analog input 2	
2 - 7	0 ... Reserved	

In addition to the status info, the analog value of the corresponding input will also be set if an error occurs (status bit = 1):

Error	Analog value (dec.)	Analog value (hex.)
Wire break	+32.767	0x7FFF
Input voltage > +10 V	+32.767	0x7FFF
Input voltage < -10 V	-32.767	0x8001
Invalid value	-32.768	0x8000

Analog input 1 / Analog input 2

These registers receive the analog value of the corresponding input.

Measurement range:

Input voltage	Register value
+10 V	+32.767
:	:
-10 V	-32.768

2.2.22 Trace functions for ACOPOSmicro models with analog inputs

Trace register

Trace control word

Bit 0 of the trace control word is used to turn on/off the trace functionality for the corresponding analog input.

Bit	Description
0	0 ... Trace function is disabled 1 ... Trace function is enabled
1 - 15	0 ... Reserved

A buffer overrun error can be cleared by resetting bit 0.

Trace status

Bit	Description
0	0 ... Trace function is disabled 1 ... Trace function is enabled
1	0 ... Reserved
2	0 ... Trace data recording not active 1 ... Trace data recording active
3	0 ... Trace data reading is not active 1 ... Trace data reading is active
4	0 ... Not ready for trigger 1 ... Ready for trigger
5	Trigger active 0 ... No trigger active, trace recording stopped 1 ... Trigger in process, trace data is being recorded
6	Trace OK 0 ... Buffer overrun or trace function disabled for analog input 1 ... No buffer overrun error
7	Trace error 0 ... No buffer overrun or trace function disabled for analog input 1 ... Buffer overrun error

Trace buffer status

This register contains the amount of buffer memory (in bytes) available for the trace recording. The buffer must be large enough for the configured number of sampling values to be recorded.

If "Infinite Trace" was selected, then the buffer must be read fast enough for analog values to be recorded at the set sampling rate.

Register description • Standard function modelNumber of triggers executed

This register contains the number of successfully executed trigger events (trace recordings).

Number of failed triggers

This register contains the number of failed trigger events (trace recordings).

This register will be increased by one when a trigger event occurs and under the following conditions:

- Trace function is not enabled
- Trace buffer is full (two trigger events in short succession)
- Pre-trigger function: If the configured number of recordings before the trigger event has not yet been reached.

Trace trigger

This register can be used to start recording separately for each trace instance. This is only possible if the trigger function has been enabled for the corresponding analog input.

Bit	Description	
0	Trigger for trace instance 1	The trigger event will be triggered at a positive edge (bit changes from 0 → 1).
1	Trigger for trace instance 2	
2 - 7	0 ... Reserved	

Trace runtime configuration

The following points describe how the trace runtime configuration is performed in Automation Studio.

Turn on/off trace for analog input X (Trace AnalogInput01/02 = on/off):

The trace function must first be turned on/off for the respective analog channel.

If the trace function was turned on for both analog inputs, then the recorded trace data will be provided by the AsIoTrc library in the following order:

Sample	Array index	Channel
1	0	Analog input 1
	1	Analog input 2
2	2	Analog input 1
	3	Analog input 2
:	:	:

Table 51: Order of trace data from two analog channels

Number of trace buffer:

The number of the local trace buffer must also be set on the CPU.

This number specifies how many trace recordings the CPU can save. If a trigger event occurs, even though all configured trace buffers are full, then the respective sampling values will be lost.

The recorded trace data buffer will be read out with the AsIoTrc library.

Number of samples:

The number samples to be recorded on the ACOPOSmicro module must then be set.

Make sure that the trace buffer is made large enough for the trace data occurring on the module to be recorded.

If "Infinite Trace" was selected, then the buffer must be read fast enough for analog values to be recorded at the set sampling rate.

The total number of samples from all channels (channel 1+2 for trace 1+2) cannot be larger than 8192.

Block sizes for asynchronous data transfer (AsynSize):

A higher value will result in faster data transfer from the module to the CPU.

Number of data blocks sent before repetition (AsynForward):

Every data block transferred must be confirmed by the X2X Link master. If a confirmation has not been received after the number of data blocks defined here, then all non-confirmed data blocks will be sent again.

Register description • Standard function model

Trace mode:

Trace mode	Description
Triggered	Trace recording will start after the trigger event (see bit 0/1 of the <i>Trace trigger</i> register on page 105).
Infinite	Infinite trace can be started with bit 0/1 of the <i>Trace trigger</i> register (see page 105). X2X must be configured in such a way as to ensure that the amount of data generated can be transferred quickly enough. The trace buffer must be read quickly to prevent an overflow in the module's local memory and to ensure that the samples can be recorded at the configured sampling rate.

Recording range of trace data (Trace Start/Stop Position):

This defines the number of samples that should be recorded before/after the trigger event.

Trigger type	Start position	Stop position	Beginning of recording	End of recording
Post trigger	$x \geq 0$	$y > 0$	x samples after trigger event	y samples after start position
Pre-trigger	$x < 0$	$y > 0$	x samples before trigger event	y samples after trigger event

Trace library

The AsIoTrc library must be used to read the recorded trace data (see the Automation Studio help for more information).

The ACOPOSmicro module with additional analog inputs has two separate trace instances that can be triggered at different times, independent of one another.

Each trace instance has a device name:

- <Hardware address of the module>.<Trace instance>

This device name is used to read the trace data from the desired channel using the AsIoTrcGet library function.

Examples:

- Trace for trace instance 1: "SS1.IF1.ST1.TRC1"
- Trace for trace instance 2: "SS1.IF1.ST1.TRC2"

3. Ramp function model

This function model was based on the CANopen communication profile DS402.

The ramp function model is used if an ACOPOSmicro module is operated on a bus controller.

3.1 Register overview

Register (dec.)		Description ¹⁾	Page	Configuration ²⁾					
Channel 1	Channel 2			DT	L	R	W	C	A
0	8	Set position/speed	110	DINT	1		●	●	
4	12	Control word	111	UINT	1		●	●	
6	14	Mode	113	SINT	1		●	●	
7	15	Motor brake	115	BOOL	1		●	●	
16		D Set digital outputs / control register counter	121	USINT	1		●	●	
0	8	Current position (cyclic)	110	DINT	1	●		●	●
4	12	Status	112	UINT	1	●		●	●
6		Input status	110	USINT	1	●		●	●
16		D Counter	121	INT	1	●		●	
18		D Counter latch	121	INT	1	●		●	
20		D Status of digital inputs and outputs	121	USINT	1	●		●	●
42		UZK DC bus voltage	115	UINT	1	●			●
44		Temperature of output stage	115	SINT	1	●			●
46		Temperature	115	SINT	1	●			●
48	112	Holding current	119	USINT	1		●		●
49	113	Rated current	119	USINT	1		●		●
50	114	Maximum current	119	USINT	1		●		●
52	116	Maximum speed	115	UINT	1		●		●
54	118	Max. acceleration	115	UINT	1		●		●
56	120	Maximum deceleration	115	UINT	1		●		●
58	122	Turn-around loop	118	INT	1		●		●
60	124	Set position 1	116	DINT	1		●		●
64	128	Set position 2	116	DINT	1		●		●
68	132	Homing speed	116	UINT	1		●		●
70	134	Homing configuration	116	SINT	1		●		●
72	136	Full step threshold	116	UINT	1		●		●
75	139	Jolt time limitation	117	USINT	1		●		●
80	144	Read back control word	117	UINT	1	●			●
82	146	Read back mode	117	SINT	1	●			●
84	148	Motor ID	120	UINT	1	●			●
86	150	Homed zero position for cyclic counter	117	DINT	1	●			●
90	154	G Current position (acyclic)	110	DINT	1	●			●

Table 52: Ramp function model

Register (dec.)		Description ¹⁾		Page	Configuration ²⁾					
Channel 1	Channel 2				DT	L	R	W	C	A
94	158	G	Homed zero position for acyclic counter	117	DINT	1	●			●
98	162		Error code	118	UINT	1	●			●
200		D	Counter configuration	122	UINT	1		●		●
1409 (16) ³⁾		A	Status of the analog inputs	123	USINT	1	●		●	
1414 (18) ³⁾		A	Analog input 1	123	INT	1	●		●	
1418 (20) ³⁾		A	Analog input 2	123	INT	1	●		●	
Trace register for ACOPOSmicro models with analog inputs ⁴⁾										
1038	1166	A	Trace control word	124	UINT	1		●	●	
1090	1218	A	Trace status	124	UINT	1	●		●	
1094	1222	A	Trace buffer status	124	UINT	1	●		●	●
1098	1226	A	Number of triggers executed	125	IN	1	●		●	●
1102	1230	A	Number of failed triggers	125	IN	1	●		●	●
1345		A	Trace trigger	125	USINT	1		●	●	

Table 52: Ramp function model

1) Some registers are only available in certain ACOPOSmicro models.

G ... These registers are only provided if the respective channel also has an encoder.

D ... These registers are only provided in models with additional digital inputs/outputs

A ... These registers are only provided in models with additional analog inputs/outputs

2) Key: DT ... Data type, L ... Length, R ... Read, W ... Write, C ... Cyclic, A ... Acyclic

3) The CANIOBusController function model is nearly identical to the ramp function model. However, the registers for the analog values are in a different range (values in parentheses).

4) Trace functionality is not provided for the CANIOBusController function model.

3.2 Register description

3.2.1 Set position/speed

This synchronous register is used to set position or speed, depending on the operating mode.

- Position mode (see the section 3.2.7 "Mode" on page 113):
Synchronous setting of the set position (signed long) in microsteps.
In ramp mode, one micro step is always 1/256 full-step.
- Speed mode (see the section 3.2.7 "Mode" on page 113):
In this mode, this register is viewed as signed set speed.

3.2.2 Current position (cyclic)

The synchronous register contains the current position (signed long) of the internal step counter.

In ACOPOSmicro models in which an encoder can also be connected for the motor, this register can be switched to the ABR counter (see bit 14 in the register Control word on page 111).

3.2.3 Current position (acyclic)

This register is only provided in ACOPOSmicro models in which an encoder can be connected for the motor.

The asynchronous register contains the current position (signed long) of the ABR counter. This register can be switched to the internal step counter (see bit 14 in the register Control word on page 111).

3.2.4 Input status

This register indicates the logical states of the digital inputs.

Bit	Description
0	Digital input 1 (connection X2, pin 1)
1	Digital input 2 (connection X2, pin 3)
2 - 7	Reserved (always 0)

These digital inputs will be used as trigger input depending on the ACOPOSmicro model.

Register description • Ramp function model

3.2.5 Control word

Depending on the module's state, this register can be used to issue commands (see the section 3.4 "Ramp function model operation" on page 128).

Bit	Description ¹⁾
0	Switch on
1	Enable voltage
2	Quick stop
3	Enable operation
4 - 6	Mode-specific
7	Fault reset
8 - 10	Mode-specific
11	0 ... No effect 1 ... Rising edge: Motor ID trigger ²⁾
12	0 ... No effect 1 ... Rising edge: Reset warnings
13	0 ... Disable undercurrent detection (default) 1 ... Enable undercurrent detection
14	G Contents of the "Current position cyclic/acyclic" register: 0 ... Register <i>Current position (cyclic)</i> : Internal position counter, cyclic Register <i>Current position (acyclic)</i> : ABR counter, acyclic 1 ... Register <i>Current position (cyclic)</i> : ABR counter, cyclic Register <i>Current position (acyclic)</i> : Internal position counter, acyclic Default setting: 0
15	0 ... Reserved

1) Some bits are only available in certain ACOPOSmicro models.

G ... This bit is only provided if the respective channel also has an encoder

2) A measurement of the motor ID can be made with this bit. Take note that the application ensures that the conditions for a measurement (see note in the section 3.2.28 "Motor ID" on page 120) are fulfilled.

3.2.6 Status

The bits in Status represent the status of the *State machine* (for a detailed description, see the sections 3.4.2 "Status register" on page 130 and 3.4.3 "State machine" on page 131).

Bit	Description
0	Ready to switch on
1	Switched on
2	Operation enabled
3	Fault (error bit)
4	Voltage enabled
5	Quick stop
6	Switch on disabled
7	Warning
8	0 ... Reserved
9	Remote (always 1 because there is no local mode for ACOPOSmicro)
10	Target reached Depending on the mode: Position reached, Homing done, Speed reached
11	Internal limit active (always 0, because this function does not exist on ACOPOSmicro; e.g.: Position range limit)
12 - 15	0 ... Reserved

Register description • Ramp function model

3.2.7 Mode

Value (dec.)	Description
1	Position mode: Approach target position as soon as target position is changed
2	Speed mode: Constant speed
-121	Remaining distance mode
-122	Set actual position
-123	Move to target position when external input is set
-124	Two position mode
-125	Move to set position 1 (position set asynchronous)
-126	Move to set position 2 (position set asynchronous)
-127	Positive homing (also see 3.2.18 "Homing configuration" on page 116)
-128	Negative homing (also see 3.2.18 "Homing configuration" on page 116)

Information:

Valid for all modes: The *Target Reached* bit is set in the register *Status* (see section 3.2.6 "Status" on page 112) when the current action is finished (depending on the mode position or speed reached).

A new position or speed can be specified even before the current action is finished.

Mode	Description of the individual modes
1	<u>Position mode</u> The new position is moved to as soon as the set position is set in the <i>Set position/speed</i> register that is not equal to the current position. This is done with a ramp function that accounts for the defined maximum speeds and accelerations. The set position can also be changed during an active positioning procedure. The set position is specified in micro steps (1/256 full-step).
2	<u>Speed mode - Constant speed (pos./neg.)</u> The value in the <i>Set position/speed</i> register is now interpreted as set speed (microsteps / 25 ms). Taking the maximum permissible acceleration into account, the motor moves with a ramp to the desired set speed and maintains this speed until a new set speed is specified. Values are allowed within the range from -65535 to +65535. When a value is entered outside of this range, it is adjusted to these limits.

Mode	Description of the individual modes
-121	<u>Remaining distance mode</u> (like mode 1) The number of steps defined in the "SetPosition 1" register are added to the current position and the resulting position is approached at a rising/falling edge on the corresponding digital input ¹⁾. Note: Not added to the target position, but rather to the actual position at the moment of the trigger. Negative values are also allowed for the offset defined in "SetPosition 1". New target positions are no longer accepted in the <i>Set position/speed</i> register after the trigger event. To do this, you must first switch to mode 0 and then back to mode -121. The <i>Target Reached</i> bit in the <i>Status</i> register is not set to 1 until the end position (after the trigger event) has been reached. The homing configuration determines whether the rising or falling edge is used as trigger on the digital input. The turn-around loop is not enabled in this mode (any configured values not equal to 0 are ignored).
-122	<u>Set actual position</u> The target position defined in the <i>Set position/speed</i> register is accepted as the current actual position in the internal step counter when the state machine is in the "Operation Enable" state. Before this mode is started, the motor must be at a standstill and be physically located at the point that is valid for the position to be set.
-123	<u>Move to target position when external input is set</u> The set position defined in the <i>Set position/speed</i> register is moved to when a rising edge occurs on the corresponding digital input ¹⁾. A new set position is not accepted until another rising edge occurs on the corresponding digital input. This can also occur during the active positioning procedure and will be applied immediately.
-124	<u>Two position mode</u> The positions "SetPosition 1" and "SetPosition 2" are defined in the asynchronous registers. When 1 is set on the corresponding digital input ¹⁾, the SetPosition 1 is moved to. When 0 is set, then SetPosition 2 is moved to. The changeover can also occur during an active positioning procedure.
-125 -126	<u>Move to set position X</u> The purpose of this is to enable a virtual switch from the speed mode to the position mode, which otherwise isn't possible because of the double use of the register for position – and speed specification. • Mode -125: Set position 1 • Mode -126: Set position 2
-127 -128	<u>Positive / negative homing</u> • The motor must be at a standstill before switching from another mode to one of the homing modes. • You must first specify in the homing configuration (see the section 3.2.18 "Homing configuration" on page 116) whether homing should occur at low/high level on the digital input, via the R-input or unconditionally. <u>Homing via R-input or digital input ¹⁾</u> Case 1: Active homing level not yet reached → Motor not yet at end position: Movement continues at homing speed in the homing direction until the active level for "homing-stop" is on the input. Case 2: Active homing level already reached → Motor at end position: Movement continues at homing speed against the homing direction until the active level for "homing-stop" is no longer the digital input. Movement continues at homing speed in the homing direction until the active level for "homing-stop" is on the input again. <u>Homing unconditional (immediate) without motor movement</u> If the homing condition occurs, then the motor stops and the values of the position counter and ABR counter ²⁾ valid at the moment when the homing condition occurs are entered in the "Homed zero position" register.

1) For motor 1: Digital input 1 (connection X2, pin 1)

For motor 2: Digital input 2 (connection X2, pin 3)

2) The ABR counter is only available in ACOPOSmicro models with encoders.

Register description • Ramp function model**3.2.8 Motor brake**

Bit	Description
0	Digital output state (motor brake)
1 - 7	0 ... Reserved

This register is used to control the digital output (motor brake).

On the 2-channel ACOPOSmicro model, bit 0 of both registers (Motor 1 and Motor 2) is linked with an OR function. This means that the digital output is activated if the bit is set on one or both motors.

3.2.9 UZK DC bus voltage

This register contains the current supply voltage of the power element.

Units: V

3.2.10 Temperature of output stage

This register contains the current module temperature of the power element.

Units: °C

3.2.11 Temperature

This register contains the current module temperature.

Units: °C

3.2.12 Maximum speed

Maximum speed for the absolute positioning modes (1, -123, -124, -125, -126).

Does not work for the speed and homing modes (2, -127, -128).

Units: Microsteps / 25 ms

3.2.13 Max. acceleration

Maximum acceleration (also applies for homing modes).

Units: Microsteps / 25 ms²

3.2.14 Maximum deceleration

Maximum deceleration (also applies for homing modes).

Units: Microsteps / 25 ms²

3.2.15 Set position 1

Position to move to in the modes -124 (when 1 is set on the digital input) and -125.

3.2.16 Set position 2

Position to move to in the modes -124 (when 0 is set on the digital input) and -126.

3.2.17 Homing speed

Speed for the homing modes (-127 and -128).

Units: Microsteps / 25 ms

3.2.18 Homing configuration

Value (dec.)	Description
-123	Homing at positive edge on the R input
-124	Homing at negative edge on the R input
-125	Homing at positive edge on the digital input ¹⁾
-126	Homing at negative edge on the digital input ¹⁾
-127	Reserved
-128	Immediate homing
Everything else	No effect

1) For motor 1: Digital input 1 (connection X2, pin 1)
For motor 2: Digital input 2 (connection X2, pin 3)

3.2.19 Full step threshold

This register configures the speed (unit: microsteps / 25 ms). When this defined speed has been reached, the drive will automatically change from micro-steps to full-step mode. This makes it possible to optimize the torque at higher speeds, while micro-step mode ensures optimum concentricity at lower speeds.

The value 65,535 causes full-step mode to be deactivated and the motor to always be run in micro-step mode.

Example

A set value of 3,200 micro-steps per 25 ms is equal to 128,000 micro-steps per second. In the ramp function model, one full-step consists of 256 micro-steps, whereby the set value is equal to 500 full-steps per second. On a motor with 200 steps per revolution, the following speed is equal to:

$$T^{-1} = \frac{500 \text{ Steps/Seconds}}{200 \text{ Steps/Revolution}} = 2.5 \frac{\text{Revolutions}}{\text{Seconds}} = 150 \text{ 1/min}$$

3.2.20 Jolt time limitation

If a value other than 0 is assigned to this register, then jolt limitation is implemented by averaging the values for the steps to be carried out (set speed) in each 25 ms cycle with a FIFO. The value of the register corresponds to the number of FIFO elements (0 to 16). If a value greater than 16 is entered, then it is internally limited to 16.

Changes made while a motor is running will be applied as soon as ...

- ... the motor has reached the set position (only positioning modes).
- ... the motor has stopped (all modes).

3.2.21 Read back control word

Read back the register Control word(see 3.2.5 "Control word" on page 111).

3.2.22 Read back mode

Read back the register Mode(see 3.2.7 "Mode" on page 113).

3.2.23 Homed zero position for cyclic counter

After a homing procedure, the homing point of the internal step counter can be read back here.

In ACOPOSmicro models in which an encoder can also be connected for the motor, this register can be switched to the ABR counter (see bit 14 in the register Control word, described in the section 3.2.5 "Control word" on page 111).

3.2.24 Homed zero position for acyclic counter

This register is only provided in ACOPOSmicro models in which an encoder can be connected for the motor.

After a homing procedure, the homing point for the acyclic position counter can be read back here (either the internal step counter or ABR counter depending on bit 14 of the *Control word* register, see section 3.2.5 "Control word" on page 111).

3.2.25 Turn-around loop

This parameter is only used in modes 1, -123, -124, -125, -126 (absolute positioning modes).

If the value for the turn-around loop is not 0, then ...

- ... the target position will be approached from the one direction.
- ... and coming from the other direction, the target position will initially be passed by the configured number of steps and then approached from the opposite direction.

This ensures that the target position is always approached from the same direction (to avoid mechanical backlash).

The sign of the defined value determines which direction the turn-around loop runs.

3.2.26 Error code

The cause can be read out here when an error or warning occurs:

Error code	Error type	Description
0x0000	-	No error
0xFF02	Error	Module not enabled (Enable input = 0)
0x3210	Error	Over-voltage on the DC bus
0x3100	Error	Module supply voltage
0x4200	Error	Over-temperature
0x2300	Warning	Overcurrent
0x3220	Error	Under-voltage on the DC bus
0xFF00	Warning	Undercurrent ¹⁾
0xFF10	Warning	Encoder supply ²⁾
0xFF11	Warning	Encoder wire break ²⁾

1) Undercurrent is only detected if bit 13 = 1 in the control word (undercurrent detection enabled).

2) Encoder error messages can only occur if the ACOPOSmicro model contains an encoder component.

Register description • Ramp function model

3.2.27 Holding current, rated current, maximum current

The holding current, rated current and maximum current registers are used for configuring the desired motor current.

- Units: Percent of the module rated current (100% is equal to 10 A)

It would make sense if:

- Holding current < rated current < maximum current

The motor rated current is entered in the rated current register according to the motor data sheet.

Register	Description
Holding current	In situations requiring less torque (e.g.: at a standstill) the device should be switched to holding current. This reduces heating of the motor.
Rated current	Current during "normal operation". Due to the lower relative inaccuracy at higher currents, a current setting of at least 15% of the module's rated current (10 A) is recommended for the rated current register.
Maximum current	Should be selected if a higher motor torque is required during acceleration phases (short-term).

When the current changes to a weaker current value (e. g. during transition from the acceleration phase to the constant speed mode), the respectively stronger current is maintained for an additional 100 ms. This is done with following priority regardless of the actual defined values: Maximum current before rated current before holding current.

3.2.28 Motor ID

This register is used for practical differentiation and identification of the connected motor types (for service purposes).

The value of this register equals the amount of time (in μs) needed to trigger a current jump of $\Delta I = 1 \text{ A}$ in a motor winding. This time depends on several factors:

- Amplitude of operating voltage (DC bus voltage)
- Inductance and resistance of the motor winding

Notes	
1.	To achieve reproducible results, the measurement must be made under the following defined conditions: a) At a standstill b) Motors must be in a half-step position (phase A with full current, phase B with no current). I.e. the internal step counter of the ACOPOSmicro (see the section 3.2.2 "Current position (cyclic)" on page 110) must appear as follows: + Full steps are divisible by 4 + Microsteps = 0
2.	The condition 1.b is present after a ACOPOSmicro reset/power-up. Immediately afterwards, when the holding current is applied to the motor for the first time (at a standstill), the duration for applying the current is measured. This is also a suitable time to read the motor identification register in the application.
3.	As operating range for determining the motor ID, a current setting of at least 15% of the module's rated current (10 A) is recommended for the rated current register (see section 3.2.27 "Holding current, rated current, maximum current" on page 119) due to the lower relative inaccuracy at higher currents.
4.	Value range of the motor ID register
Motor ID (Register 84, 148)	Meaning
0	No motor identifier available (after PowerUp as long as the measurement conditions are not met)
1 ... 32767	Valid value range of the <i>motor ID</i> register
65504 ... 65519	Ground fault: Measurement not possible
65531	Over-current: Measurement not possible
65532	Wire break: Measurement not possible
65533	Motor position incorrect: Measurement not possible
65534	Invalid value: Overflow
65535	Busy: Measurement in progress

Table 53: Notes about motor identification

Register description • Ramp function model

3.2.29 Register for ACOPOSmicro models with additional digital inputs/outputs

Status of digital inputs and outputs

Bit	Description		
0	Status of digital input 1	A	Counter input 1
1	Status of digital input 2	B	
2	Status of digital input 3	R	External counter frequency for counter input 1
3	Status of digital input 4		Reference for enable input
4	Status of digital output 1		
5	Status of digital output 2		
6	Overflow of counter 1 (only valid if gate or period measurement is set in the counter configuration with bit 3 - 6): 0 ... Counter value is within the counting range (0 to \$7FFF). The value is only valid if bit 0 of the control register counter is set. 1 ... Counter overflow This bit is reset with Bit 0 of the control register counter.		
7	LatchDone: State changes each time the counter state is successfully latched (reset value = 0)		

Set digital outputs / control register counter

Bit	Description
0	0 ... Disable detection of counter overflow and reset the overflow bit on counter 1 1 ... Enable detection of counter overflow
1	0 ... Reserved
2	Referencing (only valid in AB(R) mode, see Bit 8+9 in counter configuration) 0 ... Disable referencing 1 ... Enable referencing
3	0 ... Reset counter (to zero) 1 ... Start counter Note: Only set after counter configuration is complete!
4 - 5	0 ... Reserved
6	Set digital output 1
7	Set digital output 2

Counter

This register contains the present counter state. Different counters (event counter, ABR counter, etc) are available depending on the configuration

Counter latch

The counter will be applied to this register at the next latch event (see ABR mode in "Counter configuration" on page 122).

Counter configuration

Bit	Description
0 - 1	Counter frequency (only valid if set in gate or period measurement): 00 ... Internal counter frequency 4 MHz 01 ... Internal counter frequency 31.25 kHz 10 ... External counter frequency (digital input 3) 11 ... Reserved
2	0 ... Reserved
3 - 6	Counter mode: 0000 ...No counter 0001 ...Event counter, positive edge 0010 ...Event counter, negative edge 0011 ...Event counter, both edges 0100 ...Period measurement between two consecutive negative edges 0101 ...Period measurement between two consecutive positive edges 0110 ...Gate measurement of the LOW level 0111 ...Gate measurement of the HIGH level 1000 ...AB counter without homing 1001 ...ABR counter with homing Counter set to zero: at positive edge of R 1010 ...ABR counter with homing Counter set to zero: at positive edge of R 1011 ...ABR counter with immediate homing Counter set to zero: at positive edge of bit 2 of the counter control register 1100 ...AB counter without homing 1101 ...ABR counter with homing Counter state is applied to the counter latch register: at positive edge of R 1110 ...ABR counter with homing Counter state is applied to the counter latch register: at positive edge of R 1111 ...ABR counter with immediate homing Counter state is applied to the counter latch register: at positive edge of bit 2 of the counter control register
7	0 ... Reserved
8 - 9	Configuration of reference enable input (digital input 4) 00 ... The digital input 4 (reference enable input) does not affect the R input. The R input is always enabled. 01 ...Reserved 10 ... R input is enabled if reference enable input is LOW. 11 ... R input is enabled if reference enable input is HIGH.
10 - 15	0 ... Reserved

Register description • Ramp function model

3.2.30 Register for ACOPOSmicro models with additional analog inputs

Status of the analog inputs

Bit	Description	
0	Analog input 1	0 ... OK 1 ... Outside the valid range or open circuit
1	Analog input 2	
2 - 7	0 ... Reserved	

In addition to the status info, the analog value of the corresponding input will also be set if an error occurs (status bit = 1):

Error	Analog value (dec.)	Analog value (hex.)
Wire break	+32.767	0x7FFF
Input voltage > +10 V	+32.767	0x7FFF
Input voltage < -10 V	-32.767	0x8001
Invalid value	-32.768	0x8000

Analog input 1 / Analog input 2

These registers receive the analog value of the corresponding input.

Measurement range:

Input voltage	Register value
+10 V	+32.767
:	:
-10 V	-32.768

3.3 Trace functions for ACOPOSmicro models with analog inputs

3.3.1 Trace register

Trace control word

Bit 0 of the trace control word is used to turn on/off the trace functionality for the corresponding analog input.

Bit	Description
0	0 ... Trace function is disabled 1 ... Trace function is enabled
1 - 15	0 ... Reserved

A buffer overrun error can be cleared by resetting bit 0.

Trace status

Bit	Description
0	0 ... Trace function is disabled 1 ... Trace function is enabled
1	0 ... Reserved
2	0 ... Trace data recording not active 1 ... Trace data recording active
3	0 ... Trace data reading is not active 1 ... Trace data reading is active
4	0 ... Not ready for trigger 1 ... Ready for trigger
5	Trigger active 0 ... No trigger active, trace recording stopped 1 ... Trigger in process, trace data is being recorded
6	Trace OK 0 ... Buffer overrun or trace function disabled for analog input 1 ... No buffer overrun error
7	Trace error 0 ... No buffer overrun or trace function disabled for analog input 1 ... Buffer overrun error

Trace buffer status

This register contains the amount of buffer memory (in bytes) available for the trace recording. The buffer must be large enough for the configured number of sampling values to be recorded.

If "Infinite Trace" was selected, then the buffer must be read fast enough for analog values to be recorded at the set sampling rate.

Register description • Ramp function model**Number of triggers executed**

This register contains the number of successfully executed trigger events (trace recordings).

Number of failed triggers

This register contains the number of failed trigger events (trace recordings).

This register will be increased by one when a trigger event occurs and under the following conditions:

- Trace function is not enabled
- Trace buffer is full (two trigger events in short succession)
- In the event of a Pre-Trigger function, if the number of scans before the trigger event did not record the required (configured) number of scans.

Trace trigger

This register can be used to start recording separately for each analog input. This is only possible if the trigger function has been enabled for the corresponding analog input.

Bit	Description	
0	Trigger for analog input 1	The trigger event will be triggered at a positive edge (bit changes from 0 → 1).
1	Trigger for analog input 2	
2 - 7	0 ... Reserved	

3.3.2 Trace runtime configuration

The following points describe how the trace runtime configuration is performed in Automation Studio.

Turn on/off trace for analog input X (Trace AnalogInputXX = on/off):

The trace function must first be turned on/off for the respective analog channel.

If the trace function was turned on for both analog inputs, then the recorded trace data will be entered with the AsIoTrc library in the following order:

Sample	Array index	Channel
1	0	Analog input 1
	1	Analog input 2
2	2	Analog input 1
	3	Analog input 2
:	:	:

Table 54: Order of trace data from two analog channels

Number of trace buffer:

The number of the local trace buffer must also be set on the CPU.

This number specifies how many trace recordings the CPU can save. If a trigger event occurs, even though the trace buffer is full, then the respective sampling values will be lost.

The recorded trace data will be read out with the AsIoTrc library.

Number of samples:

The number samples to be recorded on the ACOPOSmicro module must then be set.

Make sure that the trace buffer is made large enough for the trace data occurring on the module to be recorded.

If "Infinite Trace" was selected, then the buffer must be read fast enough for analog values to be recorded at the set sampling rate.

Block sizes for asynchronous data transfer (AsynSize):

A higher value will result in faster data transfer from the module to the CPU.

Number of data blocks sent before repetition (AsynForward):

Every data block transferred must be confirmed by the X2X Link master. If a confirmation has not been received after the number of data blocks defined here, then all non-confirmed data blocks will be sent again.

Register description • Ramp function model

Trace mode:

Trace mode	Description
Triggered	Trace recording will start after the trigger event (see bit 0/1 of the <i>Trace trigger</i> register on page 125).
Infinite	Infinite trace can be started with bit 0/1 of the <i>Trace trigger</i> register (see page 125). X2X must be configured in such a way as to ensure that the amount of data generated can be transferred quickly enough. The trace buffer must be read quickly to prevent an overflow in the module's local memory and to ensure that the samples can be recorded at the configured sampling rate.

Recording range of trace data (Trace Start/Stop Position):

This defines the number of samples that should be recorded before/after the trigger event.

Trigger type	Start position	Stop position	Beginning of recording	End of recording
Post trigger	$x \geq 0$	$y > 0$	x samples after trigger event	y samples after start position
Pre-trigger	$x < 0$	$y > 0$	x samples before trigger event	y samples after trigger event

3.3.3 Trace library

The AsIoTrc library must be used to read the recorded trace data (see the Automation Studio help for more information).

The ACOPOSmicro module with additional analog inputs has two separate trace instances that can be triggered at different times, independent of one another.

Each trace instance has a device name:

- <Hardware address of the module>.<Trace instance>

This device name is used to read the trace data from the desired channel using the AsIoTrcGet library function.

Examples:

- Trace for analog input 1: "SS1.IF1.ST1.TRC1"
- Trace for analog input 2: "SS1.IF1.ST1.TRC2"

3.4 Ramp function model operation

Control was based on the CANopen communication profile DS402.

Commands are written to the control word for controlling the modules. The current module state is returned to the Status register (see section 3.2.6 "Status" on page 112). The function mode (absolute position, constant speed, homing, etc.) is set in the Mode register (see the section 3.2.7 "Mode" on page 113).

3.4.1 Control word

Control word bits and their state for the state machine commands:

Command	Reserved	ABR counter sync/async	Undercurrent detection	Warning reset	Motor ID trigger	Mode-specific (reserved)	Mode-specific (reserved)	Mode-specific (reserved)	Fault reset	Mode-specific (reserved)	Mode-specific (reserved)	Mode-specific (reserved)	Enable operation	Quick stop	Enable voltage	Switch on
Bit ¹⁾	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Shutdown	x	x	x	x	x	x	x	x	0	x	x	x	x	1	1	0
Switch on	x	x	x	x	x	x	x	x	0	x	x	x	0	1	1	1
Disable voltage	x	x	x	x	x	x	x	x	0	x	x	x	x	x	0	x
Quick stop	x	x	x	x	x	x	x	x	0	x	x	x	x	0	1	x
Disable operation	x	x	x	x	x	x	x	x	0	x	x	x	0	1	1	1
Enable operation	x	x	x	x	x	x	x	x	0	x	x	x	1	1	1	1
Fault reset	x	x	x	x	x	x	x	x	↑	x	x	x	x	x	x	x
ACOPOSmicro model ²⁾		G														

1) x ... Any; ↑ ... Rising edge

2) Some commands are only available in certain ACOPOSmicro models:

G ... This command is only available if the respective channel also has an encoder.

Information about the control word:

Bits 0, 1, 2, 3 and 7 (light gray in the previous table)	This bits control the status of the State Machine (see the section 3.4.3 "State machine" on page 131) according to the command in the table above.
Bits 4 - 6 and 8 - 10	Reserved, should only be written with 0.
Motor ID trigger	Rising edge activates the motor ID measurement.
Warning reset	Rising edge resets warnings (no effects on errors, which are reset using Fault Reset; the state machine is not effected by this bit)
Fault reset	Rising edge resets errors and warnings (see the section 3.4.3 "State machine" on page 131)
Undercurrent detection	0 ... Undercurrent detection disabled; active undercurrent warnings will be reset 1 ... Undercurrent detection enabled/disabled
ABR counter sync/async. ¹⁾	0 ... Internal position counter of the ramp generator on cyclic registers 0/8 Value of the ABR counter on acyclic registers 90/154. 1 ... Internal position counter of the ramp generator on acyclic registers 90/154 Value of the ABR counter on cyclic registers 0/8.

1) This command is only available if the respective channel also has an encoder.

3.4.2 Status register

The individual bits of the Status register (see section 3.2.6 "Status" on page 112) and its states depend on the currently enabled status of the state machine:

Status	Reserved	Reserved	Reserved	Reserved	Int. limit active	Target reached	Remote	Reserved	Warning	Switch on disabled	Quick stop	Voltage enabled	Fault	Operation enabled	Switched on	Ready to switch on
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Not ready to switch on	x	x	x	0	0	x	1	0	x	0	x	0	0	0	0	0
Switch on disabled	x	x	x	0	0	x	1	0	x	1	x	0	0	0	0	0
Ready to switch on	x	x	x	0	0	x	1	0	x	0	1	0	0	0	0	1
Switched on	x	x	x	0	0	x	1	0	x	0	1	1	0	0	1	1
Operation enable	x	x	x	0	0	x	1	0	x	0	1	1	0	1	1	1
Quick stop active	x	x	x	0	0	x	1	0	x	0	0	1	0	1	1	1
Fault Reaction active	x	x	x	0	0	x	1	0	x	0	x	0	1	1	1	1
Fault	x	x	x	0	0	x	1	0	x	0	x	0	1	0	0	0

Information about status register:

Bits 0, 1, 2, 3, 5 and 6 (light gray in the previous table)	These bits are set according to the currently enabled status of the state machine (see the section 3.4.3 "State machine" on page 131).
Voltage enabled	Becomes 1 as soon as the motor is supplied with current.
Warning	Becomes 1 if a warning is detected ("Over-current", "Under-current"). The type of warning is listed in the error code register (see the table in the section 3.2.26 "Error code" on page 118). The highest priority error / warning is shown. The priority corresponds to the order in the mentioned table. Warnings can be reset with a rising edge on the Warning Reset bit in the control word.
Remote	Always 1 because there is no local mode on ACOPOSmicro.
Target reached	<u>In modes 1, -123, -124, -125, -126 (absolute positioning):</u> Becomes 0 when positioning is started Becomes 1 when the target is reached <u>In mode 2 (constant speed):</u> Becomes 0 when the motor accelerates/decelerates Becomes 1 when the set speed is reached <u>In modes -127, -128 (homing):</u> Becomes 0 when homing is started Becomes 1 when homing is terminated <u>In mode -122 (Set actual position):</u> Changes briefly to 0 and immediately 1 again when the position is set.
Internal limit active	Always 0 because no position range limit exists on the ACOPOSmicro.

3.4.3 State machine

The motor is controlled according to the state machine illustrated below. After the module is started, the state machine automatically changes to the state, "*Not ready to switch on*". The application then operates the state machine by writing commands in the control word (see the section 3.4.1 "*Control word*" on page 128).

The state machine achieves the successive states "*Ready to switch on*", "*Switched on*" and "*Operation Enable*" by writing the consecutive commands "*Shutdown*", "*Switch on*" and "*Enable Operation*".

Information:

Motor movements are not performed until the "*Operation Enable*" state, according to the setting in the Mode register (see section 3.2.7 "*Mode*" on page 113).

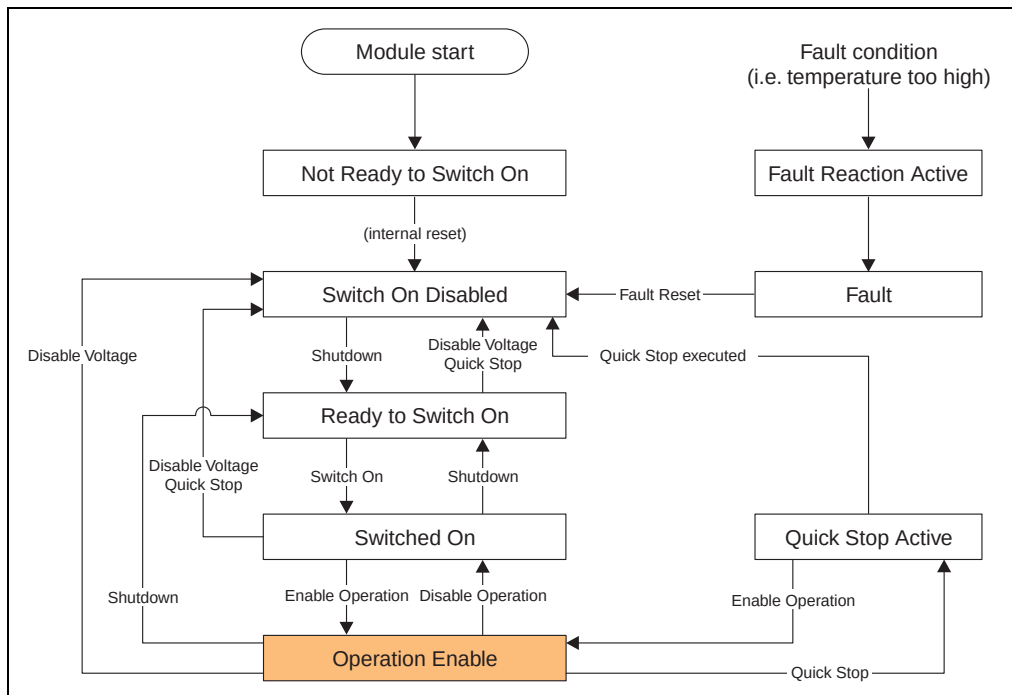


Figure 28: State machine - Flow chart

Register description • Ramp function model

State change	Description
Not ready to switch on → Switch on disabled	This state change occurs automatically after the module start and the internal initialization.
Switch on disabled → Ready to switch on	This state change is brought on by the "Shutdown" command. No others actions are performed.
Ready to Switch On → Switch on disabled	This state change is brought on by the command "Disable Voltage" or "Quick Stop". No others actions are performed.
Switched on → Switch on disabled	This state change is brought on by the command "Disable Voltage" or "Quick Stop". Motor voltage switched off immediately.
Ready to Switch On → Switched on	This state change is brought on by the "Switch on" command. Motor voltage switched on. When this state change occurs for the first time since the module start, the motor ID measurement is performed before the "Switched on" state is achieved. This can take approximately 1 second.
Switched on → Ready to switch on	This state change is brought on by the "Shutdown" command. Motor voltage switched off immediately.
Switched on → Operation enable	This state change is brought on by the "Enable operation" command. Motor movements are now performed depending on the defined mode.
Operation enable → Switched on	This state change is brought on by the "Disable operation" command. If in motion, the motor is decelerated with the configured deceleration. Motor voltage remains on in the "Switched on" state.
Operation enable → Ready to switch on	This state change is brought on by the "Shutdown" command. Motor voltage switched off immediately.
Operation enable → Switch on disabled	This state change is brought on by the "Disable voltage" command. Motor voltage switched off. It is strongly recommended to only make this state change on a stopped motor, because regeneration on a motor running idle can cause an over-voltage error on the DC bus (0x3210).
Operation enable → Quick stop active	This state change is brought on by the Quick stop command. If in motion, the motor is decelerated with the configured deceleration. During the deceleration, the state machine remains in the Quick Stop Active state. Once the motor is at standstill, the state automatically changes to the Switch on disabled state. While the state machine is in the Quick Stop Active state, then back to the Operation Enable state with the Enable operation command.
→ Fault reaction active	This state change is brought on when an error occurs and cannot be triggered with a command by the user. It can be triggered by error types listed as "Error" (see section 3.2.26 "Error code" on page 118). (Other error types listed as "Warning" only cause the Warning Bit to be set in the status word and do not cause a state change in the state machine.) Motor voltage is switched off and the state machine then changes immediately to the Fault state. The type of error is listed in the error code register (see the table in the section 3.2.26 "Error code" on page 118). The highest priority error is shown. The priority corresponds to the order in the error code table.
Fault → Switch on disabled	This state change is brought on by the Fault Reset command. However, the state only changes if no more errors are present when the command is written. All errors and warnings are reset. 0 is set again in the error code register, otherwise a warning code remains if a warning is still present.

Table 55: State machine - State change

Chapter 8 • Standards and certifications

1. Valid European guidelines

- EMC guidelines 89/336/EWG
- Low-voltage guidelines 73/23/EWG
- Machine directive 98/37/EG

2. Valid standards

Standard	Description
IEC/EN 61800-2 : 1998	Adjustable speed electrical power drive systems • Part 2: General requirements; Rating specifications for low voltage adjustable frequency AC power drive systems
IEC/EN 61800-3 : 2004	Adjustable speed electrical power drive systems • Part 3: EMC requirements including specific test methods
IEC 61800-5-1	Electrical drive systems with adjustable speed • Part 5-1: Safety requirements - Electrical, thermal and power requirements (IEC 61800-5-1:2003)
IEC/EN 61131-2 : 2003	Programmable logic controllers • Part 2: Equipment requirements and tests
UL 508 C : 2002	Industrial control equipment • Part 6: Solid-state motor controllers

Table 56: Valid standards for ACOPOSmicro stepper motor module

The limit values specified from section 3 "Environmental limits" to section 6 "Other environmental limit values according to IEC 61800-2" are taken from product standard EN 61800 (or IEC 61800) for stepper motor modules in industrial environments (2nd environment). Stricter test procedures and limit values are used during the type tests for ACOPOSmicro stepper motor modules. Additional information is available from B&R.

3. Environmental limits

3.1 Mechanical conditions according to EN 61800-2

3.1.1 Operation

IEC 60721-3-3, class 3M1	
	IEC 61800-2
Vibration during operation $2 \leq f < 9$ Hz $9 \leq f < 200$ Hz	0.3 mm amplitude 1 m/s ² acceleration

Table 57: Mechanical conditions during operation

3.1.2 Transport

IEC 60721-3-2, class 2M1	
	IEC 61800-2
Vibration during transport $2 \leq f < 9$ Hz $9 \leq f < 200$ Hz $200 \leq f < 500$ Hz	3.5 mm amplitude 10 m/s ² acceleration 15 m/s ² acceleration

Table 58: Mechanical conditions during transport

3.2 Climate conditions according to IEC 61800-2

3.2.1 Operation

IEC 60721-3-3, class 3K3	
	IEC 61800-2
Ambient temperature during operation	5 to 45°C
Relative humidity during operation	5 - 85%, non-condensing

Table 59: Climate conditions during operation

3.2.2 Bearings

IEC 60721-3-1, class 1K4	
	IEC 61800-2
Storage temperature	-25 to +55°C

Table 60: Climate conditions (temperature) during storage

IEC 60721-3-1, class 1K3	
	IEC 61800-2
Relative humidity during storage	5 - 95%, non-condensing

Table 61: Climate conditions (humidity) during storage

3.2.3 Transport

IEC 60721-3-2, class 2K3	
	IEC 61800-2
Transport temperature	-25 to +70°C
Relative humidity during transport	95% at +40°C

Table 62: Climate conditions during transport

4. Requirements for immunity to disturbances (EMC)

4.1 Evaluation criteria (performance criteria)

Criteria A Test object not influenced during test.

Criteria B Test object only temporarily influenced during test.

Criteria C The system does not reboot automatically (reset required).

4.2 High frequency disturbances according to EN 61800-3

These immunity tests are valid for industrial environments (2nd environment).

4.2.1 Electrostatic discharge

Tests according to EN 61000-4-2		
	IEC 61800-3	Performance criteria
Contact discharge to powder-coated and bare metal housing parts	4 kV	B
Discharge through the air to plastic housing parts	8 kV	

Table 63: Limits for electrical discharge

4.2.2 Electromagnetic fields

Tests according to EN 61000-4-3		
	IEC 61800-3	Performance criteria
Housing, completely wired	80 MHz - 1 GHz, 10 V/m, 80% amplitude modulation at 1 kHz	A

Table 64: Limits for electromagnetic fields

4.2.3 Burst

Tests according to EN 61000-4-4		
	IEC 61800-3	Performance criteria
Power connection	2 kV, 1 min, direct coupling	B
Lines for measurement and control functions in the process environment	2 kV, 1 min	
Signal interfaces, other lines	1 kV, 1 min	

Table 65: Limits for burst

4.2.4 Surge

Tests according to EN 61000-4-5		
	IEC 61800-3	Performance criteria
Power connection	1 kV (2Ω) ¹⁾ , DM, symmetrical 2 kV (12Ω) ¹⁾ , CM, unsymmetrical	B

Table 66: Limits for surge

1) The impedance was added from EN 61000-4-5 because it is not defined in EN 61800-3.

4.2.5 High frequency conducted disturbances

Tests according to EN 61000-4-6		
	IEC 61800-3	Performance criteria
Power connection	0.15 - 80 MHz, 10 V, 80% amplitude modulation at 1 kHz	A
Lines for measurement and control functions in the process environment		
Signal interfaces, other lines		

Table 67: Limits for conducted disturbances (radio frequency)

5. Requirements for emissions (EMC)

5.1 High frequency emissions according to EN 61800-3

These emissions tests are valid for industrial environments (2nd environment).

5.1.1 Conducted emissions on the connections of the upstream power element for power supply

Tests according to EN 55011			
Continuous current on motor	Frequency range [MHz]	Quasi-peak value	Average
I ≤ 100 A	$0.15 \leq f < 0.5$	100 dB (μV)	90 dB (μV)
	$0.5 \leq f < 5$	86 dB (μV)	76 dB (μV)
	$5 \leq f < 30$	90 dB (μV)	80 dB (μV)

Table 68: Limits for conducted emissions on the power connections of the upstream power element for power supply

5.1.2 Electromagnetic emissions

Tests according to EN 55011	
Frequency range [MHz]	Quasi-peak value
$30 \leq f \leq 230$	40 dB (μV/m), measured at distance of 30 m ¹⁾
$230 < f \leq 1000$	50 dB (μV/m), measured at distance of 30 m ¹⁾

Table 69: Limits for electromagnetic emissions

1) The limit values were increased by 10 dB (μV/m) when measuring from distances of 10 m.

6. Other environmental limit values according to IEC 61800-2

IEC 61800-2	
Degree of pollution according to IEC 61800-2, 4.1.2.1.	2 (non-conductive material)
Overvoltage cat. according to IEC 60364-4-443:1999	III
Protection according to IEC 60529	IP20
Reduction of the continuous current at installation altitudes over 500 m above sea level	10% per 1000 m
Maximum installation altitude	2000 m ¹⁾

Table 70: Additional environmental limits

1) Additional requirements are to be arranged with B&R.

7. International certifications

B&R products and services comply with applicable standards. They are international standards from organizations such as ISO, IEC and CENELEC, as well as national standards from organizations such as UL, CSA, FCC, VDE, ÖVE, etc. We give special consideration to the reliability of our products in an industrial environment.

Certifications	
USA and Canada 	All important B&R products are tested and listed by Underwriters Laboratories and checked quarterly by a UL inspector. This mark is valid for the USA and Canada and simplifies certification of your machines and systems in these areas.
Europe 	All harmonized EN standards for the valid guidelines are met.
Russian Federation 	GOST-R certification is available for the export of all ACOPOSmicro stepper motor modules in the Russian Federation.

Table 71: International certifications

Appendix A • Abbreviations

1. General information

Abbreviations appear throughout the User's Manual, for example in data tables or descriptions of pin assignments.

2. Overview

Abbreviation	Stands for	Description
NC	Normally closed	A relay contact. Also known as a normally closed (N.C.) contact.
	Not connected	Used in the description of pin assignments if a terminal or pin is not connected to a module.
ND	Not defined	In data tables, this stands for a value that has not been defined. Because a cable manufacturer does not provide certain technical data, for example.
NO	Normally open	A relay contact. Also known as a normally open (N.O.) contact.
TBD	To be defined	Used in the technical data tables when certain pieces of information are not yet available. The value will be provided later.

Table 72: Abbreviations used in the User's Manual

Appendix B • Variable assignment in Automation Studio X2X Master

1. Overview

This appendix provides an overview of the variable assignments in Automation Studio X2X Master for all ACOPOSmicro function models:

ACOPOSmicro model	Standard function model		Ramp function model	
	Documentation	Overview Variable assignments	Documentation	Overview Variable assignments
80SD100XD.C044-01	86	144	108	149
80SD100XD.C04X-13		151		157
80SD100XD.C0XX-01		159		163
80SD100XD.C0XX-21		165		171
80SD100XS.C04X-01		175		179
80SD100XS.C0XX-01		181		184

All registers are covered in detail in Chapter 7 "Register description". This description provides a summary of the functionality for all ACOPOSmicro models. The differences are noted in the respective register descriptions.

This Appendix B "Variable assignment in Automation Studio X2X Master" also provides the following information:

- Specific overview of the individual registers for each ACOPOSmicro model.
- Variable assignment in Automation Studio X2X Master for all registers and data points (individual bits within a few registers).
The descriptions for registers, whose bits have no corresponding variable assignment, were not covered again in this Appendix .

2. 80SD100XD.C044-01 Standard function model

2.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	2	Position1Sync	Position2Sync	Position
10		Error		Error/Status
12	14	Motoridentification01	Motoridentification02	Motor ID
16	24	Motor1Step0	Motor2Step0	Motor configuration MotorStep1
18	26	Motor1Step1	Motor2Step1	Motor configuration MotorStep2
20	28	Motor1Step2	Motor2Step2	Motor configuration MotorStep3
22	30	Motor1Step3	Motor2Step3	Motor configuration MotorStep4
33	36	ConfigOutput03	ConfigOutput06	Holding current
34	37	ConfigOutput04	ConfigOutput07	Rated current
35	38	ConfigOutput05	ConfigOutput08	Maximum current
46		ConfigOutput02		Module configuration
54		StatusOutput14		Error acknowledgement / Digital output motor brake
60	62	Position1Latched	Position2Latched	Position latch
68	70	usSinceTrigger01	usSinceTrigger02	usSinceTrigger
72		StepperLatchStatus		Latch trigger status
73		LifeCnt		Life cycle counter
76		StepperLatchConfig		Latch configuration
78		TriggerConfig		Trigger configuration
80		Temperature		Temperature
81		MotorIdentTrigger		Motor ID trigger
84	86	ConfigOutput09	ConfigOutput10	Full step threshold
88		CoolerTemperature		Cooler temperature
90		UZKVoltage		DC bus voltage
Register for ACOPOSmicro with encoder				
130	144	PosVal03	PosVal04	ABR counter
134	148	LatchVal03	LatchVal04	ABR counter latch
154		StatusOutput15		ABR control register
155		ConfigOutput11		ABR configuration
158		ConfigOutput13		ABR advanced configuration

Table 73: Standard function model

2.2 Error: Error/Status

Bit	Variable assignments in Automation Studio	Description	
0		0 ... Reserved	
1	OvertemperatureError01	E 0 ... No error 1 ... Error: Overtemperature	Motor 1
2	CurrentError01	E 0 ... No error 1 ... Error: Set current not reached	
3	OvercurrentError01	E 0 ... No error 1 ... Error: Overcurrent	
4		0 ... Reserved	
5	OvertemperatureError02	E 0 ... No error 1 ... Error: Overtemperature	Motor 2
6	CurrentError02	E 0 ... No error 1 ... Error: Set current not reached	
7	OvercurrentError02	E 0 ... No error 1 ... Error: Overcurrent	
8	DrvOk01	S Status of drive 1: 0 ... Not ready 1 ... Ready	Motor 1
9	DrvOk02	S Status of drive 2: 0 ... Not ready 1 ... Ready	Motor 2
10	IncrementalEncoderOk01	Status of ABR encoder 1: 0 ... Not ready 1 ... Ready	Encoder
11	EncoderPowerSupplyError	Encoder supply error 0 ... 24 VDC OK 1 ... Error: Encoder supply not in the required range	
12	IncrementalEncoderOk02	Status of ABR encoder 2: 0 ... Not ready 1 ... Ready	
13	ModulePowerSupplyError	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage	
14	Enable	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK	
15	UZKStatus	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)	

2.3 StatusOutput14: Error acknowledgement / Digital output motor brake

Bit	Variable assignments in Automation Studio	Description
0	ClearError01	Error acknowledgement - Motor 1
1	ClearError02	Error acknowledgement - Motor 2
2 - 6		0 ... Reserved
7	DigitalOutput	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

2.4 StepperLatchStatus: Latch trigger status

Bit	Variable assignments in Automation Studio	Description	
0	LatchInput01	Digital input for the latch event (level)	Stepper motor 1
1	LatchDone01	State changes each time the counter state is successfully latched (reset value = 0)	
2	LatchInput02	Digital input for the latch event (level)	Stepper motor 2
3	LatchDone02	State changes each time the counter state is successfully latched (reset value = 0)	
4	DigitalInput01 TriggerInput01	Status of digital input 1 Status of trigger input 2	
5	DigitalInput02 TriggerInput02	Status of digital input 2 Status of trigger input 2	
6	LatchDone03	State changes each time ABR counter 1 is successfully latched (reset value = 0) See ABR counter control register and ABR counter configuration register	ABR counter 1
7	LatchDone04	State changes each time ABR counter 2 is successfully latched (reset value = 0) See ABR counter control register and ABR counter configuration register	ABR counter 2

2.5 StepperLatchConfig: Latch configuration

Bit	Variable assignments in Automation Studio	Description	
0	StartLatch01	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 1
1 2	TriggerEdgePos01 TriggerEdgeNeg01	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 1 10 ... Latch position at negative edge on trigger input 1 11 ...Reserved	
3		0 ... Reserved	
4	StartLatch02	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 2
5 6	TriggerEdgePos02 TriggerEdgeNeg02	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 2 10 ... Latch position at negative edge on trigger input 2 11 ...Reserved	
7		0 ... Reserved	

2.6 TriggerConfig: Trigger configuration

Bit	Variable assignments in Automation Studio	Description
0	TriggerEdge01	0 ... Trigger 1: Trigger edge = positive 1 ... Trigger 1: Trigger edge = negative
1	StartTrigger01	Activate trigger 1 (when changes occur)
2	TriggerEdge02	0 ... Trigger 2: Trigger edge = positive 1 ... Trigger 2: Trigger edge = negative
3	StartTrigger02	Activate trigger 2 (when changes occur)
4 - 7		0 ... Reserved

2.7 StatusOutput15: ABR control register

Bit	Variable assignments in Automation Studio	Description
0	ActivateReferencing03	0 ... Deactivate referencing of ABR counter 1 1 ... Activate referencing of ABR counter 1 (positive edge activates referencing)
1 - 3		0 ... Reserved
4	ActivateReferencing04	0 ... Deactivate referencing of ABR counter 2 1 ... Activate referencing of ABR counter 2 (positive edge activates referencing)
5 - 7		0 ... Reserved

3. 80SD100XD.C044-01 Ramp function model

3.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	8	AbsPos01	AbsPos02	Set position/speed
4	12	MpGenControl01	MpGenControl02	Control word
6	14	MpGenMode01	MpGenMode02	Mode
7	15	MpGenBrake01	MpGenBrake02	Motor brake
0	8	AbsPos01ActVal	AbsPos02ActVal	Current position (cyclic)
4	12	MpGenStatus01	MpGenStatus02	Status
6		InputStatus		Input status
42		UZKVoltage		UZK DC bus voltage
44		CoolerTemperature		Temperature of output stage
46		Temperature		Temperature
48	112	ConfigOutput03a	ConfigOutput06a	Holding current
49	113	ConfigOutput04a	ConfigOutput07a	Rated current
50	114	ConfigOutput05a	ConfigOutput08a	Maximum current
52	116	MaxSpeed01pos	MaxSpeed02pos	Max. speed
54	118	MaxAcc01	MaxAcc02	Max. acceleration
56	120	MaxDec01	MaxDec02	Maximum deceleration
58	122	RevLoop01	RevLoop02	Turn-around loop
60	124	FixedPos01a	FixedPos02a	Set position 1
64	128	FixedPos01b	FixedPos02b	Set position 2
68	132	RefSpeed01	RefSpeed02	Homing speed
70	134	RefConfig01	RefConfig02	Homing configuration
72	136	FullStepThreshold01	FullStepThreshold02	Full step threshold
75	139	JoltTime01	JoltTime02	Jolt time limitation
80	144	ControlReadback01	ControlReadback02	Read back control word
82	146	ModeReadback01	ModeReadback02	Read back mode
84	148	Motoridentification01	Motoridentification02	Motor ID
86	150	RefPos01CyclicCounter	RefPos02CyclicCounter	Homed zero position for cyclic counter
90	154	AbsPos1ActValAcylic	AbsPos2ActValAcylic	G Current position (acyclic)
94	158	RefPos01AcylicCounter	RefPos02AcylicCounter	G Homed zero position for acyclic counter
98	162	ErrorCode01	ErrorCode02	Error code

Table 74: Ramp function model

3.2 Motor brake

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	MpGenBrake01	MpGenBrake02	Digital output state (motor brake)
1 - 7			0 ... Reserved

The motor brake register is used to control the digital output (motor brake).

On the 2-channel ACOPOSmicro model, bit 0 of both registers (Motor 1 and Motor 2) is linked with an OR function. This means that the digital output is activated if the bit is set on one or both motors.

4. 80SD100XD.C04X-13 Standard function model

4.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	2	Position1Sync	Position2Sync	Position
10		Error		Error/Status
12	14	Motoridentification01	Motoridentification02	Motor ID
16	24	Motor1Step0	Motor2Step0	Motor configuration MotorStep1
18	26	Motor1Step1	Motor2Step1	Motor configuration MotorStep2
20	28	Motor1Step2	Motor2Step2	Motor configuration MotorStep3
22	30	Motor1Step3	Motor2Step3	Motor configuration MotorStep4
33	36	ConfigOutput03	ConfigOutput06	Holding current
34	37	ConfigOutput04	ConfigOutput07	Rated current
35	38	ConfigOutput05	ConfigOutput08	Maximum current
46		ConfigOutput02		Module configuration
54		StatusOutput14		Error acknowledgement / Digital output motor brake
60	62	Position1Latched	Position2Latched	Position latch
68	70	usSinceTrigger01	usSinceTrigger02	usSinceTrigger
72		StepperLatchStatus		Latch trigger status
73		LifeCnt		Life cycle counter
76		StepperLatchConfig		Latch configuration
78		TriggerConfig		Trigger configuration
80		Temperature		Temperature
81		MotorIdentTrigger		Motor ID trigger
84	86	ConfigOutput09	ConfigOutput10	Full step threshold
88		CoolerTemperature		Cooler temperature
90		UZKVoltage		DC bus voltage

Table 75: Standard function model

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
Register for ACOPOSmicro with encoder				
130	-	PosVal03	-	ABR counter
134	-	LatchVal03	-	ABR counter latch
154		StatusOutput15		ABR control register
155		ConfigOutput11		ABR configuration
158		ConfigOutput13		ABR advanced configuration
Register for ACOPOSmicro models with additional digital inputs/outputs				
192		Digital_in_do_rb		Status of digital inputs and outputs
193		StatusOutput16		Set digital outputs / control register counter
194		Counter01		Counter
196		CounterLatched01		Counter latch
198		ConfigOutput12		Counter configuration

Table 75: Standard function model (cont.)

4.2 Error: Error/Status

Bit	Variable assignments in Automation Studio	Description	
0		0 ... Reserved	
1	OvertemperatureError01	E 0 ... No error 1 ... Error: Overtemperature	Motor 1
2	CurrentError01	E 0 ... No error 1 ... Error: Set current not reached	
3	OvercurrentError01	E 0 ... No error 1 ... Error: Overcurrent	
4		0 ... Reserved	
5	OvertemperatureError02	E 0 ... No error 1 ... Error: Overtemperature	Motor 2
6	CurrentError02	E 0 ... No error 1 ... Error: Set current not reached	
7	OvercurrentError02	E 0 ... No error 1 ... Error: Overcurrent	
8	DrvOk01	S Status of drive 1: 0 ... Not ready 1 ... Ready	Motor 1
9	DrvOk02	S Status of drive 2: 0 ... Not ready 1 ... Ready	Motor 2
10	IncrementalEncoderOk	Status of ABR encoder: 0 ... Not ready 1 ... Ready	Encoder
11	EncoderPowerSupplyError	E Encoder supply error 0 ... 24 VDC OK 1 ... Error: Encoder supply not in the required range	
12		0 ... Reserved	
13	ModulePowerSupplyError	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage	
14	Enable	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK	
15	UZKStatus	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)	

4.3 StatusOutput14: Error acknowledgement / Digital output motor brake

Bit	Variable assignments in Automation Studio	Description
0	ClearError01	Error acknowledgement - Motor 1
1	ClearError02	Error acknowledgement - Motor 2
2 - 6		0 ... Reserved
7	DigitalOutput	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

4.4 StepperLatchStatus: Latch trigger status

Bit	Variable assignments in Automation Studio	Description	
0	LatchInput01	Digital input for the latch event (level)	Stepper motor 1
1	LatchDone01	State changes each time the counter state is successfully latched (reset value = 0)	
2	LatchInput02	Digital input for the latch event (level)	Stepper motor 2
3	LatchDone02	State changes each time the counter state is successfully latched (reset value = 0)	
4	DigitalInput01 TriggerInput01	Status of digital input 1 Status of trigger input 2	
5	DigitalInput02 TriggerInput02	Status of digital input 2 Status of trigger input 2	
6	LatchDone03	State changes each time ABR counter 1 is successfully latched (reset value = 0) See ABR counter control register and ABR counter configuration register	ABR counter 1
7		0 ... Reserved	

4.5 StepperLatchConfig: Latch configuration

Bit	Variable assignments in Automation Studio	Description	
0	StartLatch01	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 1
1 2	TriggerEdgePos01 TriggerEdgeNeg01	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 1 10 ... Latch position at negative edge on trigger input 1 11 ...Reserved	
3		0 ... Reserved	
4	StartLatch02	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 2
5 6	TriggerEdgePos02 TriggerEdgeNeg02	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 2 10 ... Latch position at negative edge on trigger input 2 11 ...Reserved	
7		0 ... Reserved	

4.6 TriggerConfig: Trigger configuration

Bit	Variable assignments in Automation Studio	Description
0	TriggerEdge01	0 ... Trigger 1: Trigger edge = positive 1 ... Trigger 1: Trigger edge = negative
1	StartTrigger01	Activate trigger 1 (when changes occur)
2	TriggerEdge02	0 ... Trigger 2: Trigger edge = positive 1 ... Trigger 2: Trigger edge = negative
3	StartTrigger02	Activate trigger 2 (when changes occur)
4 - 7		0 ... Reserved

4.7 StatusOutput15: ABR control register

Bit	Variable assignments in Automation Studio	Description
0	ActivateReferencing03	0 ... Deactivate referencing of ABR counter 1 1 ... Activate referencing of ABR counter 1 (positive edge activates referencing)
1 - 7		0 ... Reserved

4.8 Digital_in_do_rb: Status of digital inputs and outputs

Bit	Variable assignments in Automation Studio	Description
0	DigitalInput03	Status of digital input 1
1	DigitalInput04	Status of digital input 2
2	DigitalInput05	Status of digital input 3
3	DigitalInput06	Status of digital input 4
4	DigitalOutputReadBack02	Status of digital output 1
5	DigitalOutputReadBack03	Status of digital output 2
6	CounterOverflow01	Overflow of counter 1 (only valid if gate or period measurement is set in the counter configuration with bit 3-6): 0 ... Counter value is within the counting range (0 to \$7FFF). The value is only valid if bit 0 of the control register counter is set. 1 ... Counter overflow This bit is reset with Bit 0 of the control register counter.
7	CounterLatchDone01	State changes each time the counter state is successfully latched (reset value = 0)

4.9 StatusOutput16: Set digital outputs / control register counter

Bit	Variable assignments in Automation Studio	Description
0	CounterOverflowDetectEnable01	0 ... Disable detection of counter overflow and reset the overflow bit on counter 1 1 ... Enable detection of counter overflow
1		0 ... Reserved
2	ActivateRefCounter01	Referencing (only valid in AB(R) mode, see Bit 8+9 in counter configuration) 0 ... Disable referencing 1 ... Enable referencing
3	CounterEnable01	0 ... Reset counter (to zero) 1 ... Start counter Note: Only set after counter configuration is complete!
4 - 5		0 ... Reserved
6	DigitalOutput02	Set digital output 1
7	DigitalOutput03	Set digital output 2

5. 80SD100XD.C04X-13 Ramp function model

5.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	8	AbsPos01	AbsPos02	Set position/speed
4	12	MpGenControl01	MpGenControl02	Control word
6	14	MpGenMode01	MpGenMode02	Mode
7	15	MpGenBrake01	MpGenBrake02	Motor brake
16		OptionsBoardControl		D Set digital outputs / control register counter
0	8	AbsPos01ActVal	AbsPos02ActVal	Current position (cyclic)
4	12	MpGenStatus01	MpGenStatus02	Status
6		InputStatus		Input status
16		Counter01		D Counter
18		CounterLatched01		D Counter latch
20		Digital_in_do_rb		D Status of digital inputs and outputs
42		UZKVoltage		UZK DC bus voltage
44		CoolerTemperature		Temperature of output stage
46		Temperature		Temperature
48	112	ConfigOutput03a	ConfigOutput06a	Holding current
49	113	ConfigOutput04a	ConfigOutput07a	Rated current
50	114	ConfigOutput05a	ConfigOutput08a	Maximum current
52	116	MaxSpeed01pos	MaxSpeed02pos	Max. speed
54	118	MaxAcc01	MaxAcc02	Max. acceleration
56	120	MaxDec01	MaxDec02	Maximum deceleration
58	122	RevLoop01	RevLoop02	Turn-around loop
60	124	FixedPos01a	FixedPos02a	Set position 1
64	128	FixedPos01b	FixedPos02b	Set position 2
68	132	RefSpeed01	RefSpeed02	Homing speed
70	134	RefConfig01	RefConfig02	Homing configuration
72	136	FullStepThreshold01	FullStepThreshold02	Full step threshold
75	139	JoltTime01	JoltTime02	Jolt time limitation
80	144	ControlReadback01	ControlReadback02	Read back control word
82	146	ModeReadback01	ModeReadback02	Read back mode
84	148	Motoridentification01	Motoridentification02	Motor ID
86	150	RefPos01CyclicCounter	RefPos02CyclicCounter	Homed zero position for cyclic counter
90	154	AbsPos1ActValAcyclic	AbsPos2ActValAcyclic	G Current position (acyclic)
94	158	RefPos01AcyclicCounter	-	G Homed zero position for acyclic counter
98	162	ErrorCode01	ErrorCode02	Error code
200		ConfigOutput12		D Counter configuration

Table 76: Ramp function model

5.2 Motor brake

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	MpGenBrake01	MpGenBrake02	Digital output state (motor brake)
1 - 7			0 ... Reserved

The motor brake register is used to control the digital output (motor brake).

On the 2-channel ACOPOSmicro model, bit 0 of both registers (Motor 1 and Motor 2) is linked with an OR function. This means that the digital output is activated if the bit is set on one or both motors.

6. 80SD100XD.C0XX-01 Standard function model

6.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	2	Position1Sync	Position2Sync	Position
10		Error		Error/Status
12	14	Motoridentification01	Motoridentification02	Motor ID
16	24	Motor1Step0	Motor2Step0	Motor configuration MotorStep1
18	26	Motor1Step1	Motor2Step1	Motor configuration MotorStep2
20	28	Motor1Step2	Motor2Step2	Motor configuration MotorStep3
22	30	Motor1Step3	Motor2Step3	Motor configuration MotorStep4
33	36	ConfigOutput03	ConfigOutput06	Holding current
34	37	ConfigOutput04	ConfigOutput07	Rated current
35	38	ConfigOutput05	ConfigOutput08	Maximum current
46		ConfigOutput02		Module configuration
54		StatusOutput14		Error acknowledgement / Digital output motor brake
60	62	Position1Latched	Position2Latched	Position latch
68	70	usSinceTrigger01	usSinceTrigger02	usSinceTrigger
72		StepperLatchStatus		Latch trigger status
73		LifeCnt		Life cycle counter
76		StepperLatchConfig		Latch configuration
78		TriggerConfig		Trigger configuration
80		Temperature		Temperature
81		MotorIdentTrigger		Motor ID trigger
84	86	ConfigOutput09	ConfigOutput10	Full step threshold
88		CoolerTemperature		Cooler temperature
90		UZKVoltage		DC bus voltage

Table 77: Standard function model

6.2 Error: Error/Status

Bit	Variable assignments in Automation Studio	Description	
0		0 ... Reserved	
1	OvertemperatureError01	E 0 ... No error 1 ... Error: Overtemperature	Motor 1
2	CurrentError01	E 0 ... No error 1 ... Error: Set current not reached	
3	OvercurrentError01	E 0 ... No error 1 ... Error: Overcurrent	
4		0 ... Reserved	
5	OvertemperatureError02	E 0 ... No error 1 ... Error: Overtemperature	Motor 2
6	CurrentError02	E 0 ... No error 1 ... Error: Set current not reached	
7	OvercurrentError02	E 0 ... No error 1 ... Error: Overcurrent	
8	DrvOk01	S Status of drive 1: 0 ... Not ready 1 ... Ready	Motor 1
9	DrvOk02	S Status of drive 2: 0 ... Not ready 1 ... Ready	Motor 2
10 - 12		0 ... Reserved	
13	ModulePowerSupplyError	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage	
14	Enable	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK	
15	UZKStatus	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)	

6.3 StatusOutput14: Error acknowledgement / Digital output motor brake

Bit	Variable assignments in Automation Studio	Description
0	ClearError01	Error acknowledgement - Motor 1
1	ClearError02	Error acknowledgement - Motor 2
2 - 6		0 ... Reserved
7	DigitalOutput01	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

6.4 StepperLatchStatus: Latch trigger status

Bit	Variable assignments in Automation Studio	Description	
0	LatchInput01	Digital input for the latch event (level)	Stepper motor 1
1	LatchDone01	State changes each time the counter state is successfully latched (reset value = 0)	
2	LatchInput02	Digital input for the latch event (level)	Stepper motor 2
3	LatchDone02	State changes each time the counter state is successfully latched (reset value = 0)	
4	StatusInput01 TriggerInput01	Status of digital input 1 Status of trigger input 2	
5	StatusInput02 TriggerInput02	Status of digital input 2 Status of trigger input 2	
6 - 7		0 ... Reserved	

6.5 StepperLatchConfig: Latch configuration

Bit	Variable assignments in Automation Studio	Description	
0	StartLatch01	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 1
1 2	TriggerEdgePos01 TriggerEdgeNeg01	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 1 10 ... Latch position at negative edge on trigger input 1 11 ...Reserved	
3		0 ... Reserved	
4	StartLatch02	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 2
5 6	TriggerEdgePos02 TriggerEdgeNeg02	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 2 10 ... Latch position at negative edge on trigger input 2 11 ...Reserved	
7		0 ... Reserved	

6.6 TriggerConfig: Trigger configuration

Bit	Variable assignments in Automation Studio	Description
0	TriggerEdge01	0 ... Trigger 1: Trigger edge = positive 1 ... Trigger 1: Trigger edge = negative
1	StartTrigger01	Activate trigger 1 (when changes occur)
2	TriggerEdge02	0 ... Trigger 2: Trigger edge = positive 1 ... Trigger 2: Trigger edge = negative
3	StartTrigger02	Activate trigger 2 (when changes occur)
4 - 7		0 ... Reserved

7. 80SD100XD.C0XX-01 Ramp function model

7.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	8	AbsPos01	AbsPos02	Set position/speed
4	12	MpGenControl01	MpGenControl02	Control word
6	14	MpGenMode01	MpGenMode02	Mode
7	15	MpGenBrake01	MpGenBrake02	Motor brake
0	8	AbsPos01ActVal	AbsPos02ActVal	Current position (cyclic)
4	12	MpGenStatus01	MpGenStatus02	Status
6		InputStatus		Input status
42		UZKVoltage		UZK DC bus voltage
44		CoolerTemperature		Temperature of output stage
46		Temperature		Temperature
48	112	ConfigOutput03a	ConfigOutput06a	Holding current
49	113	ConfigOutput04a	ConfigOutput07a	Rated current
50	114	ConfigOutput05a	ConfigOutput08a	Maximum current
52	116	MaxSpeed01pos	MaxSpeed02pos	Max. speed
54	118	MaxAcc01	MaxAcc02	Max. acceleration
56	120	MaxDec01	MaxDec02	Maximum deceleration
58	122	RevLoop01	RevLoop02	Turn-around loop
60	124	FixedPos01a	FixedPos02a	Set position 1
64	128	FixedPos01b	FixedPos02b	Set position 2
68	132	RefSpeed01	RefSpeed02	Homing speed
70	134	RefConfig01	RefConfig02	Homing configuration
72	136	FullStepThreshold01	FullStepThreshold02	Full step threshold
75	139	JoltTime01	JoltTime02	Jolt time limitation
80	144	ControlReadback01	ControlReadback02	Read back control word
82	146	ModeReadback01	ModeReadback02	Read back mode
84	148	Motoridentification01	Motoridentification02	Motor ID
86	150	RefPos01CyclicCounter	RefPos02CyclicCounter	Homed zero position for cyclic counter
98	162	ErrorCode01	ErrorCode02	Error code

Table 78: Ramp function model

7.2 Motor brake

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	MpGenBrake01	MpGenBrake02	Digital output state (motor brake)
1 - 7			0 ... Reserved

The motor brake register is used to control the digital output (motor brake).

On the 2-channel ACOPOSmicro model, bit 0 of both registers (Motor 1 and Motor 2) is linked with an OR function. This means that the digital output is activated if the bit is set on one or both motors.

8. 80SD100XD.C0XX-21 Standard function model

8.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	2	Position1Sync	Position2Sync	Position
10		Error		Error/Status
12	14	Motoridentification01	Motoridentification02	Motor ID
16	24	Motor1Step0	Motor2Step0	Motor configuration MotorStep1
18	26	Motor1Step1	Motor2Step1	Motor configuration MotorStep2
20	28	Motor1Step2	Motor2Step2	Motor configuration MotorStep3
22	30	Motor1Step3	Motor2Step3	Motor configuration MotorStep4
33	36	ConfigOutput03	ConfigOutput06	Holding current
34	37	ConfigOutput04	ConfigOutput07	Rated current
35	38	ConfigOutput05	ConfigOutput08	Maximum current
46		ConfigOutput02		Module configuration
54		StatusOutput14		Error acknowledgement / Digital output motor brake
60	62	Position1Latched	Position2Latched	Position latch
68	70	usSinceTrigger01	usSinceTrigger02	usSinceTrigger
72		StepperLatchStatus		Latch trigger status
73		LifeCnt		Life cycle counter
76		StepperLatchConfig		Latch configuration
78		TriggerConfig		Trigger configuration
80		Temperature		Temperature
81		MotorIdentTrigger		Motor ID trigger
84	86	ConfigOutput09	ConfigOutput10	Full step threshold
88		CoolerTemperature		Cooler temperature
90		UZKVoltage		DC bus voltage

Table 79: Standard function model

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
Register for ACOPOSmicro models with additional analog inputs				
1409				Status of the analog inputs
1414		AnalogInput01		Analog input 1
1418		AnalogInput02		Analog input 2
Register for ACOPOSmicro models with additional analog inputs				
1038	1166	TraceEnable01	TraceEnable02	Trace control word
1090	1218			Trace status
1094	1222	FreeBufferSize01	FreeBufferSize02	Trace buffer status
1098	1226	TriggerCount01	TriggerCount02	Number of triggers executed
1102	1230	TriggerFailCount01	TriggerFailCount02	Number of failed triggers
1345				Trace trigger

Table 79: Standard function model (cont.)

8.2 Error: Error/Status

Bit	Variable assignments in Automation Studio	Description	
0		0 ... Reserved	
1	OvertemperatureError01	E 0 ... No error 1 ... Error: Overtemperature	Motor 1
2	CurrentError01	E 0 ... No error 1 ... Error: Set current not reached	
3	OvercurrentError01	E 0 ... No error 1 ... Error: Overcurrent	
4		0 ... Reserved	
5	OvertemperatureError02	E 0 ... No error 1 ... Error: Overtemperature	Motor 2
6	CurrentError02	E 0 ... No error 1 ... Error: Set current not reached	
7	OvercurrentError02	E 0 ... No error 1 ... Error: Overcurrent	
8	DrvOk01	S Status of drive 1: 0 ... Not ready 1 ... Ready	Motor 1
9	DrvOk02	S Status of drive 2: 0 ... Not ready 1 ... Ready	Motor 2
10 - 12		0 ... Reserved	
13	ModulePowerSupplyError	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage	
14	Enable	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK	
15	UZKStatus	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)	

8.3 StatusOutput14: Error acknowledgement / Digital output motor brake

Bit	Variable assignments in Automation Studio	Description
0	ClearError01	Error acknowledgement - Motor 1
1	ClearError02	Error acknowledgement - Motor 2
2 - 6		0 ... Reserved
7	DigitalOutput01	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

8.4 StepperLatchStatus: Latch trigger status

Bit	Variable assignments in Automation Studio	Description	
0	LatchInput01	Digital input for the latch event (level)	Stepper motor 1
1	LatchDone01	State changes each time the counter state is successfully latched (reset value = 0)	
2	LatchInput02	Digital input for the latch event (level)	Stepper motor 2
3	LatchDone02	State changes each time the counter state is successfully latched (reset value = 0)	
4	StatusInput01 TriggerInput01	Status of digital input 1 Status of trigger input 2	
5	StatusInput02 TriggerInput02	Status of digital input 2 Status of trigger input 2	
6 - 7		0 ... Reserved	

8.5 StepperLatchConfig: Latch configuration

Bit	Variable assignments in Automation Studio	Description	
0	StartLatch01	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 1
1 2	TriggerEdgePos01 TriggerEdgeNeg01	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 1 10 ... Latch position at negative edge on trigger input 1 11 ...Reserved	
3		0 ... Reserved	
4	StartLatch02	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register	Channel 2
5 6	TriggerEdgePos02 TriggerEdgeNeg02	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 2 10 ... Latch position at negative edge on trigger input 2 11 ...Reserved	
7		0 ... Reserved	

8.6 TriggerConfig: Trigger configuration

Bit	Variable assignments in Automation Studio	Description
0	TriggerEdge01	0 ... Trigger 1: Trigger edge = positive 1 ... Trigger 1: Trigger edge = negative
1	StartTrigger01	Activate trigger 1 (when changes occur)
2	TriggerEdge02	0 ... Trigger 2: Trigger edge = positive 1 ... Trigger 2: Trigger edge = negative
3	StartTrigger02	Activate trigger 2 (when changes occur)
4 - 7		0 ... Reserved

8.7 Status of the analog inputs

Bit	Variable assignments in Automation Studio	Description
0	AnalogInput01OK	Analog input 1
1	AnalogInput02OK	Analog input 2
2 - 7		0 ... Reserved

8.8 Trace control word

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	TraceEnable01	TraceEnable02	0 ... Trace function is disabled 1 ... Trace function is enabled
1 - 15			0 ... Reserved

8.9 Trace status

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	TraceEnabled01	TraceEnabled02	0 ... Trace function is disabled 1 ... Trace function is enabled
1			0 ... Reserved
2	TraceWriteActive01	TraceWriteActive02	0 ... Trace data recording not active 1 ... Trace data recording active
3	TraceReadActive01	TraceReadActive02	0 ... Trace data sending is not active 1 ... Trace data sending is active
4	ReadyForTrigger01	ReadyForTrigger02	0 ... Not ready for trigger 1 ... Ready for trigger
5	TriggerActive01	TriggerActive02	Trigger active 0 ... No trigger active, trace recording stopped 1 ... Trigger in process, trace data is being recorded
6	TraceOK01	TraceOK02	Trace OK 0 ... Buffer overrun or trace function disabled for analog input 1 ... No buffer overrun error
7	TraceError01	TraceError02	Trace error 0 ... No buffer overrun or trace function disabled for analog input 1 ... Buffer overrun error

8.10 Trace trigger

Bit	Variable assignments in Automation Studio	Description	
0	TraceTrigger01	Trigger for trace instance 1	The trigger event will be triggered at a positive edge (bit changes from 0 → 1).
1	TraceTrigger02	Trigger for trace instance 2	
2 - 7		0 ... Reserved	

9. 80SD100XD.C0XX-21 Ramp function model

9.1 Register overview

Register (dec.)		Variable assignments in Automation Studio		Description
Channel 1	Channel 2	Channel 1	Channel 2	
0	8	AbsPos01	AbsPos02	Set position/speed
4	12	MpGenControl01	MpGenControl02	Control word
6	14	MpGenMode01	MpGenMode02	Mode
7	15	MpGenBrake01	MpGenBrake02	Motor brake
0	8	AbsPos01ActVal	AbsPos02ActVal	Current position (cyclic)
4	12	MpGenStatus01	MpGenStatus02	Status
6		InputStatus		Input status
42		UZKVoltage		UZK DC bus voltage
44		CoolerTemperature		Temperature of output stage
46		Temperature		Temperature
48	112	ConfigOutput03a	ConfigOutput06a	Holding current
49	113	ConfigOutput04a	ConfigOutput07a	Rated current
50	114	ConfigOutput05a	ConfigOutput08a	Maximum current
52	116	MaxSpeed01pos	MaxSpeed02pos	Max. speed
54	118	MaxAcc01	MaxAcc02	Max. acceleration
56	120	MaxDec01	MaxDec02	Maximum deceleration
58	122	RevLoop01	RevLoop02	Turn-around loop
60	124	FixedPos01a	FixedPos02a	Set position 1
64	128	FixedPos01b	FixedPos02b	Set position 2
68	132	RefSpeed01	RefSpeed02	Homing speed
70	134	RefConfig01	RefConfig02	Homing configuration
72	136	FullStepThreshold01	FullStepThreshold02	Full step threshold
75	139	JoltTime01	JoltTime02	Jolt time limitation
80	144	ControlReadback01	ControlReadback02	Read back control word
82	146	ModeReadback01	ModeReadback02	Read back mode
84	148	Motoridentification01	Motoridentification02	Motor ID
86	150	RefPos01CyclicCounter	RefPos02CyclicCounter	Homed zero position for cyclic counter
98	162	ErrorCode01	ErrorCode02	Error code
1409 (16) ¹⁾				A Status of the analog inputs
1414 (18) ¹⁾		AnalogInput01		A Analog input 1
1418 (20) ¹⁾		AnalogInput02		A Analog input 2

Table 80: Ramp function model

Register (dec.)		Variable assignments in Automation Studio		Description	
Channel 1	Channel 2	Channel 1	Channel 2		
Trace register for ACOPOSmicro models with analog inputs ²⁾					
1038	1166	TraceEnable01	TraceEnable02	A	Trace control word
1090	1218			A	Trace status
1094	1222	FreeBufferSize01	FreeBufferSize02	A	Trace buffer status
1098	1226	TriggerCount01	TriggerCount02	A	Number of triggers executed
1102	1230	TriggerFailCount01	TriggerFailCount02	A	Number of failed triggers
1345				A	Trace trigger

Table 80: Ramp function model

- 1) The CANIOBusController function model is nearly identical to the ramp function model. However, the registers for the analog values are in a different range (values in parentheses).
- 2) Trace functionality is not provided for the CANIOBusController function model.

9.2 Motor brake

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	MpGenBrake01	MpGenBrake02	Digital output state (motor brake)
1 - 7			0 ... Reserved

The motor brake register is used to control the digital output (motor brake).

On the 2-channel ACOPOSmicro model, bit 0 of both registers (Motor 1 and Motor 2) is linked with an OR function. This means that the digital output is activated if the bit is set on one or both motors.

9.3 Status of the analog inputs

Bit	Variable assignments in Automation Studio	Description	
0	AnalogInput01OK	Analog input 1	0 ... OK 1 ... Outside the valid range or open circuit
1	AnalogInput02OK	Analog input 2	
2 - 7		0 ... Reserved	

9.4 Trace control word

Bit 0 of the trace control word is used to turn on/off the trace functionality for the corresponding analoginput.

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	TraceEnable01	TraceEnable02	0 ... Trace function is disabled 1 ... Trace function is enabled
1 - 15			0 ... Reserved

A buffer overrun error can be cleared by resetting bit 0.

9.5 Trace status

Bit	Variable assignments in Automation Studio		Description
	Channel 1	Channel 2	
0	TraceEnabled01	TraceEnabled02	0 ... Trace function is disabled 1 ... Trace function is enabled
1			0 ... Reserved
2	TraceWriteActive01	TraceWriteActive02	0 ... Trace data recording not active 1 ... Trace data recording active
3	TraceReadActive01	TraceReadActive02	0 ... Trace data sending is not active 1 ... Trace data sending is active
4	ReadyForTrigger01	ReadyForTrigger02	0 ... Not ready for trigger 1 ... Ready for trigger
5	TriggerActive01	TriggerActive02	Trigger active 0 ... No trigger active, trace recording stopped 1 ... Trigger in process, trace data is being recorded
6	TraceOK01	TraceOK02	Trace OK 0 ... Buffer overrun or trace function disabled for analog input 1 ... No buffer overrun error
7	TraceError01	TraceError02	Trace error 0 ... No buffer overrun or trace function disabled for analog input 1 ... Buffer overrun error

9.6 Trace trigger

This register can be used to start recording separately for each trace instance. This is only possible if the trigger function has been enabled for the corresponding analog input.

Bit	Variable assignments in Automation Studio	Description	
0	TraceTrigger01	Trigger for trace instance 1	The trigger event will be triggered at a positive edge (bit changes from 0 → 1).
1	TraceTrigger02	Trigger for trace instance 2	
2 - 7		0 ... Reserved	

10. 80SD100XS.C04X-01 Standard function model

10.1 Register overview

Register (dec.)	Variable assignments in Automation Studio	Description
0	Position1Sync	Position
10	Error	Error/Status
12	Motoridentification01	Motor ID
16	Motor1Step0	Motor configuration MotorStep1
18	Motor1Step1	Motor configuration MotorStep2
20	Motor1Step2	Motor configuration MotorStep3
22	Motor1Step3	Motor configuration MotorStep4
33	ConfigOutput03	Holding current
34	ConfigOutput04	Rated current
35	ConfigOutput05	Maximum current
46	ConfigOutput02	Module configuration
54	StatusOutput14	Error acknowledgement / Digital output motor brake
60	Position1Latched	Position latch
68	usSinceTrigger01	usSinceTrigger
72	StepperLatchStatus	Latch trigger status
73	LifeCnt	Life cycle counter
76	StepperLatchConfig	Latch configuration
78	TriggerConfig	Trigger configuration
80	Temperature	Temperature
81	MotorIdentTrigger	Motor ID trigger
84	ConfigOutput09	Full step threshold
88	CoolerTemperature	Cooler temperature
90	UZKVoltage	DC bus voltage
Register for ACOPOSmicro with encoder		
130	PosVal03	ABR counter
134	LatchVal03	ABR counter latch
154	StatusOutput15	ABR control register
155	ConfigOutput11	ABR configuration
158	ConfigOutput13	ABR advanced configuration

Table 81: Standard function model

10.2 Error: Error/Status

Bit	Variable assignments in Automation Studio	Description
0		0 ... Reserved
1	OvertemperatureError01	E 0 ... No error 1 ... Error: Overtemperature
2	CurrentError01	E 0 ... No error 1 ... Error: Set current not reached
3	OvercurrentError01	E 0 ... No error 1 ... Error: Overcurrent
4 - 7		0 ... Reserved
8	DrvOk01	S Status of drive: 0 ... Not ready 1 ... Ready
9		0 ... Reserved
10	IncrementalEncoderOk	Status of ABR encoder: 0 ... Not ready 1 ... Ready
11	EncoderPowerSupplyError	E Encoder supply error 0 ... 24 VDC OK 1 ... Error: Encoder supply not in the required range
12		0 ... Reserved
13	ModulePowerSupplyError	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage
14	Enable	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK
15	UzkStatus	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)

10.3 StatusOutput14: Error acknowledgement / Digital output motor brake

Bit	Variable assignments in Automation Studio	Description
0	ClearError01	Error acknowledgement for motor
1 - 6		0 ... Reserved
7	DigitalOutput01	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

10.4 StepperLatchStatus: Latch trigger status

Bit	Variable assignments in Automation Studio	Description
0	LatchInput01	Digital input for the latch event (level)
1	LatchDone01	State changes each time the counter state is successfully latched (reset value = 0)
2 - 3		0 ... Reserved
4	DigitalInput01 TriggerInput01	Status of digital input 1 Status of trigger input 2
5	DigitalInput02	Status of digital input 2
6	LatchDone03	State changes each time ABR counter is successfully latched (reset value = 0) See ABR counter control register and ABR counter configuration register
7		0 ... Reserved

10.5 StepperLatchConfig: Latch configuration

Bit	Variable assignments in Automation Studio	Description
0	StartLatch01	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register
1 2	TriggerEdgePos01 TriggerEdgeNeg01	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 10 ... Latch position at negative edge on trigger input 11 ...Reserved
3 - 7		0 ... Reserved

10.6 TriggerConfig: Trigger configuration

Bit	Variable assignments in Automation Studio	Description
0	TriggerEdge01	0 ... Trigger: Trigger edge = positive 1 ... Trigger: Trigger edge = negative
1	StartTrigger01	Activate trigger (when changes occur)
2 - 7		0 ... Reserved

10.7 StatusOutput15: ABR control register

Bit	Variable assignments in Automation Studio	Description
0	ActivateReferencing03	0 ... Deactivate referencing of ABR counter 1 ... Activate referencing of ABR counter (positive edge activates referencing)
1 - 7		0 ... Reserved

11. 80SD100XS.C04X-01 Ramp function model

11.1 Register overview

Register (dec.)	Variable assignments in Automation Studio	Description
0	AbsPos01	Set position/speed
4	MpGenControl01	Control word
6	MpGenMode01	Mode
7	MpGenBrake01	Motor brake
0	AbsPos01ActVal	Current position (cyclic)
4	MpGenStatus01	Status
6	InputStatus	Input status
42	UZXVoltage	UZX DC bus voltage
44	CoolerTemperature	Temperature of output stage
46	Temperature	Temperature
48	ConfigOutput03a	Holding current
49	ConfigOutput04a	Rated current
50	ConfigOutput05a	Maximum current
52	MaxSpeed01pos	Max. speed
54	MaxAcc01	Max. acceleration
56	MaxDec01	Maximum deceleration
58	RevLoop01	Turn-around loop
60	FixedPos01a	Set position 1
64	FixedPos01b	Set position 2
68	RefSpeed01	Homing speed
70	RefConfig01	Homing configuration
72	FullStepThreshold01	Full step threshold
75	JoltTime01	Jolt time limitation
80	ControlReadback01	Read back control word
82	ModeReadback01	Read back mode
84	Motoridentification01	Motor ID
86	RefPos01CyclicCounter	Homed zero position for cyclic counter
90	AbsPos1ActValAcyclic	G Current position (acyclic)
94	RefPos01AcyclicCounter	G Homed zero position for acyclic counter
98	ErrorCode01	Error code

Table 82: Ramp function model

11.2 Motor brake

Bit	Variable assignments in Automation Studio	Description
0	MpGenBrake01	Digital output state (motor brake)
1 - 7		0 ... Reserved

This register is used to control the digital output (motor brake).

12. 80SD100XS.C0XX-01 Standard function model

12.1 Register overview

Register (dec.)	Variable assignments in Automation Studio	Description
0	PositionSync	Position
10	Error	Error/Status
12	Motoridentification	Motor ID
16	MotorStep0	Motor configuration MotorStep1
18	MotorStep1	Motor configuration MotorStep2
20	MotorStep2	Motor configuration MotorStep3
22	MotorStep3	Motor configuration MotorStep4
33	ConfigOutput03	Holding current
34	ConfigOutput04	Rated current
35	ConfigOutput05	Maximum current
46	ConfigOutput02	Module configuration
54	StatusOutput14	Error acknowledgement / Digital output motor brake
60	PositionLatched	Position latch
68	UsSinceTrigger	usSinceTrigger
72	StepperLatchStatus	Latch trigger status
73	LifeCnt	Life cycle counter
76	StepperLatchConfig	Latch configuration
78	TriggerConfig	Trigger configuration
80	Temperature	Temperature
81	MotorIdentTrigger	Motor ID trigger
84	ConfigOutput09	Full step threshold
88	CoolerTemperature	Cooler temperature
90	UZXVoltage	DC bus voltage

Table 83: Standard function model

12.2 Error: Error/Status

Bit	Variable assignments in Automation Studio	Description
0		0 ... Reserved
1	OvertemperatureError	E 0 ... No error 1 ... Error: Overtemperature
2	CurrentError	E 0 ... No error 1 ... Error: Set current not reached
3	OvercurrentError	E 0 ... No error 1 ... Error: Overcurrent
4 - 7		0 ... Reserved
8	DrvOk	S Status of drive: 0 ... Not ready 1 ... Ready
9 - 12		0 ... Reserved
13	ModulePowerSupplyError	E Module supply error 0 ... 24 VDC OK 1 ... Error: Under-voltage
14	Enable	S Enable status (corresponds with Enable input) 0 ... Disable 1 ... Enable OK
15	UZKStatus	S Status of supply voltage for power element (U_{ZK} ... DC bus voltage): 0 ... Supply voltage for power element not in the valid range 1 ... Supply voltage for power element OK ($18\text{ V} \leq U_{ZK} \leq 80\text{ V}$)

12.3 StatusOutput14: Error acknowledgement / Digital output motor brake

Bit	Variable assignments in Automation Studio	Description
0	ClearError	Error acknowledgement for motor
1 - 6		0 ... Reserved
7	DigitalOutput	Status of digital output / motor brake 0 ... Motor brake off 1 ... Motor brake on

12.4 StepperLatchStatus: Latch trigger status

Bit	Variable assignments in Automation Studio	Description
0	LatchInput	Digital input for the latch event (level)
1	LatchDone	State changes each time the counter state is successfully latched (reset value = 0)
2 - 3		0 ... Reserved
4	StatusInput01 TriggerInput	Status of digital input 1 Status of trigger input 2
5	StatusInput02	Status of digital input 2
6 - 7		0 ... Reserved

12.5 StepperLatchConfig: Latch configuration

Bit	Variable assignments in Automation Studio	Description
0	StartLatch	Turn on/off latch function for position register 0 ... Negative edge: Turn off latch function for position register 1 ... Positive edge: Turn on latch function for position register
1 2	TriggerEdgePos TriggerEdgeNeg	Configure latch function for position register 00 ... Unconditional latch position 01 ... Latch position at positive edge on trigger input 10 ... Latch position at negative edge on trigger input 11 ... Reserved
3 - 7		0 ... Reserved

12.6 TriggerConfig: Trigger configuration

Bit	Variable assignments in Automation Studio	Description
0	TriggerEdge	0 ... Trigger: Trigger edge = positive 1 ... Trigger: Trigger edge = negative
1	StartTrigger	Activate trigger (when changes occur)
2 - 7		0 ... Reserved

13. 80SD100XS.C0XX-01 Ramp function model

13.1 Register overview

Register (dec.)	Variable assignments in Automation Studio	Description
0	AbsPos01	Set position/speed
4	MpGenControl01	Control word
6	MpGenMode01	Mode
7	MpGenBrake01	Motor brake
0	AbsPos01ActVal	Current position (cyclic)
4	MpGenStatus01	Status
6	InputStatus	Input status
42	UZKVoltage	UZK DC bus voltage
44	CoolerTemperature	Temperature of output stage
46	Temperature	Temperature
48	ConfigOutput03a	Holding current
49	ConfigOutput04a	Rated current
50	ConfigOutput05a	Maximum current
52	MaxSpeed01pos	Max. speed
54	MaxAcc01	Max. acceleration
56	MaxDec01	Maximum deceleration
58	RevLoop01	Turn-around loop
60	FixedPos01a	Set position 1
64	FixedPos01b	Set position 2
68	RefSpeed01	Homing speed
70	RefConfig01	Homing configuration
72	FullStepThreshold01	Full step threshold
75	JoltTime01	Jolt time limitation
80	ControlReadback01	Read back control word
82	ModeReadback01	Read back mode
84	Motoridentification01	Motor ID
86	RefPos01CyclicCounter	Homed zero position for cyclic counter
98	ErrorCode01	Error code

Table 84: Ramp function model

13.2 Motor brake

Bit	Variable assignments in Automation Studio	Description
0	MpGenBrake01	Digital output state (motor brake)
1 - 7		0 ... Reserved

This register is used to control the digital output (motor brake).

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